

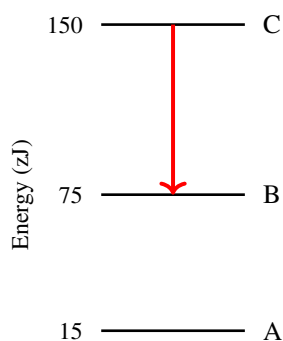
Name: _____

This practice exam covers Chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Determine which bond is expected to have the longest equilibrium bond distance from the following bonds: the P–O bond in phosphate (PO_4^{3-}), the C–C bond in ethene (C_2H_4), the S–O bond in sulfite (SO_3^{2-}), the C–C bond in propyne (C_3H_4), the S–S bond in hydrogen disulfide (H_2S_2), and the N–O bond in nitrogen dioxide (NO_2). Justify your answer from fundamental principles. Responses based on memorized trends or rules will not receive credit.

ANSWER: _____

2. This energy diagram shows the allowed energy levels of an electron in a certain atom. If the electron makes a transition from C to B, as shown by the arrow, will a photon be absorbed or emitted? You must explain your answer choice for full credit.



ANSWER: _____

3. Consider the species H_2O . Draw its Lewis structure and use it to answer the following questions. How many total valence electrons must appear in the Lewis structure of H_2O ? How many lone pairs surround the central oxygen atom? Credit is awarded if you draw the Lewis structure correctly and answer both questions correctly.

ANSWER: _____

4. Answer the following questions: is Li^+ an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: _____

5. Rank the following elements from highest metallic to lowest metallic. Support your ranking with an explanation of the underlying atomic structure; answers that simply cite a memorized periodic trend will not receive credit.

Zr, Sn, Pd, Ag

ANSWER: _____

6. There are two steps in the extraction of copper metal from chalcocite, a copper ore. In the first step, copper (I) sulfide and oxygen gas react to form copper (I) oxide and sulfur dioxide. In the second step, copper (I) oxide and carbon react to form copper and carbon monoxide. Write the net chemical equation for the production of copper from copper (I) sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

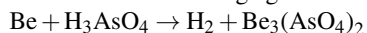
7. Sodium bromide reacts with hydronium fluoride to form hydronium bromide and sodium fluoride. Suppose 31.67 g of sodium bromide is mixed with 7.750 g of hydronium fluoride. Calculate the maximum mass of sodium fluoride that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

8. A 9.0 g sample of propane gas (C_3H_8) is held in a gas collection bulb at $12.^\circ\text{C}$ and 1.48 atm. Calculate the volume of the sample assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

9. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.



ANSWER: _____

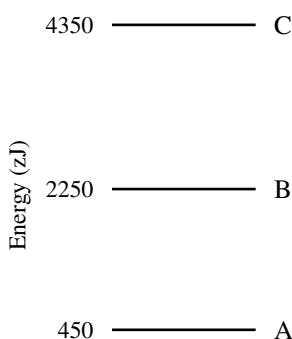
10. An unknown metal crystallizes in a FCC structure with unit cell edge length 404.95 pm and density 2.7 g/cm^3 . Calculate the molar mass of the metal and identify it.

ANSWER: _____

11. In the MO diagram of O_2 , which lies lower in energy — the σ_{2p} orbital or the π_{2p} orbitals?

ANSWER: _____

12. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the highest frequency? You must explain your answer choice for full credit.



ANSWER: _____

13. Shiv is comparing two unknown liquid samples. Sample A has a mass of 59.2 g and a volume of 7.28 mL. Sample B has a mass of 14.6 g and a volume of 27.2 mL. Which sample has the greater density? You must include correct work that supports your conclusion.

ANSWER: _____

14. Propane (C_3H_8) can be turned into hydrogen by a two-step reforming process. In the first step, propane and water react to form carbon monoxide and hydrogen. In the second step, carbon monoxide and water react to form hydrogen and carbon dioxide. Write the net chemical equation for the production of hydrogen from propane and water. Ensure that it is balanced.

ANSWER: _____

15. A particular planet has a mass of 8.3×10^{26} g and a virus has a mass of 4.62×10^{-11} mg. What is the mass of 1 mole of viruses? Ensure your answer has the correct number of significant figures.

ANSWER: _____

16. It takes 182 kJ/mol to break a bond between two particular elements. Calculate the maximum wavelength of light that will break one of these bonds.

ANSWER: _____

17. Calculate the mole fractions of methanol (CH_3OH) and water in a solution prepared by mixing 6.40 g of methanol with 90.0 g of water.

ANSWER: _____

18. Consider the reaction $\text{PCl}_5(\text{g}) \longrightarrow \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$. If $11. \text{cm}^3$ of PCl_5 gas is completely consumed at the same temperature and pressure, what volume of Cl_2 gas is produced? Justify your answer.

ANSWER: _____

19. Provide the name for the molecular compound N_2O_5 . If the compound has a common name, you may provide that instead.

ANSWER: _____

20. Blood plasma has a total dissolved-particle concentration of about 0.300 osmol/L. Classify each of the following as hypertonic, hypotonic, or isotonic with blood: 0.200 M KCl, 0.100 M Na_2SO_4 , 0.250 M $\text{C}_{12}\text{H}_{22}\text{O}_{11}$. Justify each classification.

ANSWER: _____

21. Using MO theory, determine the bond order, magnetic behavior, and whether Li_2^+ is expected to exist as a stable molecule. Briefly justify your answers.

ANSWER: _____

22. Identify the hybridization of each carbon atom in C_6H_6 (benzene). Explain your answer.

ANSWER: _____

23. The N atom in NH_4^+ has 4 regions of electron density. Determine its hybridization. Justify your answer.

ANSWER: _____

24. A physical chemist measures the width of a molecule. The width is 72 nm. What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

25. CaF_2 crystallizes in a fluorite structure with unit cell edge length 546.3 pm. Calculate the density of CaF_2 .

ANSWER: _____

26. For I_2 , the potential-energy well is 151 kJ/mol deep, measured from $\text{PE} = 0$ at large separation down to the minimum. What molecular quantity equals this value and what does it describe? Justify your answer.

ANSWER: _____

27. Consider the following compounds: nitrogen monoxide, ethylene, boron trifluoride, and carbon dioxide. Write the molecular formula for each of the listed compounds that has a dipole moment. If none have a dipole moment, write "None have a dipole moment". Draw each compound and explain your answer.

ANSWER: _____

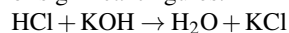
28. A civil engineer measures the length of an airport runway. The length is 4490 m. What is the length in kilometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

29. Several types of electromagnetic radiation (EMR) are: blue light, red light, yellow light, and orange light. Order these types of EMR from lowest energy to highest energy.

ANSWER: _____

30. Given this unbalanced chemical equation, report the mass of HCl needed to produce 6.71 g of KCl . Ensure your answer has the correct number of significant figures.



ANSWER: _____

31. A central atom in SO_3 has 3 bonds and 0 lone pairs. Predict its hybridization, electron-group geometry, and molecular geometry. Explain how the number of electron groups and lone pairs determines each answer.

ANSWER: _____

32. A bond prevents free rotation because rotating it would require breaking orbital overlap. Is this explained primarily by hybridization or by σ/π bonding? Justify your answer.

ANSWER: _____

33. Tyler conducted some measurements and determined that 659. g of nitrogen gas participated in a chemical reaction. Calculate the number of moles of nitrogen gas that participated in the chemical reaction.

ANSWER: _____

34. Christopher weighs out 1.164 g of carbonic acid (H_2CO_3) into a 100.-mL volumetric flask and dilutes the solution to the mark with distilled water. He plans to titrate the acid with 0.361-M bismuth hydroxide solution. Calculate the volume of bismuth hydroxide solution that Christopher must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

ANSWER: _____

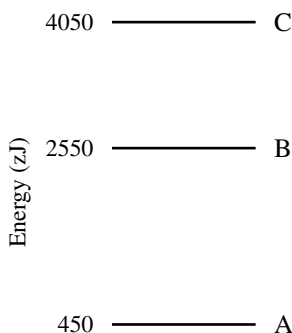
35. Equal masses (1.00 g each) of LiCl , $\text{C}_6\text{H}_{12}\text{O}_6$, AlCl_3 are each dissolved in 1.00 kg of water. Which solution has the greatest freezing point depression?

ANSWER: _____

36. Phosphoric acid, which most commonly finds uses as a rust inhibitor, a food additive, and an etching agent for dental and orthopedic use, can be synthesized using a two-step thermal process. In the first step, tetratomic phosphorus and diatomic oxygen react to form diphosphorus pentoxide. In the second step, diphosphorus pentoxide and water react to form phosphoric acid. Write the net chemical equation for the production of phosphoric acid from tetratomic phosphorus, diatomic oxygen, and water. Ensure that it is balanced.

ANSWER: _____

37. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the shortest wavelength? You must explain your answer choice for full credit.



ANSWER: _____

38. Consider the reaction $2\text{C}_2\text{H}_2(\text{g}) + 5\text{O}_2(\text{g}) \longrightarrow 4\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$. If 8.11 cm^3 of C_2H_2 gas is completely consumed at the same temperature and pressure, what volume of CO_2 gas is produced? Justify your answer.

ANSWER: _____

39. Draw the valence molecular-orbital energy-level diagram for F_2 and fill in its electrons. Determine the bond order and whether F_2 is paramagnetic or diamagnetic. Justify your conclusions.

ANSWER: _____

40. Of the following labels, identify all that can be applied to the chemical compound GeS_2 : ionic, molecular, acid, and base. Clearly explain why you chose your label(s).

ANSWER: _____

41. You know a solution's molarity and need its molality. What additional property must you measure, and why is it required?

ANSWER: _____

42. A student concludes that hybrid orbitals must have the same shapes as the original s and p orbitals because they are made from them. Is the conclusion correct? Justify your answer.

ANSWER: _____

43. A flexible gas bag contains 2.82 L of neon at -14.3°C while indoors. It is then taken outside, where the air temperature is 29.0°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

ANSWER: _____

44. A solution contains exactly the maximum amount of solute at a given temperature. Classify the solution as saturated, unsaturated, or super-saturated. You must explain your answer choice for credit.

ANSWER: _____

45. For germanium (Ge), the solid is less dense than the liquid. Does increasing the external pressure raise or lower its melting point? Justify.

ANSWER: _____

46. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur (S_8) reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide, oxygen gas, and water combine to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.

ANSWER: _____

47. Classify KOH (KOH) as a strong electrolyte, weak electrolyte, or nonelectrolyte in aqueous solution. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

48. Two students draw different valid Lewis structures for O_3 . One assigns the central O as sp^2 ; the other claims the assignment depends on which Lewis structure is used. Which student is correct? Justify your answer. Justify your answer completely.

ANSWER: _____

49. A geophysicist measures the pressure in a borehole. The pressure is 6.079×10^7 Pa. What is the pressure in megapascals? Report your answer with units. Report your answer in scientific notation.

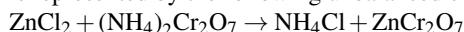
ANSWER: _____

50. Given this unbalanced chemical equation, calculate and report the percent yield of H_3OI if 4.200 g of H_3OI is produced from the reaction of 8.361 g of AlI_3 and 1.91 g of H_3OF . Ensure that your answer has the correct number of significant figures.

$AlI_3 + H_3OF \rightarrow H_3OI + AlF_3$

ANSWER: _____

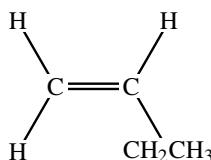
51. A way to produce solid zinc dichromate is by combining aqueous zinc chloride with aqueous ammonium dichromate. The chemical reaction is represented by the following unbalanced chemical equation:



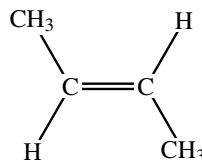
Sophia wants to produce 13.29 mg of zinc dichromate. She pours 12.77 mL of a 0.4967-M solution of aqueous zinc chloride into a beaker. What volume of a 0.051-M solution of aqueous ammonium dichromate is needed to react with the zinc chloride solution to produce 13.29 mg of zinc dichromate? If you find that 12.77 mL of the zinc chloride solution is insufficient, report the required volumes of both reactants required to produce the 13.29 mg of zinc dichromate.

ANSWER: _____

52. Two molecular representations are shown. Decide whether both drawings could represent the same sample. A memorized trend will not receive credit; your reasoning must account for the molecular structure.



Structure A



Structure B

ANSWER: _____

53. Andrew must dilute 63.0 mL of a 4.5-mM solution of aqueous ammonium sulfide $(\text{NH}_4)_2\text{S}$ until the concentration falls to 3.0 mM. He will do this by adding distilled water to the solution. How much distilled water must he add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

54. Ice at a constant 0.003 atm is heated from -20°C to 50°C . Name the phase transition(s) that occur, in order, and justify your answer using the phase diagram.

ANSWER: _____

55. State the SI base unit used to measure mass and give its symbol.

ANSWER: _____

56. A sample of 25 g of potassium chlorate decomposes completely to produce potassium chloride and oxygen gas in a sealed container. The container initially contains 0.260 mol of neon gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of neon after the reaction is complete.

ANSWER: _____

57. A 416-mL sample of ethane (C_2H_6) is initially at 283 K. Its boiling point is $-89.^\circ\text{C}$ (184.15 K). The pressure is then reduced to 66% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest $^\circ\text{C}$. Justify your answer.

ANSWER: _____

58. Using MO theory, determine the bond order of B_2^- . Briefly justify your answer.

ANSWER: _____

59. The enthalpy of solution of NaCl (NaCl) is $\Delta H_{\text{sol}} = +3.9 \text{ kJ/mol}$. Is dissolution endothermic or exothermic? Will the temperature of the solution increase or decrease when NaCl dissolves in water?

ANSWER: _____

60. Calculate the mass of a sample of antimony triiodide (SbI_3) that contains 1.27 iodine atoms.

ANSWER: _____

61. Sam measures the amount of strontium phosphate produced during a chemical reaction. He finds that 912.05 g is produced. Calculate the number of moles of strontium phosphate produced. Ensure your answer has the correct number of significant figures.

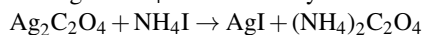
ANSWER: _____

62. Fill in the information missing from the following table. For each subshell, you must provide the principal quantum number n , the angular momentum quantum number l , and the maximum number of electrons that can fit in the subshell N_e . Credit is awarded only if the table is completed correctly. You may fill in the table directly; there is no need to redraw it.

Subshell	n	l	N_e
3p			
4d			
3d			

ANSWER: _____

63. Given this unbalanced chemical equation, calculate and report the theoretical yield of AgI formed by the mixing of 4.687 g of $\text{Ag}_2\text{C}_2\text{O}_4$ with 7.678 g of NH_4I . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

64. A gas is reported to be He. Its density is measured as $280 \frac{\text{g}}{\text{m}^3}$ at 162 kPa and 7.2°C . Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

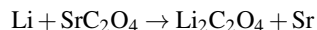
65. Provide the name for the molecular compound NF_3 . If the compound has a common name, you may provide that instead.

ANSWER: _____

66. Determine the number of σ bonds and the number of π bonds present in HCN. Justify your answer.

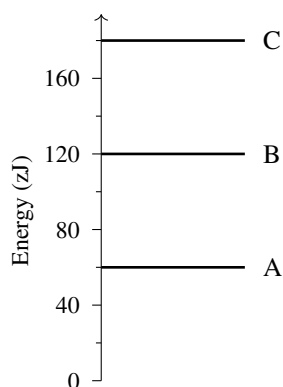
ANSWER: _____

67. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.



ANSWER: _____

68. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its ground state? Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

69. For the central atom in CCl_4 , determine its hybridization, electron-group geometry, molecular geometry, and the ideal bond angle(s). Justify your answer.

ANSWER: _____

70. Several types of electromagnetic radiation (EMR) are: orange light, green light, gamma rays, and microwaves. Order these types of EMR from lowest frequency to highest frequency.

ANSWER: _____

71. Rachel adds 148 mL of a 0.5605-M solution of calcium chlorate ($\text{Ca}(\text{ClO}_3)_2$) to a reaction flask. Calculate the amount of calcium chlorate that she has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

72. The mass enthalpy of vaporization of water (H_2O) is 2260 J/g and $M = 18.015 \text{ g/mol}$. Find the molar enthalpy of vaporization of water.

ANSWER: _____

73. Provide the element symbol that represents iron.

ANSWER: _____

74. Aaron needs to determine the mass of a sample of motor oil. The density of the sample is $0.933 \frac{\text{g}}{\text{mL}}$ and its volume is 7.97 mL. What is the mass of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

75. Sophia measures the amount of chromium (III) oxide produced during a chemical reaction. She finds that 0.638 mol is produced. Calculate the mass of chromium (III) oxide produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

76. Two solutions are prepared: solution A is a 0.200 m solution of NaCl (NaCl) in water, and solution B is a 0.200 m solution of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in water. Which solution has the greater boiling point elevation? Calculate both values. (Assume complete dissociation for ionic solutes.)

ANSWER: _____

77. When a narrow wax (paraffin) capillary tube is dipped into water, does the water rise or become depressed in the tube? Justify using cohesive and adhesive forces.

ANSWER: _____

78. Provide the element symbol that represents nitrogen.

ANSWER: _____

79. Rank the following ionic compounds in order of increasing lattice energy (smallest to largest). Briefly justify your ranking using Coulomb's law.

NaCl, MgO, RbI

ANSWER: _____

80. The species O_3 exhibits resonance. Draw all of its equivalent resonance structures. What is the bond order of each O–O bond in the resonance hybrid?

ANSWER: _____

81. A high-pressure gas cylinder is allowed to fill a large empty collection bag. The cylinder initially contains gas at 3.03 atm. The valve is opened until the pressure inside the cylinder becomes atmospheric pressure and is then closed. Determine the fraction of the original gas that remains inside the cylinder. Justify your answer.

ANSWER: _____

82. Rank the following substances from lowest to highest vapor pressure at room temperature: $(C_2H_5)_2O$, C_2H_5OH , H_2O . For credit you must justify your answer.

ANSWER: _____

83. How many hydrogen-bond donors does one molecule of CH_3COOH have?

ANSWER: _____

84. A gas sample is trapped inside a syringe with its tip sealed. Its initial volume is 12 L. After the piston is moved, the gas occupies 1.79 L and its pressure is measured to be 603 kPa. The temperature remains constant throughout. Determine the initial pressure. Justify your answer.

ANSWER: _____

85. Dissolving sodium hydroxide (NaOH) is exothermic ($\Delta H_{sol} = -44.5 \text{ kJ/mol}$). Identify the two factors that decide whether it dissolves spontaneously. Explain.

ANSWER: _____

86. Draw a valid Lewis structure for AsH_3 . You must show all existing lone pairs and lone electrons.

ANSWER: _____

87. Predict what happens to solubility when KNO_3 is dissolved in hot water and then cooled. Identify whether temperature or pressure is the key factor and explain the observation.

ANSWER: _____

88. Provide the chemical symbol of the most commonly formed ion of copper and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: _____

89. Nicholas is instructed to dissolve 0.994 g of ammonium fluoride in 168 mL of a 279.3-mM aqueous solution of aluminum iodide. Calculate the molarity of the aluminum cation in the solution after the reaction has occurred. Assume the volume of the solution does not change when aluminum is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

90. Suppose that there is heat transfer of 33 J to a system, while the system does 35 J of work. Later, there is heat transfer of 49 J out of the system while 37 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: _____

91. A pharmacist measures the mass of a powdered drug. The mass is 9533 mg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

92. Determine the electron-group arrangement, molecular shape, and ideal bond angle(s) for carbon monoxide.

ANSWER: _____

93. Propane (C_3H_8) can be turned into hydrogen by a two-step reforming process. In the first step, propane and water react to form carbon monoxide and hydrogen. In the second step, carbon monoxide and water react to form hydrogen and carbon dioxide. Write the net chemical equation for the production of hydrogen from propane and water. Ensure that it is balanced.

ANSWER: _____

94. Arrange the following substances from highest to lowest boiling point: SiCl_4 , CO_2 , CS_2 , CCl_4 . Justify your ranking. Responses based only on memorized trends will not receive credit.

ANSWER: _____

95. Which concentration unit is a dimensionless ratio of particle counts, free of any volume or temperature dependence, and where is that property most useful?

ANSWER: _____

96. A Canadian meteorologist converts a temperature reading for an American audience. The outdoor temperature is -20°C . Convert this temperature to degrees Fahrenheit and round to the nearest integer.

ANSWER: _____

97. A fixed amount of gas changes from 22.8°C to -31°C while its pressure changes from 4.65 atm to 2.28 atm. Determine the final volume as a percentage of the initial volume. Justify your answer.

ANSWER: _____

98. Two parallel unhybridized p orbitals are rotated until they are perpendicular to one another while the atoms remain the same distance apart. Predict what happens to the additional bond between the atoms. Justify your answer.

ANSWER: _____

99. Of the following labels, identify all that can be applied to the chemical compound LiOH : ionic, molecular, acid, and base. Clearly explain why you chose your label(s).

ANSWER: _____

100. A solution of HCl (HCl) is 20.0% by mass. Calculate the molality of the solution. ($\mathcal{M}(\text{HCl}) = 36.46 \text{ g/mol}$)

ANSWER: _____

101. In the MO diagram of B_2 , which lies lower in energy — the σ_{2p} orbital or the π_{2p} orbitals?

ANSWER: _____

102. What is the oxidation number of chlorine in sulfur dichloride?

ANSWER: _____

103. A solution contains more than the equilibrium maximum at a given temperature. Classify the solution as saturated, unsaturated, or supersaturated. You must explain your answer choice for credit.

ANSWER: _____

104. Rank the following liquids from highest to lowest surface tension at 20°C : H_2O , C_6H_{14} , CH_4 , Hg. Offer brief explanation for your ranking.

ANSWER: _____

105. An unknown solid has a melting point of 3550°C , is a non-conductor as a solid, and is a non-conductor. Classify it as ionic, metallic, covalent-network, or molecular.

ANSWER: _____

106. The reaction $2\text{H}_2\text{O}_2(\text{aq}) \longrightarrow 2\text{H}_2\text{O}(\text{l}) + \text{O}_2(\text{g})$ is carried out in a sealed, initially evacuated vessel with a volume of 19.3 L. After 17.1 g of H_2O_2 completely reacts while all other reactants are present in excess at 34.3°C , determine the pressure of O_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

107. A rigid 11.4-L tank at $101.^\circ\text{C}$ contains only chlorine (Cl_2) and dinitrogen monoxide (N_2O). Assume both gases behave ideally. The tank contains 11.4 g of chlorine and 11 g of dinitrogen monoxide. All of the chlorine is removed without changing the temperature or the volume. Determine the decrease in the tank pressure. Justify your answer.

ANSWER: _____

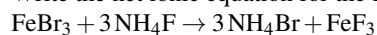
108. A 0.100m aqueous solution of NaCl (NaCl) shows a freezing point depression of 0.372°C ($K_f = 1.86^{\circ}\text{C kg/mol}$). Determine the effective van't Hoff factor and state what it reveals about the solute in solution.

ANSWER: _____

109. How many hydrogen-bond acceptors does one molecule of NH_3 have?

ANSWER: _____

110. Write the net ionic equation for the following chemical reaction, which takes place in aqueous solution:

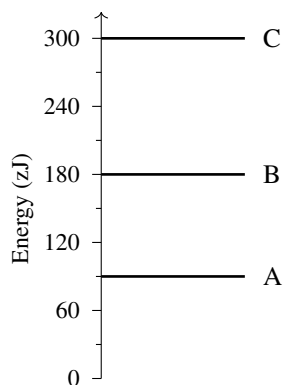


ANSWER: _____

111. Linda puts a 0.577-kg aluminum pot on the stove and fills it with 0.14 liters of water. Both the pot and the water are initially at 1.0°C . How much heat is required to bring them both to 61.0°C ? The specific heat value of water is $4186 \text{ J/kg}^{\circ}\text{C}$ and for aluminum it is $900 \text{ J/kg}^{\circ}\text{C}$. Recall the density of water is 1.000 g/cc .

ANSWER: _____

112. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its ground state? Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

113. Daniel needs to know the mass of a sample of CuCrO_4 to 3 significant figures. He puts the sample on a digital scale. This is what the scale shows: 109.478 g. If this measurement is precise enough for Daniel, round it to 3 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: _____

114. The reaction $2\text{C}_2\text{H}_2(\text{g}) + 5\text{O}_2(\text{g}) \longrightarrow 4\text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 3.07 cm^3 of water vapor is consumed, what volume of oxygen is produced or consumed? Justify your answer.

ANSWER: _____

115. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is $7.61 \times 10^{10}\text{ m}$. What is the distance in gigameters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

116. The enthalpy of solution of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is $\Delta H_{\text{sol}} = +10.9\text{ kJ/mol}$. Is dissolution endothermic or exothermic? Will the temperature of the solution increase or decrease when glucose dissolves in water?

ANSWER: _____

117. Solid I_2 (triple point 114°C , 0.12 atm) is heated at a constant 1.0 atm from 25°C to 200°C . Name the phase transition(s) that occur, in order, and justify your answer using the phase diagram.

ANSWER: _____

118. A chemist carefully measures the amount of heat needed to raise the temperature of a 2.75-kg sample of a pure substance from 1.6°C to $15.^\circ\text{C}$. The experiment shows that 1.3 kJ of heat is needed. What can the chemist report as the value of specific heat capacity of the substance?

ANSWER: _____

119. An unknown compound is completely vaporized at 120°C . At a pressure of 1.26 atm , the gas has a density of 1.7 g/L . Determine the molar mass of the compound. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

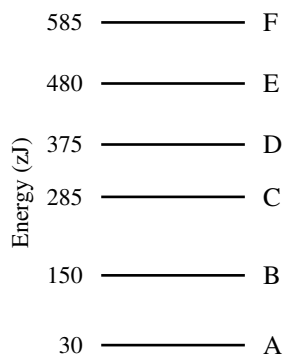
120. Does the following process produce a new substance? A copper wire is bent into a coil. You must explain your answer with valid reasoning.

ANSWER: _____

121. The lattice energy of LiCl (LiCl) is $+853\text{ kJ/mol}$ (energy to separate ions) and the hydration energy is -890 kJ/mol . Calculate ΔH_{sol} . Is dissolution endothermic or exothermic?

ANSWER: _____

122. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. How many lines are in the absorption line spectrum?



ANSWER: _____

123. A pharmacist measures the mass of a powdered drug. The mass is 8642 mg . What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

124. It takes 131 kJ/mol to break a bond between two particular elements. Calculate the maximum wavelength of light that will break one of these bonds.

ANSWER: _____

125. An ideal gas is confined beneath a freely moving piston. The gas is heated slowly while the external pressure remains constant. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

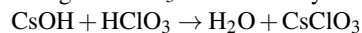
126. The reaction $2\text{CO} + \text{O}_2 \longrightarrow 2\text{CO}_2$ occurs with all gaseous substances at the same temperature and pressure. If the reaction consumes 7.07 mL of O_2 , determine the volume of CO . Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

127. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.

ANSWER: _____

128. Given this unbalanced chemical equation, calculate and report the theoretical yield of CsClO_3 formed by the mixing of 10.4 g of CsOH with 7.89 g of HClO_3 . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

129. A geologist measures the weight of an ore sample. The weight is 4.2371 kg. What is the weight in grams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

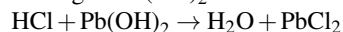
130. A beam of light is passed through blood. Will the Tyndall effect be observed? Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

131. A geologist measures the temperature of a hot spring. The temperature of the spring is -4.500°C . Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

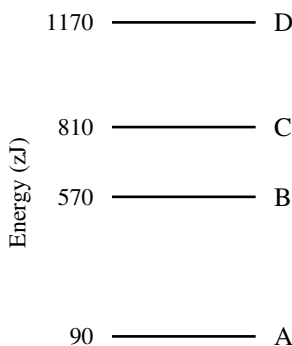
ANSWER: _____

132. Given this unbalanced chemical equation, calculate and report the theoretical yield of PbCl_2 formed by the mixing of 6.044 g of HCl with 27.79 g of $\text{Pb}(\text{OH})_2$. Ensure that your answer has the correct number of significant figures.



ANSWER: _____

133. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the longest wavelength? You must explain your answer choice for full credit.



ANSWER: _____

134. Ammonium fluoride reacts with barium permanganate to form barium fluoride and ammonium permanganate. Write the balanced chemical equation for this reaction using the smallest possible whole-number stoichiometric coefficients.

ANSWER: _____

135. For the Cl atom in ClF_3 , determine its hybridization, molecular geometry about that atom, and ideal bond angle. Justify your answer.

ANSWER: _____

136. Compare He_2 and He_2^+ using molecular-orbital (MO) theory. For each species, write the MO electron configuration, determine the bond order and magnetic character, and then explain how removing one electron affects the bond order and the magnetic behavior.

ANSWER: _____

137. Phosphoric acid, which most commonly finds uses as a rust inhibitor, a food additive, and an etching agent for dental and orthopedic use, can be synthesized using a two-step thermal process. In the first step, tetratomic phosphorus and diatomic oxygen react to form diphosphorus pentoxide. In the second step, diphosphorus pentoxide and water react to form phosphoric acid. Write the net chemical equation for the production of phosphoric acid from tetratomic phosphorus, diatomic oxygen, and water. Ensure that it is balanced.

ANSWER: _____

138. Suppose Mary heats a sample of a certain sodium cyanide hydrate, $\text{NaCN} \cdot 8\text{H}_2\text{O}$ until all the water is removed and pure NaCN is left behind. Find the percent by which the mass of the sample decreases in this process.

ANSWER: _____

139. A chemist carefully measures the amount of heat needed to raise the temperature of a 1.3-kg sample of a pure substance from 3.54°C to 28.3°C . The experiment shows that 3.2 kJ of heat is needed. What can the chemist report as the value of specific heat capacity of the substance?

ANSWER: _____

140. The molar enthalpy of sublimation of naphthalene (C_{10}H_8) is 62.36 kJ/mol and its molar enthalpy of fusion is 19.06 kJ/mol. Calculate the molar enthalpy of vaporization.

ANSWER: _____

141. Suppose that there is heat transfer of 34 J to a system, while the system does 53 J of work. Later, there is heat transfer of 41 J out of the system while 52 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: _____

142. Jennifer conducted some measurements and determined that 498. g of chlorine gas participated in a chemical reaction. Calculate the number of moles of chlorine gas that participated in the chemical reaction.

ANSWER: _____

143. Ashley prepares a solution of rubidium chloride (RbCl) by measuring out 0.0659 g of rubidium chloride into a 25.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

144. For the reaction $\text{CH}_4 + 2\text{O}_2 \longrightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, use average bond energies to determine the total bond energy of all bonds in the products, according to the balanced equation. Average bond energies (kJ/mol): C–H = 414, C=O = 799, O–H = 464, O=O = 498.

ANSWER: _____

145. An unknown solid has the following properties: low melting point, soft, does not conduct electricity, volatile. Classify this solid as ionic, metallic, covalent-network, or molecular. For credit you must justify your answer choice.

ANSWER: _____

146. A strip of solid antimony metal is put into a beaker of 0.015-M AuClO_2 solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

147. Provide the number of protons and the number of neutrons there are in an atom of chlorine-35.

ANSWER: _____

148. Classify the density of lead as a physical property or a chemical property. Explain your answer.

ANSWER: _____

149. Nicholas needs to find the volume of a 7.55 g sample of mercury. The density of the sample is $5.6 \frac{\text{g}}{\text{mL}}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

150. Calculate the vapor pressure of a solution prepared by dissolving 18.0 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in 180. g of water at 80°C . The vapor pressure of pure water at 80°C is 355.1 mmHg.

ANSWER: _____

151. Determine the number of σ bonds and the number of π bonds present in CO_2 . Justify your answer.

ANSWER: _____

152. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 974 GHz. What is the frequency in hertz (Hz)? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

153. Calculate the mass of a sample of strontium bicarbonate ($\text{Sr}(\text{HCO}_3)_2$) that contains 4.2×10^{13} carbon atoms.

ANSWER: _____

154. Without using memorized boiling-point trends, determine which substance has the higher boiling point: NH_3 or PH_3 . State your answer and justify it.

ANSWER: _____

155. A student assigns the hybridization of the central O in O_3 from the Lewis structure $\text{O}=\text{O}-\text{O}$. A second student instead uses the Lewis structure $\text{O}-\text{O}=\text{O}$. Must both students assign the same hybridization or different hybridizations? State the correct hybridization of the central atom and justify your answer. Justify your answer completely.

ANSWER: _____

156. Daniel conducted some measurements and determined that 1.1642 g of nobelium participated in a chemical reaction. Calculate the number of moles of nobelium that participated in the chemical reaction.

ANSWER: _____

157. Graphite and graphene are both composed entirely of carbon. Determine which would be expected to function better as a dry lubricant. State your answer and justify it.

ANSWER: _____

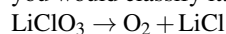
158. A vertical cylinder has an internal diameter of 8.7-cm and contains gas beneath a movable piston. The gas column is 5.74-cm tall, and the gas pressure is measured to be 6.68 kPa. Determine the force exerted by the gas on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

159. Calculate the freezing point of a solution prepared by dissolving 30.0 g of KCl (KCl) in 200. g of water. $K_f(\text{H}_2\text{O}) = 1.86^\circ\text{C kg/mol}$. Assume complete dissociation ($i = 2$).

ANSWER: _____

160. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). You must justify your answer choice. If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

161. Rank the following equimolal aqueous solutions from smallest to largest boiling point elevation: $\text{CH}_4\text{N}_2\text{O}$, Na_2SO_4 , MgBr_2 . All solutions have the same molality.

ANSWER: _____

162. A shipwright measures the length of a bolt. The length is 73.1 cm. What is the length in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

163. A semipermeable membrane separates a 0.40 M NaCl solution from a 0.10 M NaCl solution. In which direction will water flow by osmosis? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

164. A geologist measures the height of a mountain. The height is 3.0080 km. What is the height in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

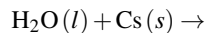
165. Calculate the boiling point of a solution prepared by dissolving 25.0 g of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in 250. g of water. $K_b(\text{H}_2\text{O}) = 0.512^\circ\text{C kg/mol}$.

ANSWER: _____

166. Suppose a popular FM-radio station broadcasts radio waves with a frequency of 89.1 MHz. Calculate the wavelength of these radio waves.

ANSWER: _____

167. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".

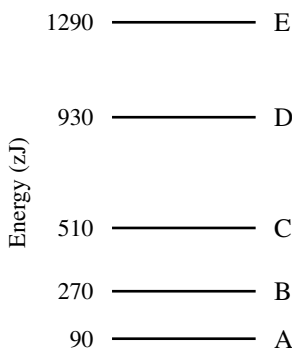


ANSWER: _____

168. An atom of a particular element has a radius of 220 pm and the average orbital speed of the electrons in it is approximately 5.52×10^7 m/s. Calculate the least possible uncertainty in a measurement of the speed of an electron in this atom. Write your answer as a percentage of the average speed and round it to two significant figures.

ANSWER: _____

169. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the lowest frequency? You must explain your answer choice for full credit.



ANSWER: _____

170. A particular planet has a mass of 8.291×10^{27} g and a virus has a mass of 9.780×10^{-12} mg. How many moles of viruses have a mass equal to the mass of this particular planet? Ensure your answer has the correct number of significant figures.

ANSWER: _____

171. Is O_2^- paramagnetic or diamagnetic? Justify your conclusion using its molecular-orbital electron configuration.

ANSWER: _____

172. Matthew weighs out 1.83 g of an unknown triprotic acid into a 250.-mL volumetric flask and dilutes the solution to the mark with distilled water. He then titrates this solution with 0.3010-M copper (II) hydroxide solution. When the titration reaches the equivalence point, Matthew finds that he has added 93.063 mL of the copper (II) hydroxide solution. Calculate the molar mass of the unknown triprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

173. Rank the following ionic compounds from least negative to most negative (strongest) lattice energy: MgO, CaO, NaCl.

ANSWER: _____

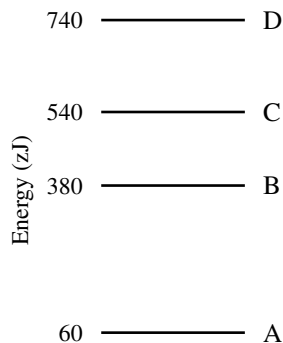
174. Draw the Lewis electron-dot symbol for an atom of As.

ANSWER: _____

175. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

ANSWER: _____

176. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the lowest frequency? You must explain your answer choice for full credit.



ANSWER: _____

177. Classify a solid of magnesium and oxygen (MgO) as ionic, metallic, covalent network, or molecular, and predict its melting point (high or low) and electrical conductivity. Justify your answer.

ANSWER: _____

178. The molar enthalpy of vaporization of benzene (C_6H_6) is 30.8kJ/mol and $M = 78.114\text{g/mol}$. Find the mass enthalpy of vaporization of benzene.

ANSWER: _____

179. A solution is prepared by dissolving 0.500g of lysozyme in enough water to make $200.\text{mL}$ of solution. The osmotic pressure at 25°C is $4.30 \times 10^{-3}\text{atm}$. Calculate the molar mass of lysozyme. Use $\Pi = MRT$, so $M = \Pi/(RT)$, then $\mathcal{M} = (m/V)/M$.

ANSWER: _____

180. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is $2.1447 \times 10^{-7}\text{m}$. What is the wavelength in nanometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

181. Classify iron as a molecular (covalent) solid, a covalent network solid, an ionic solid, or a metallic solid, and predict whether it has a relatively high or low melting point. Explain.

ANSWER: _____

182. How does increasing temperature affect the solubility of CO_2 in water? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

183. Explain how an emulsifier such as lecithin lets oil and water form a stable mayonnaise. A complete answer must reference molecular structure or charge; assertion alone earns no credit.

ANSWER: _____

184. Classify the following as an element, a compound, a homogeneous mixture, or a heterogeneous mixture: distilled water, which always contains hydrogen and oxygen in a fixed mass ratio. You must explain your answer with definitions.

ANSWER: _____

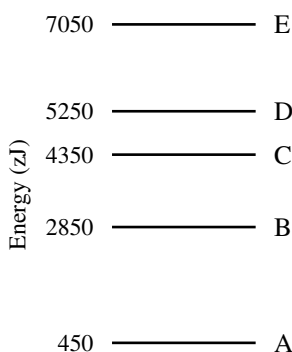
185. Graphite and graphene are both composed entirely of carbon. Determine which would be expected to function better as a dry lubricant. State your answer and justify it.

ANSWER: _____

186. A liquid is boiling at 40°C in a chamber where the external pressure is 91 kPa. It occupies a volume of 4 L. Determine whether the liquid's vapor pressure at this temperature is less than, equal to, or greater than the external pressure. Estimate the vapor pressure in SI units and justify your answer.

ANSWER: _____

187. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the lowest frequency? You must explain your answer choice for full credit.



ANSWER: _____

188. On an intermolecular potential-energy curve $u(r)$, what is the physical significance of the separation r at which $u(r)$ reaches its minimum? Justify.

ANSWER: _____

189. There are two steps in the extraction of copper metal from chalcocite, a copper ore. In the first step, copper (I) sulfide and oxygen gas react to form copper (I) oxide and sulfur dioxide. In the second step, copper (I) oxide and carbon react to form copper and carbon monoxide. Write the net chemical equation for the production of copper from copper (I) sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

190. Determine the number of σ bonds and the number of π bonds present in C_2H_4 . Justify your answer.

ANSWER: _____

191. A physical chemist measures the width of a molecule. The width is 70 nm. What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

192. Measurements show that an unknown compound has the following composition:

Element	Mass %
Boron	54.5 %
Carbon	45.5 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

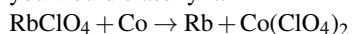
193. Identify the main group to which X belongs, given the ionic compound H_3OX . Explain your answer.

ANSWER: _____

194. Paul needs to make 218.88 mL of aqueous potassium phosphate (K_3PO_4) working solution. He plans to do this by adding distilled water to a 101.11-mL stock solution of 0.058-M aqueous potassium phosphate. What will be the concentration of his working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

195. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). You must justify your answer choice. If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

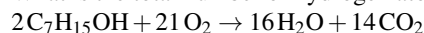
196. Provide the name and empirical formula of an ionic compound that is formed by the chromium (III) cation and the chloride anion.

ANSWER: _____

197. An unknown metal crystallizes in a BCC structure with unit cell edge length 316.52 pm and density 19.25 g/cm³. Calculate the molar mass of the metal and identify it.

ANSWER: _____

198. What is the total number of hydrogen atoms shown on the product side of this chemical equation?



ANSWER: _____

199. Elizabeth measures the speed of a moving object and records 31 $\frac{\text{km}}{\text{h}}$. Convert this speed to $\frac{\text{m}}{\text{s}}$. Ensure your answer has the correct number of significant figures.

ANSWER: _____

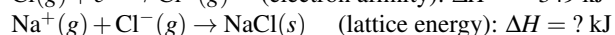
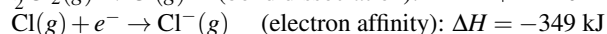
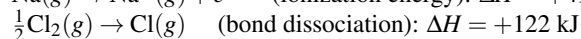
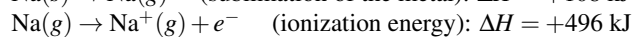
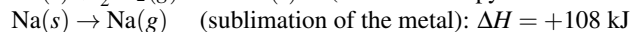
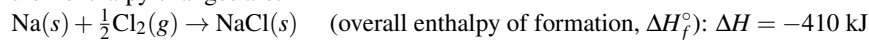
200. List every type of intermolecular force present in pure PH₃. Justify your answer.

ANSWER: _____

201. Rank C₂⁻, C₂, and C₂⁺ from least to most strongly attracted to a magnetic field. Justify your ranking by writing the molecular-orbital electron configuration of each species.

ANSWER: _____

202. Use a Born-Haber cycle (Hess's law) to determine the missing enthalpy change for the formation of NaCl(s) from its elements. The steps and their enthalpy changes are:



ANSWER: _____

203. Determine the hybridization of each atom (besides hydrogen) in H_2O . Briefly justify each assignment using the local bonding environment of that atom.

ANSWER: _____

204. When a narrow copper capillary tube is dipped into mercury, does the mercury rise or become depressed in the tube? Justify using cohesive and adhesive forces.

ANSWER: _____

205. The molar enthalpy of sublimation of ethanol ($\text{C}_2\text{H}_5\text{OH}$) is 43.20kJ/mol and its molar enthalpy of fusion is 4.6kJ/mol . Calculate the molar enthalpy of vaporization.

ANSWER: _____

206. Provide the chemical formula for the molecular compound nitrogen monoxide.

ANSWER: _____

207. A particular tin atom has a total of 4 bonds to other atoms and has 0 lone pairs of electrons around it. Calculate the formal charge on this tin atom.

ANSWER: _____

208. A sample of C_2H_2 occupies a sealed rigid vessel having a measured volume of 11.0L . After the gas reaches 4.22°C , the pressure inside is measured to be 0.58atm . Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

209. Determine the total number of σ and π bonds in $\text{CH}_2=\text{CHOCH}_3$ (methyl vinyl ether). For credit you must justify your answer by explaining the source of the σ and π bonds.

ANSWER: _____

210. What is the oxidation number of oxygen in potassium oxide?

ANSWER: _____

211. Calculate the mass of a sample of carbon diselenide (CSe_2) that contains 3.60×10^{18} selenium atoms.

ANSWER: _____

212. To help farmers and gardeners, commercial fertilizers have a big three-number "NPK" label on the bag that gives the amounts of three key plant nutrients: nitrogen, phosphorus, and potassium. For example, a bag of "10-10-10" fertilizer has 10% by mass nitrogen, 10% by mass potassium as potash (K_2O) and 10% by mass phosphorus as phosphate (P_2O_5).

ingredient	percent by mass
calcium nitrate ($\text{Ca}(\text{NO}_3)_2$)	23.0%
urea ($(\text{NH}_2)_2\text{CO}$)	37.0%
potassium sulfate (K_2SO_4)	4.0%
ammonium phosphate ($(\text{NH}_4)_3\text{PO}_4$)	6.0%
inert ingredients	30.0%

What should the first (nitrogen) number on the bag's label be? In other words, calculate the percent by mass of nitrogen in this fertilizer. Round your answer to the nearest tenth of a percent.

ANSWER: _____

213. A gas is contained beneath a square piston with sides of 6.4-cm. The gas pressure is measured to be 8.5 kPa. Determine the force exerted on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

214. Kyle weighs out 2.17 g of arsenic acid (H_3AsO_4) into a 250.-mL volumetric flask and dilutes the solution to the mark with distilled water. He plans to titrate the acid with 0.3132-M lithium hydroxide solution. Calculate the volume of lithium hydroxide solution that Kyle must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

ANSWER: _____

215. How many grams of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) must be dissolved in 200. g of water to raise the boiling point by 1.02°C ? $K_b(\text{H}_2\text{O}) = 0.512^\circ\text{C kg/mol}$.

ANSWER: _____

216. A rigid 9.6-L tank at 75.1°C contains only methane (CH₄) and neon (Ne). Assume both gases behave ideally. The tank contains 7.2 g of methane. The total amount of gas is 0.88 mol. Determine the pressure contributed by the second gas. Justify your answer.

ANSWER: _____

217. Dylan conducted some measurements and determined that 0.797 moles of nitrogen gas participated in a chemical reaction. Calculate the mass of nitrogen gas that participated in the chemical reaction.

ANSWER: _____

218. Experimental measurements show that the bond angle(s) in O₃ are <120°. Explain why this observation led chemists to describe the bonding on the O atom using hybrid orbitals.

ANSWER: _____

219. The reaction $\text{H}_2(\text{g}) + \text{Br}_2(\text{g}) \longrightarrow 2\text{HBr}(\text{g})$ occurs at constant temperature and pressure. If 10.0 cm³ of HBr gas reacts completely, what volume of H₂ gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

220. Calculate the heat required to warm 26.0-g of liquid water by 55°C. $c_{\text{water}} = 4.184 \text{ J g}^{-1} \text{ }^\circ\text{C}^{-1}$.

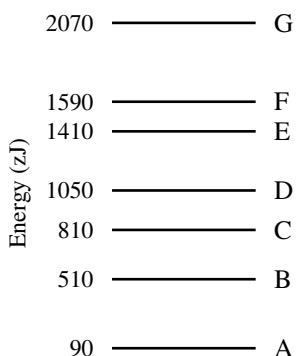
ANSWER: _____

221. As part of her duties as an analytical chemist, Julia analyzes a sample of an unknown pure substance X and finds that it contains 6.9 g of element E₁ and 5.8 g of element E₂, and no other elements. She then analyzes three other substances that also contain only elements E₁ and E₂ and records the results in a table in her lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E ₁	mass of E ₂
X	6.9 g	5.8 g
A	34.6 g	14.7 g
B	14.2 g	12.0 g
C	13.8 g	5.0 g

ANSWER: _____

222. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the highest frequency? You must explain your answer choice for full credit.



ANSWER: _____

223. What major chemical species are present in an aqueous solution of 0.094-M silver iodide?

ANSWER: _____

224. An unknown chemical compound is composed of two unknown elements, A and B. The chemical formula is A_2B_x , where x represents a whole number. Measurements of a sample of the compound show that it contains 0.60 moles of A and 1.195 moles of B. Write the complete chemical formula for the unknown compound.

ANSWER: _____

225. Solid carbon dioxide (CO_2) has its triple point at $-56.6^\circ C$ and 5.11 atm. It is held at a constant external pressure of 2.4568 atm and slowly warmed. Does the solid melt or sublime? Justify your answer.

ANSWER: _____

226. An atom of a particular element has a radius of 250 pm and the average orbital speed of the electrons in it is approximately 1.1×10^7 m/s. Calculate the least possible uncertainty in a measurement of the speed of an electron in this atom. Write your answer as a percentage of the average speed and round it to two significant figures.

ANSWER: _____

227. Rank the following elements from highest metallic to lowest metallic. Support your ranking with an explanation of the underlying atomic structure; answers that simply cite a memorized periodic trend will not receive credit.
Rb, Cs, Na, K

ANSWER: _____

228. Ammonium chloride reacts with rubidium sulfide to form ammonium sulfide and rubidium chloride. If 10.50 g of ammonium sulfide is produced from the reaction of 27.8 g of ammonium chloride and 94.9 g of rubidium sulfide, calculate the percent yield of ammonium sulfide. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

229. MO theory predicts the same bond order for the following species: C_2^{2+} , B_2 , N_2^{4+} . What is that bond order? Justify your answer using MO theory.

ANSWER: _____

230. Identify the main group to which X belongs, given the ionic compound XCl . Explain your answer.

ANSWER: _____

231. Consider the following compounds: nitrogen trifluoride, carbon tetrachloride, hydrosulfuric acid, and hydrogen cyanide. Write the molecular formula for each of the listed compounds that has a dipole moment. If none have a dipole moment, write "None have a dipole moment". Draw each compound and explain your answer.

ANSWER: _____

232. As part of her duties as an analytical chemist, Olivia analyzes a sample of an unknown pure substance X and finds that it contains 9.2 g of element E_1 and 2.9 g of element E_2 , and no other elements. She then analyzes three other substances that also contain only elements E_1 and E_2 and records the results in a table in her lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E_1	mass of E_2
X	9.2 g	2.9 g
A	40.6 g	6.4 g
B	40.2 g	5.6 g
C	31.7 g	5.0 g

ANSWER: _____

233. Calculate the freezing point of a solution prepared by dissolving 20.0 g of glucose ($C_6H_{12}O_6$) in 200. g of water. $K_f(H_2O) = 1.86^\circ C \text{ kg/mol}$.

ANSWER: _____

234. $\text{H}_3\text{OC}_2\text{H}_3\text{O}_2$ is the formula for hydronium acetate. Nick, a chemist, measures a sample of hydronium acetate and determines that it contains 1.52 moles of hydrogen. How many moles of oxygen are in the sample?

ANSWER: _____

235. Classify the following as an element or a compound: a lump of pure phosphorus. You must explain your answer using correct reasoning.

ANSWER: _____

236. State the type(s) of bonding present in Fe. Explain your answer.

ANSWER: _____

237. Benjamin must prepare 814.2 mL of a 0.657-mM working solution of aqueous ammonium bromide (NH_4Br). He will do this by pouring some 0.027802-M aqueous ammonium bromide stock solution into a graduated cylinder and diluting it with distilled water. How much of the aqueous-ammonium bromide stock solution must Benjamin pour into the graduated cylinder? Report your answer in mL. Be sure your answer has the correct number of significant figures.

ANSWER: _____

238. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$100 \text{ mol} \div 225. \text{L}$$

ANSWER: _____

239. How many grams of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) must be dissolved in 100.g of water to raise the boiling point by 0.256°C ? $K_b(\text{H}_2\text{O}) = 0.512^\circ\text{C kg/mol}$.

ANSWER: _____

240. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is $7.81 \times 10^{10}\text{m}$. What is the distance in gigameters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

241. A brick has a mass of 7.5×10^0 kg and a cooking pot has a mass of 2.118×10^0 kg. How many moles of cooking pots have a mass equal to the mass of a brick? Ensure your answer has the correct number of significant figures.

ANSWER: _____

242. What is the coordination number of each atom in a simple cubic (SC) structure?

ANSWER: _____

243. An unknown metal crystallizes in a FCC structure with unit cell edge length 404.95 pm and density 2.7 g/cm^3 . Calculate the molar mass of the metal and identify it.

ANSWER: _____

244. Given this unbalanced chemical equation, report the maximum mass of FePO_4 that could be produced by the mixing of 7.353 g of FeCl_3 with 10.60 g of $(\text{NH}_4)_3\text{PO}_4$. Ensure that your answer has the correct number of significant figures.
 $\text{FeCl}_3 + (\text{NH}_4)_3\text{PO}_4 \rightarrow \text{FePO}_4 + \text{NH}_4\text{Cl}$

ANSWER: _____

245. A civil engineer measures the length of an airport runway. The length is 5249 m. What is the length in kilometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

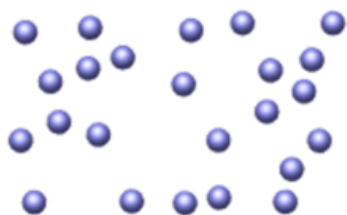
246. A physical chemist measures the width of a molecule. The width is 38 nm. What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

247. As part of his routine quality-control checks, Caleb, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of hypochlorous acid solution. He takes a 25.00-mL sample of hypochlorous acid solution and titrates it with a barium hydroxide solution, which he made by dissolving 2.81 g of solid barium hydroxide in 1.19 L of distilled water. If Caleb finds that it takes 349.8 mL of the barium hydroxide solution for the titration to reach the equivalence point, what molarity of the hypochlorous acid solution does Caleb calculate?

ANSWER: _____

248. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? You must justify your answer choice.



ANSWER: _____

249. Predict the products that form when hydrosulfuric acid is mixed with rubidium hydroxide. Write the balanced chemical equation for this reaction.

ANSWER: _____

250. Use the number of unpaired electrons to classify O_2^+ as paramagnetic or diamagnetic, and state whether it is attracted to or repelled by a magnetic field. Justify your answer.

ANSWER: _____

251. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 318.6 MPa. The cover of a circular observation port can safely withstand a maximum force of 3.2 MN. Determine the largest allowable diameter of the observation port. Report your answer in SI units and justify your answer.

ANSWER: _____

252. Identify whether the following describes a random error or a systematic error, and explain your answer: Wind gusts cause repeated outdoor temperature measurements to vary randomly around the true value.

ANSWER: _____

253. A semipermeable membrane separates a 0.05 M glucose solution from a 0.20 M glucose solution. In which direction will water flow by osmosis? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

254. A nuclear chemist measures the mass of an artificial element. The mass is 88 pg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

255. An unknown compound is completely vaporized at 78.°C. At a pressure of 1.2atm, the gas has a density of 1.1 g/L. Determine the molar mass of the compound. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

256. A lab technician measures the weight of a reagent. The weight is 4474 g. What is the weight in kilograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

257. Predict what happens to solubility when deep-sea fish are brought to the surface quickly. Identify whether temperature or pressure is the key factor and explain the observation.

ANSWER: _____

258. Samantha weighs out 2.62g of an unknown triprotic acid into a 50.-mL volumetric flask and dilutes the solution to the mark with distilled water. She then titrates this solution with 0.3616-M bismuth hydroxide solution. When the titration reaches the equivalence point, Samantha finds that she has added 73.94mL of the bismuth hydroxide solution. Calculate the molar mass of the unknown triprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

259. The C atom in C₂H₂ is *sp* hybridized. What is the greatest number of π bonds that atom can participate in simultaneously? Justify your answer.

ANSWER: _____

260. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is 7.32×10^{-14} m. What is the wavelength in femtometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

261. A mixture of propylene and methane gas is expanded from a volume of 144L to a volume of 170L while the pressure is held constant at 52 atm. Calculate the work done on the gas mixture. Report your answer in SI unit of work and to 3 significant figures.

ANSWER: _____

262. Ammonium fluoride reacts with magnesium oxide to form ammonium oxide and magnesium fluoride. Suppose 31.00 g of ammonium fluoride is mixed with 13.63 g of magnesium oxide. Calculate and report the theoretical yield of ammonium oxide formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

263. Identify the dominant intermolecular force acting between iodine (I_2) and hexane (C_6H_{14}) when they are mixed. Justify your answer.

ANSWER: _____

264. The extraction of aluminum metal from the aluminum hydroxide in bauxite ore by the Hall-Héroult process is one of the most remarkable success stories of late-19th-century chemistry, turning aluminum from a rare and precious metal into the cheap commodity it is today. In the first step, aluminum hydroxide reacts to form alumina (which is aluminum oxide) and water. In the second step, alumina and carbon react to form aluminum and carbon dioxide. Write the net chemical equation for the production of aluminum from aluminum hydroxide and carbon. Ensure that it is balanced.

ANSWER: _____

265. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur (S_8) reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide, oxygen gas, and water combine to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.

ANSWER: _____

266. Classify the following as an element or a compound: a chunk of zinc metal. You must explain your answer using correct reasoning.

ANSWER: _____

267. Classify melting point as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: _____

268. In an ionic crystal a cation (Na^+ , radius 102 pm) sits among anions (Cl^- , radius 181 pm). Predict which type of hole the cation occupies and its coordination number. Justify with the radius ratio.

ANSWER: _____

269. As part of his routine quality-control checks, William, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of selenous acid solution. He takes a 40.-mL sample of selenous acid solution and titrates it with a bismuth hydroxide solution, which he made by dissolving 2.856 g of solid bismuth hydroxide in 1.402 L of distilled water. If William finds that it takes 164.2 mL of the bismuth hydroxide solution for the titration to reach the equivalence point, what molarity of the selenous acid solution does William calculate?

ANSWER: _____

270. A shipwright measures the length of a bolt. The length is 23.4 cm. What is the length in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

271. A cell biologist measures the width of a cell. The width is $87\ \mu\text{m}$. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

272. Adding one electron to a previously unoccupied molecular orbital causes the molecule to become less stable. Was the added electron placed into a bonding or an antibonding molecular orbital? Justify your answer.

ANSWER: _____

273. A millworker measures the width of a plank. The width is 3.6 dm. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

274. A student draws benzene with alternating short and long carbon-carbon bonds and claims this represents the actual molecule. Is the claim correct? Justify your answer.

ANSWER: _____

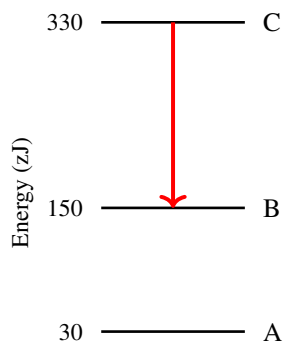
275. Provide the name and empirical formula of the ionic compound that is formed by the Mg^{2+} cation and the $\text{C}_2\text{H}_3\text{O}_2^-$ anion.

ANSWER: _____

276. A bond is formed by sideways overlap of two unhybridized p orbitals. Is this explained primarily by hybridization or by σ/π bonding? Justify your answer.

ANSWER: _____

277. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from C to B, as shown by the arrow. Calculate the wavelength of the photon that would be absorbed or emitted. Report your answer to three significant figures. Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

278. Calculate the heat absorbed when 19.9-g of sodium chloride (NaCl) melts completely at its melting point of 801°C . $\Delta H_{\text{fus}} = 28.16 \text{ kJ/mol}$.

ANSWER: _____

279. Measurements show that the enthalpy of a mixture of gaseous reactants has decreased by 226 kJ during a certain chemical reaction, which was carried out at a constant pressure. Evidenced by the volume change, -128.7 kJ of work must have been done on the mixture during the reaction. Calculate the change in energy of the gas mixture during the reaction and report whether the reaction is endothermic, exothermic, or neither. You must explain your answer choice for full credit.

ANSWER: _____

280. An engineer measures the capacity of an electricity-generating plant. The capacity is 666 MW. What is the capacity in watts? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

281. Jessica is performing a combustion analysis on a compound with a molecular molar mass of 140.27 g/mol that is composed of carbon, hydrogen, and perhaps oxygen. A sample of 4.27 g of the compound is completely burned in excess oxygen and the mass of the products is carefully measured:

Product	Mass (g)
Carbon Dioxide	13.4 g
Water	5.48 g

Use this information to find the molecular formula of the compound.

ANSWER: _____

282. List every type of intermolecular force present in pure CH_3F . Justify your answer.

ANSWER: _____

283. A sealed flask is evacuated and then filled with an unknown gas. The sample has a mass of 0.116 g, occupies 1.500 L at 100.0 kPa and 313.15 K. Possible identities are: H_2O , CH_4 , CO , NO , C_2H_2 , Ne , He , SO_2 , F_2 , H_2 , Cl_2 , Ar , Kr , SF_6 , CO_2 , HBr , H_2S , Xe , UF_6 . Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

284. Jeremy conducted some measurements and determined that 0.88106 g of meitnerium participated in a chemical reaction. Calculate the number of moles of meitnerium that participated in the chemical reaction.

ANSWER: _____

285. Rank the following liquids from highest to lowest surface tension at 20°C: $\text{C}_3\text{H}_8\text{O}_3$, C_6H_6 , C_6H_{14} , H_2O . Offer brief explanation for your ranking.

ANSWER: _____

286. Solid sulfur dioxide (SO_2) has its triple point at -75.5°C and 0.017 atm. It is held at a constant external pressure of 0.06 atm and slowly warmed. Does the solid melt or sublime? Justify your answer.

ANSWER: _____

287. A botanist measures the thickness of a leaf. The thickness is 7.2 mm. What is the thickness in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

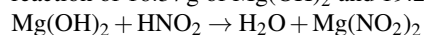
288. Calculate the heat required to completely vaporize 38.1-g of water (H_2O) at its normal boiling point of 100°C . $\Delta H_{\text{vap}} = 40.7\text{ kJ/mol}$, $M = 18.015\text{ g/mol}$.

ANSWER: _____

289. Provide the chemical symbol of the most commonly formed ion of silver and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: _____

290. Given this unbalanced chemical equation, calculate and report the percent yield of $\text{Mg}(\text{NO}_2)_2$ if 19.9 g of $\text{Mg}(\text{NO}_2)_2$ is produced from the reaction of 16.37 g of $\text{Mg}(\text{OH})_2$ and 19.2 g of HNO_2 . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

291. A colloid consists of a gas dispersed in a liquid. Name this class of colloid and give an everyday example.

ANSWER: _____

292. A sample reported to be N_2O has a mass of $1.0 \times 10^2\text{ g}$ and occupies 7.8 L at 93 kPa and $21.^\circ\text{C}$. Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

293. Daniel looks up the density of a material and finds it listed as $3.79 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.

ANSWER: _____

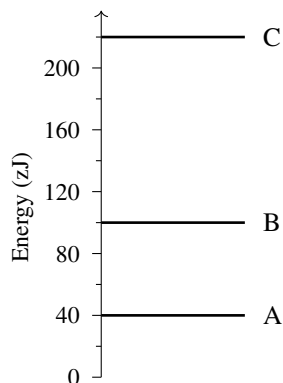
294. Suppose Dylan heats a sample of a certain chromium (III) bromide hydrate, $\text{CrBr}_3 \cdot 7\text{H}_2\text{O}$ until all the water is removed and pure CrBr_3 is left behind. Find the percent by which the mass of the sample decreases in this process.

ANSWER: _____

295. H_3OMnO_4 is the formula for hydronium permanganate. Emma, a lab technician, measures a sample of hydronium permanganate and determines that it contains 5.5 moles of manganese. How many moles of hydrogen are in the sample?

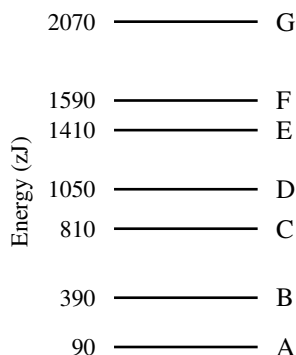
ANSWER: _____

296. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its ground state? Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

297. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the shortest wavelength? You must explain your answer choice for full credit.



ANSWER: _____

298. A solution is prepared by dissolving 1.50 g of albumin in enough water to make 50.0 mL of solution. The osmotic pressure at 25°C is 0.0111 atm. Calculate the molar mass of albumin. Use $\Pi = MRT$, so $M = \Pi/(RT)$, then $\mathcal{M} = (m/V)/M$.

ANSWER: _____

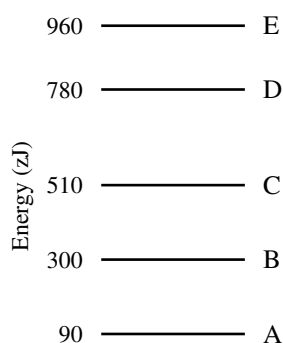
299. A 23 g sample of neon gas (Ne) is contained in a sealed reaction vessel at 22.°C and 1.3 atm. Determine the sample volume assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

300. Sophia measures the amount of boron monophosphide produced during a chemical reaction. She finds that 866.g is produced. Calculate the number of moles of boron monophosphide produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

301. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. How many lines are in the absorption line spectrum?



ANSWER: _____

302. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 535 GHz. What is the frequency in hertz (Hz)? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

303. A mercury barometer reads 773.2 torr. Determine the atmospheric pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

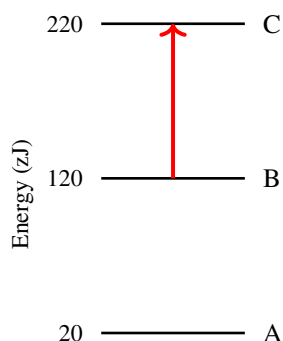
304. Brandon prepares a solution of hydronium phosphate ((H₃O)₃PO₄) by measuring out 52.66 mmol of hydronium phosphate into a 25.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

305. Use molecular-orbital theory to determine the bond order and whether N_2 is paramagnetic or diamagnetic. Justify your answer.

ANSWER: _____

306. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from B to C, as shown by the arrow. Calculate the energy of the photon that would be absorbed or emitted. Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

307. A gas pushes against a rectangular piston measuring 7.62-cm by 8.0-cm. The gas pressure is 6.5 kPa. Determine the force exerted on the piston. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

308. Identify the state of matter in which water exists at 120 °C and 1 atm.

ANSWER: _____

309. Classify the bond between carbon (C) and oxygen (O) as nonpolar covalent, polar covalent, or ionic. If the bond is polar or ionic, identify which atom carries the partial negative charge (δ^-). Justify your answer using the electronegativity difference.

ANSWER: _____

310. William puts a small sample of HF in water and lets it dissolve. After equilibrium is reached, he determines that several chemical species remain in the solution. These include H_3O^+ , F^- , HF, and H_2O . Which of the following labels apply to the compound HF? Write all that apply. Possible labels include ionic, molecular, strong acid, weak acid, strong base, and weak base. Explain why you chose your answer.

ANSWER: _____

311. For nitrogen (N_2): normal melting point -210°C , normal boiling point -196°C , triple point (-210°C , 0.124 atm), critical point (-147°C , 34 atm). What phase is present at -89°C and 79 atm? Justify your answer.

ANSWER: _____

312. Describe the layer stacking sequence of hexagonal closest packing (HCP) and explain what makes it distinct. Justify.

ANSWER: _____

313. A closed-end mercury manometer connected to a gas sample has a height difference of 101 mmHg. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

314. Does the following process produce a new substance? A copper wire is bent into a coil. You must explain your answer with valid reasoning.

ANSWER: _____

315. A substance has only paired electrons. Is it paramagnetic or diamagnetic, and is it attracted to or repelled by an external magnetic field? Explain.

ANSWER: _____

316. Chloe prepares a solution of potassium sulfide (K_2S) by measuring out 11.66 g of potassium sulfide into a 25.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

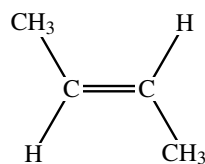
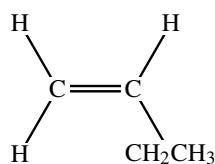
317. A sample of H_2 is collected over water. The partial pressure of the dry gas is 736.1 torr, and the vapor pressure of water is 21.9 torr. Determine the total pressure of the gas sample. Report your answer in SI units. Justify your answer.

ANSWER: _____

318. A solution shows a boiling point elevation of 0.186°C when 100. g of water is used as the solvent. How many grams of anthracene ($\text{C}_{14}\text{H}_{10}$, $M = 178.23 \text{ g/mol}$) are dissolved? ($K_b(\text{H}_2\text{O}) = 0.512^\circ\text{C kg/mol}$)

ANSWER: _____

319. Examine Structures A and B. Determine whether they could represent the same sample without breaking any covalent bonds. Justify your conclusion.



ANSWER: _____

320. A 37.0-g sample of ice at 0°C absorbs 20.9 kJ of heat. Determine the final state and temperature. ($\Delta H_{\text{fus}} = 334 \text{ J/g}$, $c_{\text{water}} = 4.18 \text{ J g}^{-1}\text{C}^{-1}$.)

ANSWER: _____

321. A nurse in a U.S. hospital records a patient's temperature as 149°F . Convert this temperature to degrees Celsius. Round your answer to the nearest integer.

ANSWER: _____

322. A gas mixture contains 0.20 mol of sulfur dioxide (SO_2) and 0.452 mol of propane (C_3H_8). Determine the mole fraction of sulfur dioxide in the mixture. Justify your answer.

ANSWER: _____

323. Vegetable oil (vegetable oil, density 0.920 g/mL) is shaken with an equal volume of water (1.00 g/mL). Will two layers form? If so, which liquid ends up on top? Explain.

ANSWER: _____

324. An unknown chemical compound is composed of two unknown elements, A and B. The chemical formula is A_3B_x , where x represents a whole number. Measurements of a sample of the compound show that it contains 0.3174 moles of A and 0.529 moles of B. Write the complete chemical formula for the unknown compound.

ANSWER: _____

325. Determine which bond is expected to be stronger: the $\text{C}\equiv\text{N}$ bond in CN^- or the $\text{C}-\text{N}$ bond in CH_3NH_2 . Justify your answer from fundamental principles. Responses based only on memorized trends will not receive credit.

ANSWER: _____

326. W crystallizes in a body-centered cubic structure with unit cell edge length 316.5pm. Calculate the density of W.

ANSWER: _____

327. A meteorologist records the outdoor air temperature. The air temperature is 70°C. Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

ANSWER: _____

328. The normal boiling point of water (H₂O) is 100.00°C ($P_{\text{vap}} = 1.000\text{ atm}$) and its molar enthalpy of vaporization is 40.7 kJ/mol. Calculate the vapor pressure of water at 71.00°C.

ANSWER: _____

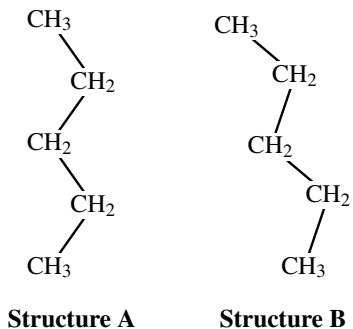
329. Propane (C₃H₈) can be turned into hydrogen by a two-step reforming process. In the first step, propane and water react to form carbon monoxide and hydrogen. In the second step, carbon monoxide and water react to form hydrogen and carbon dioxide. Write the net chemical equation for the production of hydrogen from propane and water. Ensure that it is balanced.

ANSWER: _____

330. A particular polonium atom has a total of 3 bonds to other atoms and has 1 lone pair of electrons around it. Calculate the formal charge on this polonium atom.

ANSWER: _____

331. Consider Structures A and B. Determine whether one representation could become the other without breaking any covalent bonds. Support your conclusion.



ANSWER: _____

332. Sketch the valence molecular-orbital energy-level diagram for O_2 and fill in its electrons. Determine the bond order and whether O_2 is paramagnetic or diamagnetic. Justify your conclusions.

ANSWER: _____

333. What multiplicative factor does the metric prefix 'milli' (symbol: m) represent?

ANSWER: _____

334. How many hydrogen-bond donors does one molecule of CH_3OH have?

ANSWER: _____

335. Which atom has the higher electronegativity: lithium (Li) or fluorine (F)? You must briefly explain why. Credit is awarded only for explaining the cause of the trend based on atomic structure, not for merely stating the trend.

ANSWER: _____

336. Which substance evaporates more slowly under identical conditions: methanol (CH_3OH) or butane (C_4H_{10})? Justify your answer. Unsupported trend-recitation or memorized rules receive no credit.

ANSWER: _____

337. A sample of CS_2 occupies a volumetric gas flask having a measured volume of 6.8 L. After the gas reaches $13^\circ C$, the pressure inside is measured to be 0.348 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

338. Draw a valid Lewis structure for HI. You must show all existing lone pairs and lone electrons.

ANSWER: _____

339. Classify NaCl (NaCl) as a strong electrolyte, weak electrolyte, or nonelectrolyte in aqueous solution. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

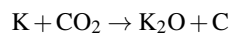
340. A sample of 13 g of sodium sulfide reacts completely with excess hydrochloric acid to produce sodium chloride and hydrogen sulfide gas in a sealed container. The container initially contains 0.352 mol of neon gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of neon after the reaction is complete.

ANSWER: _____

341. What is the coordination number of each atom in a body-centered cubic (BCC) structure?

ANSWER: _____

342. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.



ANSWER: _____

343. An open-end mercury manometer is connected to a gas sample. The atmospheric pressure is 745 torr and the mercury is 52.0 mmHg higher on the end open to the atmosphere. Determine the gas pressure. Report your answer in SI units. Justify your answer.

ANSWER: _____

344. Write the complete MO electron configuration for N_2^+ . Determine its bond order and magnetic character. Justify each answer.

ANSWER: _____

345. Classify sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) as a strong electrolyte, weak electrolyte, or nonelectrolyte in aqueous solution. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

346. Classify a solid of magnesium and oxygen (MgO) as ionic, metallic, covalent network, or molecular, and predict its melting point (high or low) and electrical conductivity. Justify your answer.

ANSWER: _____

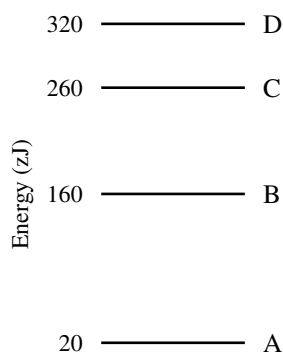
347. A cell biologist measures the width of a cell. The width is $83\ \mu\text{m}$. What is the width in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

348. Provide the name of the element that the element symbol F represents.

ANSWER: _____

349. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the shortest wavelength? You must explain your answer choice for full credit.

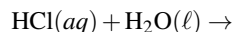


ANSWER: _____

350. List the following electron subshells in order of increasing energy. Assume these subshells are in an atom with many electrons.
3d, 4s, 2s, 4d, 3p

ANSWER: _____

351. Predict the products of the reaction below by completing the right-hand side of the chemical equation. Provide the full balanced equation with state symbols.



ANSWER: _____

352. Explain why adding an electrolyte such as alum to a colloid makes the particles coagulate and settle. A complete answer must reference molecular structure or charge; assertion alone earns no credit.

ANSWER: _____

353. A nurse measures the weight of his patient's medicine. The weight is 0.0041 g. What is the weight in milligrams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

354. Rank the following liquids from highest to lowest viscosity at 20°C: H₂O, motor oil, CH₃COCH₃, honey.

ANSWER: _____

355. A geologist measures the weight of an ore sample. The weight is 5.6312 kg. What is the weight in grams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

356. A 0.0500m aqueous solution of urea (CH₄N₂O) shows a freezing point depression of 0.0930°C ($K_f = 1.86^\circ\text{C kg/mol}$). Determine the effective van't Hoff factor and state what it reveals about the solute in solution.

ANSWER: _____

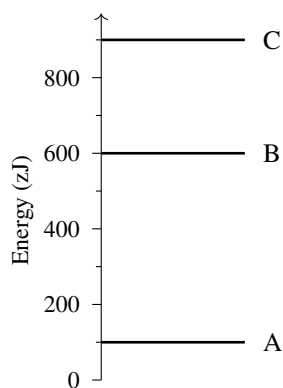
357. The industrial synthesis of acetic acid (CH₃COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH₃OH) reacts with carbon monoxide to form methyl formate (HCOOCH₃). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

358. Draw the Lewis electron-dot symbol for an atom of Li.

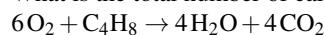
ANSWER: _____

359. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its second excited state? Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

360. What is the total number of carbon atoms shown on the reactant side of this chemical equation?



ANSWER: _____

361. Provide the name for the molecular compound Cl_2O_7 . If the compound has a common name, you may provide that instead.

ANSWER: _____

362. A sample reported to be HBr has a mass of 27g and occupies 6.9L at 116kPa and 15.6°C . Determine whether the reported identity is chemically consistent with the data. Justify your answer.

ANSWER: _____

363. Julia needs to make 403.15 mL of aqueous ammonium oxide $((\text{NH}_4)_2\text{O})$ working solution. She plans to do this by adding distilled water to a 93.0-mL stock solution of 0.0354-M aqueous ammonium oxide. What will be the concentration of her working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

364. A gas cylinder is opened into a large empty flexible reservoir. The cylinder has an internal diameter of 6.58-cm and an internal height of 29.1-cm. After the gas is released until atmospheric pressure is reached, the total volume occupied by the gas is 4.90L. Determine the initial pressure inside the cylinder. Justify your answer.

ANSWER: _____

365. Oxygen gas is collected over water. The measured pressure is 755 torr, and the vapor pressure of water is 18.1 torr. Determine the partial pressure of the oxygen gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

366. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is 5.8795×10^{-7} m. What is the wavelength in nanometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

367. Two solutions are prepared: solution A is a 0.500m solution of NaCl (NaCl) in water, and solution B is a 0.500m solution of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in water. Which solution has the greater boiling point elevation? Calculate both values. (Assume complete dissociation for ionic solutes.)

ANSWER: _____

368. Classify surface area as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: _____

369. A semipermeable membrane separates a 0.2 M sucrose solution from pure water. In which direction will water flow by osmosis? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

370. Silver sulfide reacts with hydronium chloride to form hydronium sulfide and silver chloride. Write the balanced chemical equation for this reaction using the smallest possible whole-number stoichiometric coefficients.

ANSWER: _____

371. Gabriel must note the concentration of a solution of K_2O in his lab notebook. He knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Gabriel weighs an empty graduated cylinder and finds its mass to be 6 g. He then puts some solid K_2O into the graduated cylinder and weighs it. With the K_2O added, the cylinder weighs 12 g. He adds water to the graduated cylinder and dissolves the K_2O completely. Then he reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 25.018 mL. What concentration should Gabriel write down in his lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

372. Predict how the vapor pressure of ethanol changes as its temperature increases. Justify your answer.

ANSWER: _____

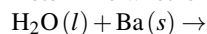
373. Use the Born-Haber cycle to calculate the lattice energy of KBr. $\Delta H_f^\circ(\text{KBr}) = -394 \text{ kJ/mol}$, $\Delta H_{\text{sub}}(\text{K}) = 89 \text{ kJ/mol}$, $IE_1(\text{K}) = 419 \text{ kJ/mol}$, $\frac{1}{2}D(\text{Br}_2) = 112 \text{ kJ/mol}$, $EA(\text{Br}) = -325 \text{ kJ/mol}$.

ANSWER: _____

374. Classify the following as a homogeneous or heterogeneous mixture: a salad containing lettuce, croutons, and tomatoes. You must explain your answer and use definitions to support your conclusion.

ANSWER: _____

375. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".



ANSWER: _____

376. The species NO_2^- exhibits resonance. Draw all of its equivalent resonance structures. What is the bond order of each N–O bond in the resonance hybrid?

ANSWER: _____

377. The lattice energy of KCl (KCl) is $+717 \text{ kJ/mol}$ (energy to separate ions) and the hydration energy is -681 kJ/mol . Calculate ΔH_{sol} . Is dissolution endothermic or exothermic?

ANSWER: _____

378. Sarah is a physicist reporting an experimental temperature in kelvin after recording it in Fahrenheit. The temperature is 14°F . Convert this temperature to kelvin. Round to the nearest integer.

ANSWER: _____

379. Aluminum crystallizes in a FCC unit cell with an atomic radius of 143 pm . Calculate the edge length of the unit cell in pm.

ANSWER: _____

380. Calculate the heat absorbed when 13.6-g of ethanol ($\text{C}_2\text{H}_5\text{OH}$) melts completely at its melting point of -114°C . $\Delta H_{\text{fus}} = 4.6\text{kJ/mol}$.

ANSWER: _____

381. Ammonium phosphate forms several hydrates with the general form



where x is an integer. If the hydrate is heated, the water can be driven off, leaving pure $(\text{NH}_4)_3\text{PO}_4$ behind. Suppose a sample of a certain hydrate of ammonium phosphate is heated until all the water is removed, and a chemist finds that the mass of the sample decreases by 42%. Give the value of x to indicate which hydrate the sample was.

ANSWER: _____

382. A fixed amount of gas is initially at 6.32°C . After its temperature is decreased to -67°C and its volume becomes 59% smaller than its initial volume, its pressure is measured to be 13.0 atm. Determine the initial pressure. Report your answer in SI units and justify your answer.

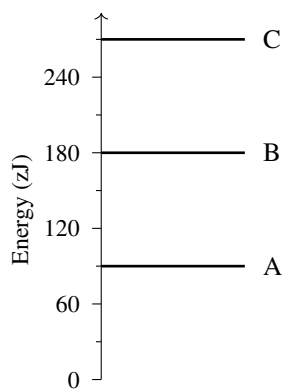
ANSWER: _____

383. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$600\text{mol} \div 130\text{L}$$

ANSWER: _____

384. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its first excited state? Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

385. Describe the layer stacking sequence of hexagonal closest packing (HCP) and explain what makes it distinct. Justify.

ANSWER: _____

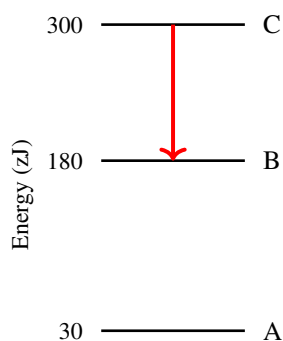
386. The industrial synthesis of acetic acid (CH_3COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH_3OH) reacts with carbon monoxide to form methyl formate (HCOOCH_3). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

387. A 59.7-g sample of liquid water at 0°C releases 6.24 kJ of heat. Determine the final state and temperature. ($\Delta H_{\text{fus}} = 334 \text{ J/g}$.)

ANSWER: _____

388. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from C to B, as shown by the arrow. Calculate the wavelength of the photon that would be absorbed or emitted. Report your answer to three significant figures. Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

389. A gas mixture has a total pressure of 0.896 atm. The mole fraction of O_2 is 0.209. Determine the partial pressure of the gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

390. Matthew weighs out 1.87 g of an unknown monoprotic acid into a 40.-mL volumetric flask and dilutes the solution to the mark with distilled water. He then titrates this solution with 0.3154-M rubidium hydroxide solution. When the titration reaches the equivalence point, Matthew finds that he has added 162.6 mL of the rubidium hydroxide solution. Calculate the molar mass of the unknown monoprotic acid. Ensure your answer has the correct number of significant figures.

ANSWER: _____

391. Classify the following as a homogeneous or heterogeneous mixture: a solution of salt dissolved in water. You must explain your answer and use definitions to support your conclusion.

ANSWER: _____

392. When 1.56 g of an unknown nonelectrolyte is dissolved in 200. g of water (H_2O), the freezing point decreases by 0.186°C . $K_f(\text{water}) = 1.86^\circ\text{C kg/mol}$. Calculate the molar mass of the unknown solute.

ANSWER: _____

393. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

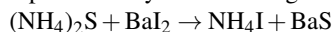
$$1000\text{ mol} \div 144.\text{ L}$$

ANSWER: _____

394. Consider the species BF_3 . Draw its Lewis structure and use it to answer the following questions. How many total valence electrons must appear in the Lewis structure of BF_3 ? How many lone pairs surround the central boron atom? Credit is awarded if you draw the Lewis structure correctly and answer both questions correctly.

ANSWER: _____

395. A way to produce solid barium sulfide is by combining aqueous ammonium sulfide with aqueous barium iodide. The chemical reaction is represented by the following unbalanced chemical equation:



Ian wants to produce 13.70 mg of barium sulfide. He pours 10.53 mL of a 0.060-M solution of aqueous ammonium sulfide into a beaker. What volume of a 0.3669-M solution of aqueous barium iodide is needed to react with the ammonium sulfide solution to produce 13.70 mg of barium sulfide? If you find that 10.53 mL of the ammonium sulfide solution is insufficient, report the required volumes of both reactants required to produce the 13.70 mg of barium sulfide.

ANSWER: _____

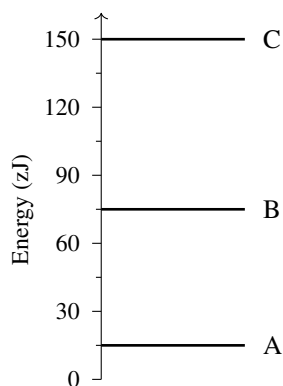
396. An unknown solid has a melting point of 1710°C , is a non-conductor as a solid, and is a non-conductor. Classify it as ionic, metallic, covalent-network, or molecular.

ANSWER: _____

397. CO_2 is collected over water. The total pressure of the gas sample is 747 torr, and the vapor pressure of water is 30. torr. Determine the partial pressure of the dry gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

398. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its first excited state? Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

399. A solution of KCl (KCl) is 25.0% by mass. Calculate the molality of the solution. ($M(KCl) = 74.55 \text{ g/mol}$)

ANSWER: _____

400. A 0.0500m aqueous solution of KCl (KCl) shows a freezing point depression of 0.186°C ($K_f = 1.86^{\circ}\text{C kg/mol}$). Determine the effective van't Hoff factor and state what it reveals about the solute in solution.

ANSWER: _____

401. Rank the following equimolal aqueous solutions from smallest to largest boiling point elevation: Na_2SO_4 , MgBr_2 , $\text{CH}_4\text{N}_2\text{O}$. All solutions have the same molality.

ANSWER: _____

402. Use molecular-orbital theory to determine the bond order and whether O_2^{2-} (peroxide) is paramagnetic or diamagnetic. Justify your answer.

ANSWER: _____

403. Which compound has a higher lattice energy, AlCl_3 or AlF_3 ? Explain your choice.

ANSWER: _____

404. Measurements show that an unknown compound has the following composition:

Element	Mass %
Ca	13.64 %
I	86.36 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

405. Determine whether aluminum and oxygen can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

ANSWER: _____

406. Suppose Ryan heats a sample of a certain iron (III) cyanide hydrate, $\text{Fe}(\text{CN})_3 \cdot x\text{H}_2\text{O}$ until all the water is removed and pure $\text{Fe}(\text{CN})_3$ is left behind. Find the percent by which the mass of the sample decreases in this process.

ANSWER: _____

407. Write the name of the inorganic acid H_2SO_3 .

ANSWER: _____

408. Experimental measurements show that the bond angle(s) in O_3 are $<120^\circ$. Explain why this observation led chemists to describe the bonding on the O atom using hybrid orbitals.

ANSWER: _____

409. A molecule's central atom is surrounded by 2 bonding groups and 3 lone pairs of electrons. Determine the electron-group arrangement, the molecular shape, and the ideal bond angle(s).

ANSWER: _____

410. Using MO theory, rank the following diatomic molecules in order of increasing bond order: C_2 , B_2 , F_2 . Justify your ranking by explaining how the MO electron occupancies lead to the observed trend.

ANSWER: _____

411. A student claims that two p orbitals always form a π bond when they overlap. Evaluate this claim by explaining what actually determines whether the bond is σ or π . Explain your reasoning completely. Answers without a correct explanation will not receive full credit.

ANSWER: _____

412. Calculate the mass percent of germanium in germanium tetrabromide (GeBr_4). Round your answer to the nearest percentage.

ANSWER: _____

413. Draw a valid Lewis structure for HI . You must show all existing lone pairs and lone electrons.

ANSWER: _____

414. Rank the following liquids from highest to lowest surface tension at 20°C : C_6H_{14} , CH_4 , Hg , H_2O . Offer brief explanation for your ranking.

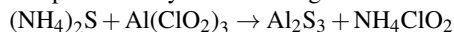
ANSWER: _____

415. As part of her duties as an analytical chemist, Elizabeth measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X . She then records these masses in a table in her lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Elizabeth would find in a new 10.0-g sample of X . Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1	mass of E_2
1	13.6 g	6.0 g
2	7.0 g	3.1 g
3	8.9 g	4.0 g
4	7.6 g	3.4 g

ANSWER: _____

416. A way to produce solid aluminum sulfide is by combining aqueous ammonium sulfide with aqueous aluminum chlorite. The chemical reaction is represented by the following unbalanced chemical equation:



Brian wants to produce 31.86 mg of aluminum sulfide. He pours 7.217 mL of a 0.70-M solution of aqueous ammonium sulfide into a beaker. What volume of a 0.050-M solution of aqueous aluminum chlorite is needed to react with the ammonium sulfide solution to produce 31.86 mg of aluminum sulfide? If you find that 7.217 mL of the ammonium sulfide solution is insufficient, report the required volumes of both reactants required to produce the 31.86 mg of aluminum sulfide.

ANSWER: _____

417. Determine which bond is expected to be longer: the $\text{C}\equiv\text{N}$ bond in CN^- or the $\text{C}-\text{N}$ bond in CH_3NH_2 . Justify your answer from fundamental principles. Responses based only on memorized trends will not receive credit.

ANSWER: _____

418. A chemist isolates an unknown gas in a sealed vessel. The sample has a mass of 5.044 g, occupies 1.500 L at 101.3 kPa and 293.15 K. Possible identities are: CO_2 , CO , Xe , He , Ar , Ne , CH_4 , SF_6 , UF_6 , C_2H_2 , H_2S , SO_2 , NO , F_2 , H_2O , H_2 , Kr , Cl_2 , HBr . Determine the identity of the unknown gas from the given choices. Justify your answer.

ANSWER: _____

419. Nick must note the concentration of a solution of CrCl_3 in his lab notebook. He knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Nick weighs an empty graduated cylinder and finds its mass to be 10 g. He then puts some solid CrCl_3 into the graduated cylinder and weighs it. With the CrCl_3 added, the cylinder weighs 21 g. He adds water to the graduated cylinder and dissolves the CrCl_3 completely. Then he reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 69.18 mL. What concentration should Nick write down in his lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

420. Calculate the heat absorbed when 7.82-g of benzene (C_6H_6) melts completely at its melting point of 6°C . $\Delta H_{\text{fus}} = 9.87 \text{ kJ/mol}$.

ANSWER: _____

421. What is the name and symbol of the metric prefix that represents a factor of 10^{-12} ?

ANSWER: _____

422. A pharmacist measures the mass of a powdered drug. The mass is 7398 mg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

423. Measurements show that an unknown compound has the following composition:

Element	Mass %
Nitrogen	43.7 %
Hydrogen	6.30 %
Oxygen	50.0 %

Write the empirical chemical formula of the unknown compound.

ANSWER: _____

424. Write the name of the inorganic acid H_3AsO_4 .

ANSWER: _____

425. Classify the following as an element, a compound, a homogeneous mixture, or a heterogeneous mixture: a bowl of cereal with milk, in which solid pieces can be seen in the liquid. You must explain your answer with definitions.

ANSWER: _____

426. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is 9.23×10^{-5} g. What is the mass in micrograms? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

427. Provide the name of the element that the element symbol S represents.

ANSWER: _____

428. Measurements show that an unknown compound has the following composition:

Element	Mass %
Hydrogen	2.63 %
Oxygen	27.8 %
Bromine	69.5 %

Write the empirical chemical formula of the unknown compound.

ANSWER: _____

429. Nitric acid is often manufactured in a two-step process from the atmospheric gases nitrogen and oxygen plus hydrogen, prepared by reforming of natural gas. In the first step, nitrogen and hydrogen react to form ammonia. In the second step, ammonia and oxygen react to form nitric acid and water. Write the net chemical equation for the production of nitric acid from nitrogen, hydrogen, and oxygen. Ensure that it is balanced.

ANSWER: _____

430. Using MO theory, determine the magnetic behavior of N_2^+ . Briefly justify your answer.

ANSWER: _____

431. Identify the state of matter in which water exists at 25 °C and 1 atm.

ANSWER: _____

432. Shiv must note the concentration of a solution of $NaMnO_4$ in his lab notebook. He knows that the concentration of a solution equals the mass of what is dissolved divided by the total volume of the solution. Shiv weighs an empty graduated cylinder and finds its mass to be 10 g. He then puts some solid $NaMnO_4$ into the graduated cylinder and weighs it. With the $NaMnO_4$ added, the cylinder weighs 20.98 g. He adds water to the graduated cylinder and dissolves the $NaMnO_4$ completely. Then he reads the total volume of the solution from the markings on the graduated cylinder. The total volume of the solution is 24.34 mL. What concentration should Shiv write down in his lab notebook? Be sure your answer has the correct number of significant figures.

ANSWER: _____

433. Calculate the wavelength of the line in the emission line spectrum of hydrogen caused by the transition of the electron from an orbital with $n = 13$ to an orbital with $n = 2$. Round your answer to 3 significant figures.

ANSWER: _____

434. Zinc bromide reacts with ammonium oxide to form ammonium bromide and zinc oxide. Suppose 57.47 g of zinc bromide is mixed with 20.07 g of ammonium oxide. Calculate and report the theoretical yield of zinc oxide formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

435. On the P - T phase diagram of sodium chloride (NaCl), the solid-liquid boundary has a positive slope. Which is denser, the solid or the liquid? Justify.

ANSWER: _____

436. Nick must dilute 73.88 mL of a 2.63-mM solution of aqueous aluminum bromide (AlBr_3) until the concentration falls to 0.654 mM. He will do this by adding distilled water to the solution. How much distilled water must he add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

437. Bismuth hydroxide reacts with hydroiodic acid to form water and bismuth iodide. If 5.014 g of water is produced from the reaction of 29.45 g of bismuth hydroxide and 52.68 g of hydroiodic acid, calculate the percent yield of water. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

438. Given this unbalanced chemical equation, report the maximum mass of $\text{Zn}_3(\text{PO}_4)_2$ that could be produced by the mixing of 11.3 g of K_3PO_4 with 10.6 g of ZnBr_2 . Ensure that your answer has the correct number of significant figures.

$$\text{K}_3\text{PO}_4 + \text{ZnBr}_2 \rightarrow \text{KBr} + \text{Zn}_3(\text{PO}_4)_2$$

ANSWER: _____

439. Classify the bond between carbon (C) and fluorine (F) as nonpolar covalent, polar covalent, or ionic. If the bond is polar or ionic, identify which atom carries the partial negative charge (δ^-). Justify your answer using the electronegativity difference.

ANSWER: _____

440. A plant root cell (internal particle concentration ≈ 0.300 osmol/L) is placed in a 0.200 M solution of KCl (KCl). Which way does water move across the membrane, and what happens to the cell? Explain your reasoning; answers without justification earn no credit.

ANSWER: _____

441. A brick has a mass of 8.775×10^0 kg and a virus has a mass of 4.8×10^{-12} mg. How many moles of viruses have a mass equal to the mass of a brick? Ensure your answer has the correct number of significant figures.

ANSWER: _____

442. Calculate the mass percent of silver in silver sulfide (Ag_2S). Round your answer to the nearest percentage.

ANSWER: _____

443. Provide the name of the element that the element symbol Cu represents.

ANSWER: _____

444. At 25°C and 1.00 atm, the solubility of O_2 (O_2) in water is $1.3 \times 10^{-3} \text{ mol/L}$. What partial pressure of O_2 is required to achieve a solubility of $5.2 \times 10^{-3} \text{ mol/L}$? Use Henry's law: $P = C/K_{\text{H}}$.

ANSWER: _____

445. You must prepare a solution for a precise freezing-point-depression measurement. Which concentration unit should express the concentration, and why?

ANSWER: _____

446. Two orbitals approach each other directly along the line joining two nuclei. Predict whether the bond formed is σ or π . Justify your answer.

ANSWER: _____

447. Why does the temperature stay constant along the melting plateau of a heating curve even though heat is continuously added? Justify.

ANSWER: _____

448. The reaction $\text{N}_2 + 3\text{H}_2 \longrightarrow 2\text{NH}_3$ occurs with all gaseous substances at the same temperature and pressure. If 3.01 mL of H_2 is completely consumed, what volume of N_2 is consumed or produced, as appropriate? Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

449. Ammonium sulfide reacts with chromium (III) oxide to form chromium (III) sulfide and ammonium oxide. Calculate the moles of chromium (III) oxide needed to produce 2.3 moles of chromium (III) sulfide. Ensure your answer has the correct number of significant figures.

ANSWER: _____

450. A movable piston traps a fixed amount of gas inside a cylinder. Initially the gas has a pressure of 170. kPa and occupies 8.87 L. The pressure is then increased to 360 kPa while the temperature remains constant. Determine the final volume. Justify your answer.

ANSWER: _____

451. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$21.8 \frac{\text{mol}}{\text{L}} \times 53. \text{L}$$

ANSWER: _____

452. An unknown chemical compound is composed of two unknown elements, A and B. The chemical formula is A_2B_x , where x represents a whole number. Measurements of a sample of the compound show that it contains 2.79 moles of A and 5.576 moles of B. Write the complete chemical formula for the unknown compound.

ANSWER: _____

453. Justin must dilute 18.9 mL of a 2.846-mM solution of aqueous ammonium iodide (NH_4I) until the concentration falls to 2.082 mM. He will do this by adding distilled water to the solution. How much distilled water must he add? Be sure your answer has the correct number of significant figures.

ANSWER: _____

454. An ionic compound crystallizes with a simple cubic array of anions and the cation in the cubic (body-center) hole. What is the cation:anion ratio (the formula unit) in the compound? Justify by counting ions per cell.

ANSWER: _____

455. A student assigns the hybridization of the central S in SO_3 from the Lewis structure $\text{O}=\text{S}(-\text{O})-\text{O}$. A second student instead uses the Lewis structure $\text{O}-\text{S}(=\text{O})-\text{O}$. Must both students assign the same hybridization or different hybridizations? State the correct hybridization of the central atom and justify your answer. Justify your answer completely.

ANSWER: _____

456. Classify P_2Se_3 as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: _____

457. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 45.8 g.

Compound	Initial Mass	Final Mass
A	46.3 g	22.2 g
B	33.9 g	0 g

ANSWER: _____

458. A nuclear chemist measures the mass of an artificial element. The mass is 30 pg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

459. Brandon measures the amount of antimony trifluoride produced during a chemical reaction. He finds that 0.0921 mol is produced. Calculate the mass of antimony trifluoride produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

460. A colloid consists of a liquid dispersed in a solid. Name this class of colloid and give an everyday example.

ANSWER: _____

461. Predict the products that form when boric acid is mixed with zinc hydroxide. Write the balanced chemical equation for this reaction.

ANSWER: _____

462. A rigid 13-L tank at 93.°C contains only dinitrogen monoxide (N₂O) and ethane (C₂H₆). Assume both gases behave ideally. The tank contains 10. g of dinitrogen monoxide and 8.58 g of ethane. Determine the total pressure. Justify your answer.

ANSWER: _____

463. Calculate the boiling point of a solution prepared by dissolving 15.0 g of sucrose (C₁₂H₂₂O₁₁) in 250. g of water. $K_b(\text{H}_2\text{O}) = 0.512^\circ\text{C kg/mol}$.

ANSWER: _____

464. For the central atom in SF_6 , determine its hybridization, electron-group geometry, molecular geometry, and the ideal bond angle(s). Justify your answer.

ANSWER: _____

465. A particular carrot has an energy content of 68.8 kcal. What is this in kilojoules?

ANSWER: _____

466. Two hybrid orbitals on the same atom are proposed to have different energies even though they came from the same hybridization scheme. Is this consistent with the standard hybridization model? Justify your answer.

ANSWER: _____

467. When a narrow glass capillary tube is dipped into water, does the water rise or become depressed in the tube? Justify using cohesive and adhesive forces.

ANSWER: _____

468. Two molecular orbitals are formed from the same pair of atomic orbitals. One has less electron density between the nuclei than the other. Which molecular orbital has the higher energy? Justify your answer.

ANSWER: _____

469. Use the number of unpaired electrons to classify C_2 as paramagnetic or diamagnetic, and state whether it is attracted to or repelled by a magnetic field. Justify your answer.

ANSWER: _____

470. An atom of a particular element has a radius of 193 pm and the average orbital speed of the electrons in it is approximately 1.08×10^7 m/s. Calculate the least possible uncertainty in a measurement of the speed of an electron in this atom. Write your answer as a percentage of the average speed and round it to two significant figures.

ANSWER: _____

471. Ashley puts a small sample of SCl_2 in water and lets it dissolve. After equilibrium is reached, she determines that several chemical species remain in the solution. These include SCl_2 and H_2O . Which of the following labels apply to the compound SCl_2 ? Write all that apply. Possible labels include ionic, molecular, strong acid, weak acid, strong base, and weak base. Explain why you chose your answer.

ANSWER: _____

472. Two Period-2 atoms each contribute one $2s$ orbital and three $2p$ orbitals to molecular-orbital formation. How many molecular orbitals are produced? Justify your answer.

ANSWER: _____

473. Dylan needs to determine the mass of a sample of ethanol. The density of the sample is $7.2 \frac{\text{g}}{\text{mL}}$ and its volume is 57.19 mL. What is the mass of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

474. Using the Gibbs phase rule, determine the degrees of freedom for a sample of pure liquid ethanol. Report the value, state what it means, and justify your answer.

ANSWER: _____

475. A rigid container is evacuated and then refilled to the same temperature and pressure with a different ideal gas whose molar mass is larger. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

476. Which compound has a higher lattice energy, AlCl_3 or AlF_3 ? Explain your choice.

ANSWER: _____

477. The vapor pressure of methanol (CH_3OH) is 0.130 atm at 20.00°C and 1.000 atm at 64.60°C . Calculate ΔH_{vap} in kJ/mol. $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.

ANSWER: _____

478. CO_2 is collected over water. The total pressure of the gas sample is 766.6 torr, and the vapor pressure of water is 27.0 torr. Determine the partial pressure of the dry gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

479. Several types of electromagnetic radiation (EMR) are: x-rays, red light, yellow light, and ultraviolet light. Order these types of EMR from lowest energy to highest energy.

ANSWER: _____

480. Nick has a 0.146-M solution of caesium carbonate (Cs_2CO_3). If he needs to add 0.295 g of caesium carbonate to a reaction flask, what volume of the solution should he pour? Be sure your answer has the correct number of significant figures.

ANSWER: _____

481. Several types of electromagnetic radiation (EMR) are: red light, microwaves, gamma rays, and infrared light. Order these types of EMR from shortest wavelength to longest wavelength.

ANSWER: _____

482. A geologist measures the weight of an ore sample. The weight is 4.2641 kg. What is the weight in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

483. A lab technician measures the weight of a reagent. The weight is 4790 g. What is the weight in kilograms? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

484. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$130.5\text{ mL} + 150\text{ mL}$$

ANSWER: _____

485. A particular polonium atom has a total of 3 bonds to other atoms and has 1 lone pair of electrons around it. Calculate the formal charge on this polonium atom.

ANSWER: _____

486. Two atoms form the same number of bonds but differ in whether one bond can be rotated without changing the bonding interaction. Is this difference explained primarily by hybridization or by σ/π bonding? Justify your answer.

ANSWER: _____

487. Several types of electromagnetic radiation (EMR) are: blue light, yellow light, x-rays, and ultraviolet light. Order these types of EMR from lowest energy to highest energy.

ANSWER: _____

488. Eric prepares a solution of sodium oxide (Na_2O) by measuring out 275.3 mmol of sodium oxide into a 25.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

489. Determine the total number of σ and π bonds in $\text{CH}_2=\text{CH}_2$ (ethene). For credit you must justify your answer by explaining the source of the σ and π bonds.

ANSWER: _____

490. Kyle conducted some measurements and determined that 0.211 moles of fluorine gas participated in a chemical reaction. Calculate the mass of fluorine gas that participated in the chemical reaction.

ANSWER: _____

491. Using MO theory, rank the following diatomic molecules in order of increasing bond order: F_2 , B_2 , C_2 . Justify your ranking by explaining how the MO electron occupancies lead to the observed trend.

ANSWER: _____

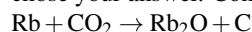
492. On the potential-energy curve for Cl_2 , what happens to the potential energy at internuclear distances much shorter than 199 pm? Why does this happen? Justify your answer.

ANSWER: _____

493. Isabella adds 533.1 mL of a 0.0127-M solution of potassium chlorite (KClO_2) to a reaction flask. Calculate the amount of potassium chlorite that she has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

494. As CO_2 is converted into C in the following reaction, determine whether carbon is oxidized, reduced, or neither. Clearly explain why you chose your answer. Consider only the carbon found in CO_2 and C.

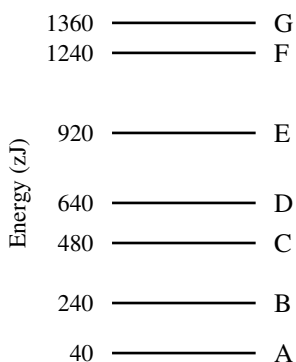


ANSWER: _____

495. An unknown solid has the following properties: variable melting point, malleable/ductile, conducts electricity as solid. Classify this solid as ionic, metallic, covalent-network, or molecular. For credit you must justify your answer choice.

ANSWER: _____

496. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the longest wavelength? You must explain your answer choice for full credit.



ANSWER: _____

497. A solution shows a boiling point elevation of 0.186°C when 100. g of water is used as the solvent. How many grams of naphthalene (C_{10}H_8 , $M = 128.17 \text{ g/mol}$) are dissolved? ($K_b(\text{H}_2\text{O}) = 0.512^{\circ}\text{C kg/mol}$)

ANSWER: _____

498. Each week, Deskins Chemicals, Inc., a manufacturer of fine chemicals, uses 128 kg of compound A and 455 kg of compound B in a reaction that produces only one product, compound C. If the average percent yield of C in the company's reactors is 74.6%, calculate the average mass of compound C that the company actually ships each week.

ANSWER: _____

499. When two s orbitals overlap head-on, what type of bond forms? Explain how the orientation of the orbital overlap determines the bond type. Explain your reasoning completely. Answers without a correct explanation will not receive full credit.

ANSWER: _____

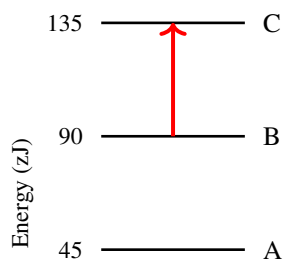
500. Use molecular-orbital theory to determine the bond order and whether C_2 is paramagnetic or diamagnetic. Justify your answer.

ANSWER: _____

501. The vapor pressure of chloroform (CHCl_3) is 0.086 atm at 0.00°C and 1.000 atm at 61.00°C . Calculate ΔH_{vap} in kJ/mol. $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.

ANSWER: _____

502. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from B to C, as shown by the arrow. Calculate the energy of the photon that would be absorbed or emitted. Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

503. Provide the name and empirical formula of the ionic compound that is formed by the Ag^+ cation and the $\text{Cr}_2\text{O}_7^{2-}$ anion.

ANSWER: _____

504. An astronomer measures the distance from Earth to a newly discovered asteroid. The distance is $7.66 \times 10^{10}\text{m}$. What is the distance in gigameters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

505. Provide the chemical symbol of the most commonly formed ion of oxygen and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: _____

506. Determine the electron-group arrangement, molecular shape, and ideal bond angle(s) for boron trifluoride.

ANSWER: _____

507. The normal boiling point of water (H_2O) is 100.00°C ($P_{\text{vap}} = 1.000\text{atm}$) and its molar enthalpy of vaporization is 40.7 kJ/mol . Calculate the vapor pressure of water at 80.00°C .

ANSWER: _____

508. Imagine an alternate universe in which the value of the Planck constant is 8.416×10^{-34} J·s. In that universe, is quantum mechanics necessary to accurately describe a 1.4×10^{-12} -m-wide object that has a mass of 4.8×10^{-19} kg and is moving at 9.1×10^{-5} m/s? Write "yes" if quantum mechanics is necessary and "no" if it is not, then explain why you made your choice. Answers lacking supporting explanation will earn no credit.

ANSWER: _____

509. When 0.125 g of an unknown nonelectrolyte is dissolved in 100. g of camphor (camphor), the freezing point decreases by 0.512°C . $K_f(\text{camphor}) = 37.7^\circ\text{C kg/mol}$. Calculate the molar mass of the unknown solute.

ANSWER: _____

510. A geologist measures the height of a mountain. The height is 5.8449 km. What is the height in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

511. Benjamin needs to know the mass of a sample of ZnI_2 to 4 significant figures. He puts the sample on a digital scale. This is what the scale shows: 69.37 g. If this measurement is precise enough for Benjamin, round it to 4 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: _____

512. Calculate the heat required to completely vaporize 41.9-g of methanol (CH_3OH) at its normal boiling point of 65°C . $\Delta H_{\text{vap}} = 35.2 \text{ kJ/mol}$, $M = 32.042 \text{ g/mol}$.

ANSWER: _____

513. CaF_2 crystallizes in a fluorite structure with unit cell edge length 546.3 pm. Calculate the density of CaF_2 .

ANSWER: _____

514. Mark measures the mass of an aluminum cylinder using a balance. The accepted mass of the object is 44.0 g. He records a measurement of 45.28 g. What is the percent error of his measurement? Report your answer to two significant figures.

ANSWER: _____

515. Methanol (CH_3OH , density 0.790 g/mL) is shaken with an equal volume of water (1.00 g/mL). Will two layers form? If so, which liquid ends up on top? Explain.

ANSWER: _____

516. Grace is comparing two unknown liquid samples. Sample A has a mass of 15 g and a volume of 13 mL . Sample B has a mass of 20.7 g and a volume of 13 mL . Which sample has the greater density? You must include correct work that supports your conclusion.

ANSWER: _____

517. Sophia measures the amount of strontium bromide produced during a chemical reaction. She finds that 0.0812 mol is produced. Calculate the mass of strontium bromide produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

518. David measures the mass of a copper disc using a balance. The accepted mass of the object is 40.4 g . He records a measurement of 41.41 g . What is the percent error of his measurement? Report your answer to two significant figures.

ANSWER: _____

519. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.363 m . What is the thickness in centimeters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

520. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 47.2 g .

Compound	Initial Mass	Final Mass
A	78.7 g	0 g
B	66.1 g	9.9 g

ANSWER: _____

521. Write the full electron configuration of a neutral strontium atom. Then, using the octet rule, predict the charge of the monatomic ion that strontium forms and identify the noble gas with which that ion is isoelectronic. Explain your reasoning.

ANSWER: _____

522. Ammonium chloride reacts with aluminum bromide to form ammonium bromide and aluminum chloride. Suppose 14.0g of ammonium chloride is mixed with 14.4g of aluminum bromide. Calculate the maximum mass of ammonium bromide that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

523. Rank the following bonds in order of increasing bond strength (bond energy): $\text{C}\equiv\text{N}$, $\text{C}=\text{N}$, $\text{C}-\text{N}$. Support your ranking with a brief explanation for credit. Credit is not awarded for simply citing a memorized trend.

ANSWER: _____

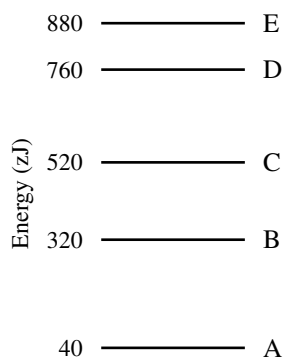
524. Calculate the osmotic pressure of 0.90% saline at 25°C (modeled as a 0.154M electrolyte solution with $i = 2$). What minimum pressure must be applied to achieve reverse osmosis?

ANSWER: _____

525. Calculate the osmotic pressure of a 0.200M aqueous solution of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) at 25°C .

ANSWER: _____

526. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the longest wavelength? You must explain your answer choice for full credit.



ANSWER: _____

527. On an intermolecular potential-energy curve $u(r)$, what is the physical significance of the separation r at which $u(r)$ reaches its minimum? Justify.

ANSWER: _____

528. Use average bond energies to estimate ΔH° for the reaction $\text{C}_2\text{H}_2 + 2\text{H}_2 \longrightarrow \text{C}_2\text{H}_6$, and state whether the reaction is endothermic or exothermic. Average bond energies (kJ/mol): $\text{C}\equiv\text{C} = 839$, $\text{C}-\text{C} = 347$, $\text{C}-\text{H} = 414$, $\text{H}-\text{H} = 436$.

ANSWER: _____

529. How does increasing temperature affect the solubility of KNO_3 in water? Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

530. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.

ANSWER: _____

531. Determine the hybridization of each atom in CO_2 . Briefly justify each assignment using the local bonding environment of that atom.

ANSWER: _____

532. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

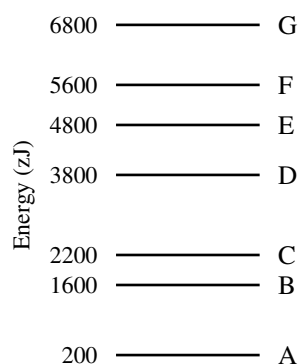
$$21. \frac{\text{mol}}{\text{L}} \times 38. \text{L}$$

ANSWER: _____

533. At 25°C and 1.00 atm, the solubility of N_2 (N_2) in water is $6.1 \times 10^{-4} \text{ mol/L}$. What partial pressure of N_2 is required to achieve a solubility of $1.8 \times 10^{-3} \text{ mol/L}$? Use Henry's law: $P = C/K_{\text{H}}$.

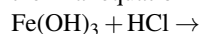
ANSWER: _____

534. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the shortest wavelength? You must explain your answer choice for full credit.



ANSWER: _____

535. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.



ANSWER: _____

536. A vertical cylinder containing gas has a piston of diameter 12-cm. The cylinder is 14-cm tall. A constant downward force of 58-N acts on the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

537. As part of her duties as an analytical chemist, Victoria measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. She then records these masses in a table in her lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Victoria would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1	mass of E_2
1	16.4 g	8.2 g
2	19.6 g	8.7 g
3	8.7 g	6.2 g
4	13.5 g	6.8 g

ANSWER: _____

538. In a certain chemical reaction, compound A combines with compound B to produce compound C (and no other products). Measurements were taken of the amounts of A and B present before and after a reaction that produced some C and are presented in the following table. Report the percent yield of C in this reaction to the nearest percent if the actual amount of C that was isolated at the end of the reaction was 45.1 g.

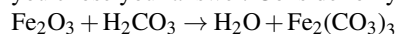
Compound	Initial Mass	Final Mass
A	77.6 g	34.9 g
B	75.9 g	0 g

ANSWER: _____

539. Calculate the osmotic pressure of seawater at 25°C (modeled as a 0.600M electrolyte solution with $i = 2$). What minimum pressure must be applied to achieve reverse osmosis?

ANSWER: _____

540. As H_2CO_3 is converted into H_2O in the following reaction, determine whether oxygen is oxidized, reduced, or neither. Clearly explain why you chose your answer. Consider only the oxygen found in H_2CO_3 and H_2O .



ANSWER: _____

541. Identify the dominant intermolecular force acting between methanol (CH_3OH) and water (H_2O) when they are mixed. Justify your answer.

ANSWER: _____

542. A cooking pot has a mass of 6.616×10^{-1} kg and a particular planet has a mass of 2.0×10^{26} g. What is the mass of 1 mole of cooking pots? Ensure your answer has the correct number of significant figures.

ANSWER: _____

543. Calculate the mole fractions of methanol (CH_3OH) and water in a solution prepared by mixing 18.4 g of methanol with 100. g of water.

ANSWER: _____

544. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$5.55 \text{ cm} - 0.30168 \text{ cm}$$

ANSWER: _____

545. Calculate the freezing point of a solution prepared by dissolving 20.0 g of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in 200. g of water. $K_f(\text{H}_2\text{O}) = 1.86^\circ\text{C kg/mol}$.

ANSWER: _____

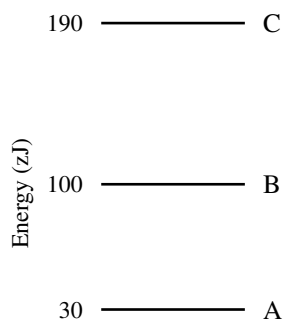
546. An atom forms both an sp^3 hybrid orbital and an sp^2 hybrid orbital. Without calculating any numerical energies, determine which hybrid orbital has the higher expectation value of energy and justify your answer.

ANSWER: _____

547. Answer the following questions: is Ba an atom, a cation, or an anion? How many protons does it have? How many electrons does it have?

ANSWER: _____

548. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. How many lines are in the emission line spectrum?



ANSWER: _____

549. Rewrite the following chemical equation as a balanced equation using the smallest possible whole-number stoichiometric coefficients.
 $(\text{H}_3\text{O})_2\text{S} + \text{AlCl}_3 \rightarrow \text{H}_3\text{OCl} + \text{Al}_2\text{S}_3$

ANSWER: _____

550. A 0.500M solution of HCl (HCl) has a density of 1.02 g/mL. Calculate the molality of the solution. ($\mathcal{M} = 36.46 \text{ g/mol}$)

ANSWER: _____

551. What is the oxidation number of iron in iron (III) sulfide?

ANSWER: _____

552. Compare He_2^+ and He_2 using molecular-orbital (MO) theory. For each species, write the MO electron configuration, determine the bond order and magnetic character, and then explain how removing one electron affects the bond order and the magnetic behavior.

ANSWER: _____

553. An X-ray diffraction experiment on a crystal produces a strong reflected beam under the conditions below. A crystal with plane spacing 394 pm produces the first-order diffraction maximum at a glancing angle of $\theta = 16.9^\circ$. Determine the wavelength of the incident X-rays. Calculate the requested quantity and justify your answer.

ANSWER: _____

554. Nick looks up the density of a material and finds it listed as $7.628 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.

ANSWER: _____

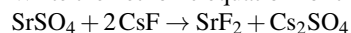
555. Linda looks up the density of a material and finds it listed as $13 \frac{\text{g}}{\text{cm}^3}$. Convert this density to $\frac{\text{kg}}{\text{m}^3}$. Show your work. Ensure your answer has the correct number of significant figures.

ANSWER: _____

556. Compare He_2^+ and He_2 using molecular-orbital (MO) theory. For each species, write the MO electron configuration, determine the bond order and magnetic character, and then explain how removing one electron affects the bond order and the magnetic behavior.

ANSWER: _____

557. Write the net ionic equation for the following chemical reaction, which takes place in aqueous solution:

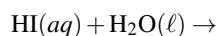


ANSWER: _____

558. Use average bond energies to estimate ΔH° for the reaction $\text{CH}_4 + 2\text{O}_2 \longrightarrow \text{CO}_2 + 2\text{H}_2\text{O}$, and state whether the reaction is endothermic or exothermic. Average bond energies (kJ/mol): C–H = 414, C=O = 799, O–H = 464, O=O = 498.

ANSWER: _____

559. Predict the products of the reaction below by completing the right-hand side of the chemical equation. Provide the full balanced equation with state symbols.



ANSWER: _____

560. Stephen is a chemical engineer who needs to convert an oven temperature for a thermodynamic calculation. The temperature is 95 °F. Convert this temperature to kelvin. Round to the nearest integer.

ANSWER: _____

561. Joseph prepares a solution of ammonium chloride (NH_4Cl) by measuring out 11.49 g of ammonium chloride into a 300.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

562. One of NH_3 and PH_3 has a substantially higher boiling point than the other. Determine which one and justify your answer.

ANSWER: _____

563. Caleb puts a small sample of $\text{Ba}(\text{OH})_2$ in water and lets it dissolve. After equilibrium is reached, he determines that several chemical species remain in the solution. These include Ba^{2+} , OH^- , and H_2O . Which of the following labels apply to the compound $\text{Ba}(\text{OH})_2$? Write all that apply. Possible labels include ionic, molecular, strong acid, weak acid, strong base, and weak base. Explain why you chose your answer.

ANSWER: _____

564. A gas cylinder is opened into a large empty flexible reservoir. The cylinder has an internal diameter of 4.1-cm, an internal height of 23-cm, and initially contains gas at 5.1 atm. The valve is opened until the gas reaches atmospheric pressure, after which the valve is closed. Determine the total volume occupied by the gas, including both the gas remaining in the cylinder and the gas in the reservoir. Justify your answer.

ANSWER: _____

565. A solution is prepared by dissolving 10.0 g of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$, $M = 342.3 \text{ g/mol}$) in 500. g of water. Calculate the mole fraction of sucrose.

ANSWER: _____

566. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$0.08877 \text{ cm} - 0.070 \text{ cm}$$

ANSWER: _____

567. Potassium bromide reacts with hydronium fluoride to form potassium fluoride and hydronium bromide. Calculate the mass of hydronium fluoride needed to produce 5.38 g of potassium fluoride. Ensure your answer has the correct number of significant figures.

ANSWER: _____

568. A strip of solid potassium metal is put into a beaker of 0.026-M NaHCO_3 solution. Does a chemical reaction occur? Explain why it does or why it does not. If it does occur, write the net ionic equation that describes it. Be sure your equation is balanced and has physical-state symbols.

ANSWER: _____

569. William puts a 0.567-kg aluminum pot on the stove and fills it with 0.162 liters of water. Both the pot and the water are initially at 10.1°C. How much heat is required to bring them both to 78.°C? The specific heat value of water is 4186. J/kg°C and for aluminum it is 900. J/kg°C. Recall the density of water is 1.000 g/cc.

ANSWER: _____

570. The reaction $2\text{NO}(\text{g}) + 2\text{H}_2(\text{g}) \longrightarrow \text{N}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 6.49 mL of hydrogen gas reacts completely, what volume of nitrogen gas is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

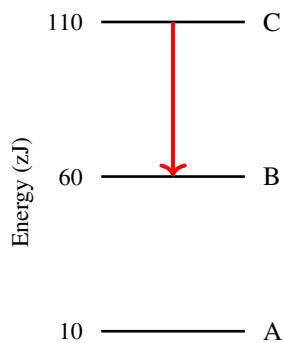
571. Write the complete MO electron configuration for B_2 . Determine its bond order and magnetic character. Justify each answer.

ANSWER: _____

572. Which element is 16% (by mass) of diantimony trioxide (Sb_2O_3)? Show your work.

ANSWER: _____

573. This energy diagram shows the allowed energy levels of an electron in a certain atom. If the electron makes a transition from C to B, as shown by the arrow, will a photon be absorbed or emitted? You must explain your answer choice for full credit.



ANSWER: _____

574. A solution of CO_2 (CO_2) in water at 25°C is prepared at a partial pressure of 2.0 atm. If the partial pressure is increased to 4.0 atm, by what factor does the solubility change? Use Henry's law.

ANSWER: _____

575. Zoe needs to find the volume of a 9.67 g sample of seawater. The density of the sample is $0.52 \frac{\text{g}}{\text{mL}}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

576. An optician measures the diameter of her patient's pupil. The diameter is 0.0047 m. What is the diameter in millimeters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

577. Predict the products that form when hydrofluoric acid is mixed with lead (II) hydroxide. Write the balanced chemical equation for this reaction.

ANSWER: _____

578. Determine whether oxygen and bromine can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

ANSWER: _____

579. The reaction $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \longrightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 14.5 cm^3 of methane is consumed, what volume of carbon dioxide is produced or consumed? Justify your answer.

ANSWER: _____

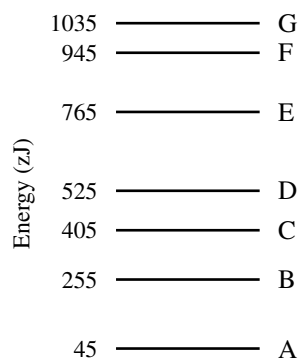
580. Describe the layer stacking sequence of cubic closest packing (CCP) and explain what makes it distinct. Justify.

ANSWER: _____

581. Hydroiodic acid reacts with rubidium hydroxide to form water and rubidium iodide. Suppose 3.826 g of hydroiodic acid is mixed with 4.098 g of rubidium hydroxide. Calculate and report the theoretical yield of rubidium iodide formed by this chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

582. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the lowest frequency? You must explain your answer choice for full credit.



ANSWER: _____

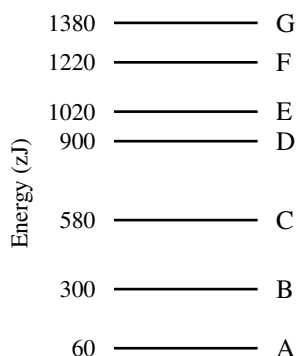
583. Consider the species CO_2 . Draw its Lewis structure and use it to answer the following questions. How many total valence electrons must appear in the Lewis structure of CO_2 ? How many lone pairs surround the central carbon atom? Credit is awarded if you draw the Lewis structure correctly and answer both questions correctly.

ANSWER: _____

584. An atom is sp^3 hybridized. What is the greatest number of π bonds it could participate in? Justify your answer.

ANSWER: _____

585. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the shortest wavelength? You must explain your answer choice for full credit.

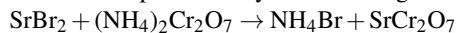


ANSWER: _____

586. Predict what happens to solubility when KNO_3 is dissolved in hot water and then cooled. Identify whether temperature or pressure is the key factor and explain the observation.

ANSWER: _____

587. A way to produce solid strontium dichromate is by combining aqueous strontium bromide with aqueous ammonium dichromate. The chemical reaction is represented by the following unbalanced chemical equation:



Knowing this, Lauren pours 12.89 mL of a 0.0307-M solution of aqueous strontium bromide and then adds 10.21 mL of a 0.0189-M solution of aqueous ammonium dichromate. She then stirs the mixture and notices strontium dichromate precipitating out of the solution. What is the mass of strontium dichromate that she should expect the reaction to yield?

ANSWER: _____

588. Fe_2O_3 is the formula for iron (III) oxide. Kyle, a lab technician, measures a sample of iron (III) oxide and determines that it contains 0.220 moles of iron. How many moles of oxygen are in the sample?

ANSWER: _____

589. Samantha is performing a combustion analysis on a compound with a molecular molar mass of 46.07 g/mol that is composed of carbon, hydrogen, and perhaps oxygen. A sample of 3.14 g of the compound is completely burned in excess oxygen and the mass of the products is carefully measured:

Product	Mass (g)
Carbon Dioxide	6.0 g
Water	3.68 g

Use this information to find the molecular formula of the compound.

ANSWER: _____

590. On a heating curve for water, the boiling plateau is much longer than the melting plateau. What does this tell you? Justify.

ANSWER: _____

591. A geologist measures the height of a mountain. The height is 9.5169 km. What is the height in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

592. As part of her duties as an analytical chemist, Anna analyzes a sample of an unknown pure substance X and finds that it contains 12.7 g of element E₁ and 4.5 g of element E₂, and no other elements. She then analyzes three other substances that also contain only elements E₁ and E₂ and records the results in a table in her lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E ₁	mass of E ₂
X	12.7 g	4.5 g
A	22.9 g	8.7 g
B	17.5 g	3.1 g
C	29.9 g	5.8 g

ANSWER: _____

593. How many grams of Fe can melt when 6.19-kJ of heat is absorbed during fusion at 1538°C? Assume $\Delta H = 247 \text{ J/g}$.

ANSWER: _____

594. A 4.57 g sample of dinitrogen monoxide gas (N₂O) is contained in a sealed reaction vessel at 13.0°C and 1.34 atm. Determine the sample volume assuming ideal-gas behavior. Report your answer in SI units. Justify your answer.

ANSWER: _____

595. A gas mixture contains CO₂, CH₄, and N₂. The total pressure is 0.985 atm. The mole fractions of CO₂ and CH₄ are 0.250 and 0.100, respectively. Determine the partial pressure of N₂. Report your answer in SI units. Justify your answer.

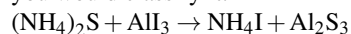
ANSWER: _____

596. Multiply the following measurements, ensuring that your answer has the correct number of significant figures.

$$42. \frac{\text{mol}}{\text{L}} \times 15.8 \text{ L}$$

ANSWER: _____

597. Classify this reaction as one of the following types: (I) combination, (II) decomposition, (III) single-displacement, (IV) double-displacement (AKA metathesis). You must justify your answer choice. If you believe that the reaction cannot be classified as one of these types, write how you would classify it.



ANSWER: _____

598. Measurements show that an unknown compound has the following composition:

Element	Mass %
Aluminum	7.69 %
Chromium	44.4 %
Oxygen	47.9 %

Write the empirical chemical formula of the unknown compound.

ANSWER: _____

599. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 285 GHz. What is the frequency in hertz (Hz)? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

600. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 686 MPa. The piston of a circular circular piston can safely withstand a maximum force of 3.4 MN. Determine the largest allowable diameter of the circular piston. Report your answer in SI units and justify your answer.

ANSWER: _____

601. Explain why X-rays, rather than visible light, are used to determine the spacing of atoms in crystals. Justify.

ANSWER: _____

602. A bond allows unrestricted rotation about the internuclear axis without decreasing orbital overlap. Determine whether the bond is σ or π . Justify your answer.

ANSWER: _____

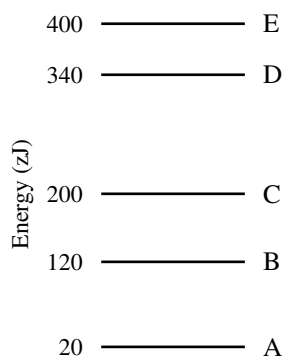
603. Calculate the molar solubility of CO_2 (CO_2) in water at 25°C when the partial pressure of CO_2 above the solution is 0.25 atm. Use $K_{\text{H}} = 3.4 \times 10^{-2} \text{ mol}/(\text{L}\cdot\text{atm})$ (Henry's law: $C = K_{\text{H}}P$).

ANSWER: _____

604. Suppose a popular FM-radio station broadcasts radio waves with a frequency of 90.7 MHz. Calculate the wavelength of these radio waves.

ANSWER: _____

605. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the longest wavelength? You must explain your answer choice for full credit.



ANSWER: _____

606. Blood plasma has a total dissolved-particle concentration of about 0.300 osmol/L. Classify each of the following as hypertonic, hypotonic, or isotonic with blood: 0.150M NaCl, 0.150M Na₂SO₄, 0.100M MgCl₂. Justify each classification.

ANSWER: _____

607. Write the chemical formula of silver sulfide decahydrate.

ANSWER: _____

608. Benzene (C₆H₆) has a normal boiling point of 80°C and $\Delta H_{\text{vap}} = 30.8 \text{ kJ/mol}$. At what temperature does it boil inside a pressure cooker, where the external pressure is 2.13 atm? ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.)

ANSWER: _____

609. Calculate the number of phosphorus atoms in a 4.943-g sample of phosphorus triiodide (PI₃).

ANSWER: _____

610. Calculate the vapor pressure of a solution prepared by dissolving 68.4 g of sucrose (C₁₂H₂₂O₁₁) in 100. g of water at 60°C. The vapor pressure of pure water at 60°C is 149.4 mmHg.

ANSWER: _____

611. Two pure substances, HCl and HF, are compared. Predict which requires the higher temperature to boil. Justify your prediction. Unsupported trend statements receive no credit.

ANSWER: _____

612. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.217 m. What is the thickness in centimeters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

613. Which atom has the higher electronegativity: sodium (Na) or magnesium (Mg)? You must briefly explain why. Credit is awarded only for explaining the cause of the trend based on atomic structure, not for merely stating the trend.

ANSWER: _____

614. The reaction $2\text{NH}_3(\text{g}) + 3\text{Cl}_2(\text{g}) \longrightarrow \text{N}_2(\text{g}) + 6\text{HCl}(\text{g})$ occurs at constant temperature and pressure. If 3.2cm^3 of Cl_2 gas reacts completely, what volume of HCl gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

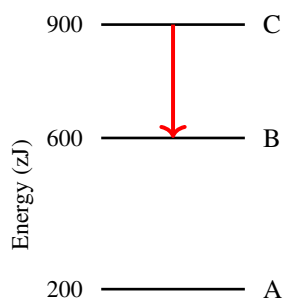
615. A chemistry graduate student is designing a pressure vessel for an experiment. The vessel will contain gases at pressures up to 571.8MPa . The piston of a circular circular piston can safely withstand a maximum force of 5.4MN . Determine the largest allowable diameter of the circular piston. Report your answer in SI units and justify your answer.

ANSWER: _____

616. Calculate the vapor pressure of a solution prepared by dissolving 34.2g of urea ($\text{CH}_4\text{N}_2\text{O}$) in $200.\text{g}$ of water at 80°C . The vapor pressure of pure water at 80°C is 355.1mmHg .

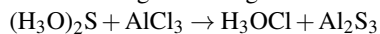
ANSWER: _____

617. This energy diagram shows the allowed energy levels of an electron in a certain atom. If the electron makes a transition from C to B, as shown by the arrow, will a photon be absorbed or emitted? You must explain your answer choice for full credit.



ANSWER: _____

618. Given this unbalanced chemical equation, report the mass of $(\text{H}_3\text{O})_2\text{S}$ needed to produce 3.21g of H_3OCl . Ensure your answer has the correct number of significant figures.



ANSWER: _____

619. An atom forms both an sp^3 hybrid orbital and an sp hybrid orbital. Without calculating any numerical energies, determine which hybrid orbital has the higher expectation value of energy and justify your answer.

ANSWER: _____

620. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is 7.43×10^{-5} g. What is the mass in micrograms? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

621. Graphite and graphene both consist entirely of carbon. Determine which material would be expected to conduct electricity more effectively within its carbon layer(s). State your answer and justify it.

ANSWER: _____

622. Determine the total number of σ and π bonds in $\text{CH}_2=\text{CHNH}_2$ (vinylamine). For credit you must justify your answer by explaining the source of the σ and π bonds.

ANSWER: _____

623. Calculate the boiling point of a solution prepared by dissolving 15.0 g of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in 250. g of water. $K_b(\text{H}_2\text{O}) = 0.512^\circ\text{C kg/mol}$.

ANSWER: _____

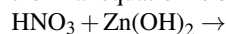
624. A mixture of ethane and sulfur tetrafluoride gas is expanded from a volume of 1.0×10^2 L to a volume of 110 L while the pressure is held constant at 77.3 atm. Calculate the work done on the gas mixture. Report your answer in SI unit of work and to 3 significant figures.

ANSWER: _____

625. Rank the following elements from highest metallicity to lowest metallicity. Support your ranking with an explanation of the underlying atomic structure; answers that simply cite a memorized periodic trend will not receive credit.
Mg, Sr, Ca, Ba

ANSWER: _____

626. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.



ANSWER: _____

627. Which element is 29% (by mass) of potassium sulfide (K_2S)? Show your work.

ANSWER: _____

628. A gas mixture contains 11 g of ethylene (C_2H_4) and 8.83 g of dinitrogen difluoride (N_2F_2). Determine the mole fraction of ethylene in the mixture. Justify your answer.

ANSWER: _____

629. The industrial synthesis of acetic acid (CH_3COOH), the acid found in vinegar, via the methanol carbonylation process involves two key steps. In the first step, methanol (CH_3OH) reacts with carbon monoxide to form methyl formate (HCOOCH_3). In the second step, methyl formate reacts with carbon monoxide and water to produce acetic acid, carbon monoxide, and water. Write the net chemical equation for the production of acetic acid from methanol, carbon monoxide, and water. Ensure that it is balanced.

ANSWER: _____

630. Provide the name and empirical formula of an ionic compound that is formed by the calcium cation and the oxide anion.

ANSWER: _____

631. A mixture contains visible particles that settle on standing. Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

632. A particular planet has a mass of 3.393×10^{27} g and a cooking pot has a mass of 8.41×10^{-1} kg. How many moles of this particular planet have a mass equal to the mass of a cooking pot? Ensure your answer has the correct number of significant figures.

ANSWER: _____

633. The reaction $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If $14. \text{cm}^3$ of hydrogen is consumed, what volume of chlorine is produced or consumed? Justify your answer.

ANSWER: _____

634. For the Be atom in BeCl_2 , determine its hybridization, molecular geometry about that atom, and ideal bond angle. Justify your answer.

ANSWER: _____

635. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is $7.7648 \times 10^{-7} \text{ m}$. What is the wavelength in nanometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

636. Equal masses (1.00 g each) of LiCl , MgCl_2 , CH_3OH are each dissolved in 1.00 kg of water. Which solution has the greatest boiling point elevation?

ANSWER: _____

637. A solid shows identical properties in every direction (isotropic) and has no sharp melting point. Is it more likely crystalline or amorphous? Justify.

ANSWER: _____

638. List the following electron subshells in order of increasing energy. Assume these subshells are in an atom with many electrons.
3s, 2s, 5s, 3p, 2p

ANSWER: _____

639. The lattice energy of NaF (NaF) is $+929 \text{ kJ/mol}$ (energy to separate ions) and the hydration energy is -904 kJ/mol . Calculate ΔH_{sol} . Is dissolution endothermic or exothermic?

ANSWER: _____

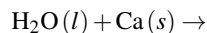
640. Supercritical CO_2 is used to decaffeinate coffee. Suggest why it is preferred over a liquid organic solvent, based on its physical properties.

ANSWER: _____

641. Identify the dominant type of intermolecular force present in pure H_2O . You must justify your answer.

ANSWER: _____

642. Determine whether this reaction will occur. If it does, write the complete balanced equation. If it does not, write "no reaction occurs".



ANSWER: _____

643. Which atom has the greater metallic character: sodium (Na) or phosphorus (P)? You must briefly explain why. Credit is awarded only for explaining the cause of the trend based on atomic structure, not for merely stating the trend.

ANSWER: _____

644. Does the central atom in BeF_2 obey the octet rule? If not, classify the exception as electron-deficient or expanded octet, and state the number of electrons around the central atom.

ANSWER: _____

645. Classify the following as a physical or chemical change: Mixing red and blue paint. Explain your answer.

ANSWER: _____

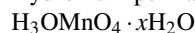
646. Rank the following liquids from highest to lowest viscosity at 20°C : C_6H_{14} , $\text{C}_2\text{H}_5\text{OH}$, H_2O , $\text{C}_3\text{H}_8\text{O}_3$.

ANSWER: _____

647. List every type of intermolecular force present in pure NH_3 . Justify your answer.

ANSWER: _____

648. Hydronium permanganate forms several hydrates with the general form



where x is an integer. If the hydrate is heated, the water can be driven off, leaving pure H_3OMnO_4 behind. Suppose a sample of a certain hydrate of hydronium permanganate is heated until all the water is removed, and a chemist finds that the mass of the sample decreases by 44%. Give the value of x to indicate which hydrate the sample was.

ANSWER: _____

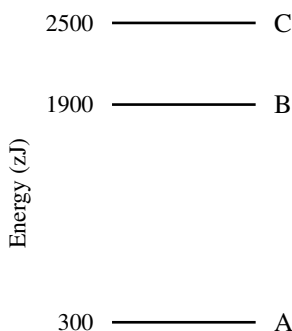
649. Rank these primary amines from most to least soluble in water: $\text{C}_2\text{H}_5\text{NH}_2$, $\text{C}_3\text{H}_7\text{NH}_2$, $\text{C}_8\text{H}_{17}\text{NH}_2$, $\text{C}_6\text{H}_{13}\text{NH}_2$. Explain the trend.

ANSWER: _____

650. An optician measures the diameter of her patient's pupil. The diameter is 0.0015 m. What is the diameter in millimeters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

651. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the lowest frequency? You must explain your answer choice for full credit.



ANSWER: _____

652. Classify the following as a pure substance or a mixture: a diamond gemstone. You must explain your answer and include definitions.

ANSWER: _____

653. A solution shows a freezing point depression of 0.512°C when 200. g of water is used as the solvent. How many grams of anthracene ($\text{C}_{14}\text{H}_{10}$, $\mathcal{M} = 178.23 \text{ g/mol}$) are dissolved? ($K_f(\text{H}_2\text{O}) = 1.86^\circ\text{C kg/mol}$)

ANSWER: _____

654. Emily conducted some measurements and determined that 0.0220 moles of nitrogen gas participated in a chemical reaction. Calculate the mass of nitrogen gas that participated in the chemical reaction.

ANSWER: _____

655. Nathan is a chemical engineer who needs to convert an oven temperature for a thermodynamic calculation. The temperature is 140°F . Convert this temperature to kelvin. Round to the nearest integer.

ANSWER: _____

656. Strontium sulfide reacts with ammonium phosphate to form strontium phosphate and ammonium sulfide. Calculate the moles of ammonium phosphate needed to produce 0.1427 moles of ammonium sulfide. Ensure your answer has the correct number of significant figures.

ANSWER: _____

657. Benjamin measures the speed of a moving object and records $24.7 \frac{\text{km}}{\text{h}}$. Convert this speed to $\frac{\text{m}}{\text{s}}$. Ensure your answer has the correct number of significant figures.

ANSWER: _____

658. Rank these primary amines from most to least soluble in water: CH_3NH_2 , $\text{C}_4\text{H}_9\text{NH}_2$, $\text{C}_6\text{H}_{13}\text{NH}_2$, $\text{C}_8\text{H}_{17}\text{NH}_2$. Explain the trend.

ANSWER: _____

659. Provide the number of protons and the number of neutrons there are in an atom of ^4He .

ANSWER: _____

660. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 69 fs. What is the time in seconds? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

661. To help farmers and gardeners, commercial fertilizers have a big three-number "NPK" label on the bag that gives the amounts of three key plant nutrients: nitrogen, phosphorus, and potassium. For example, a bag of "10-10-10" fertilizer has 10% by mass nitrogen, 10% by mass potassium as potash (K_2O) and 10% by mass phosphorus as phosphate (P_2O_5).

ingredient	percent by mass
calcium nitrate ($\text{Ca}(\text{NO}_3)_2$)	26.0%
urea ($(\text{NH}_2)_2\text{CO}$)	23.0%
potassium sulfate (K_2SO_4)	15.0%
ammonium phosphate ($(\text{NH}_4)_3\text{PO}_4$)	25.0%
inert ingredients	11.0%

What should the second (potassium) number on the bag's label be? In other words, calculate the percent by mass of potassium in this fertilizer. Round your answer to the nearest tenth of a percent.

ANSWER: _____

662. Titanium is extracted from its ore, rutile (titanium dioxide), via the two-step Kroll process. In the first step, titanium dioxide is converted to titanium tetrachloride by reacting it with chlorine and carbon. In the second step, titanium tetrachloride is reduced using molten magnesium to form titanium metal and magnesium chloride (which is then often refined back to magnesium). Write the net chemical equation for the production of titanium from titanium dioxide, chlorine, carbon, and magnesium. Ensure that it is balanced.

ANSWER: _____

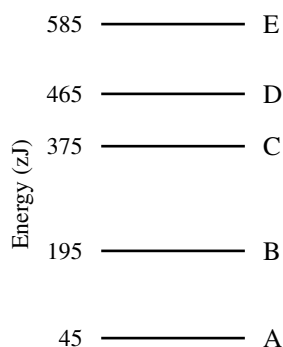
663. Calculate the osmotic pressure of seawater at 25°C (modeled as a 0.600M electrolyte solution with $i = 2$). What minimum pressure must be applied to achieve reverse osmosis?

ANSWER: _____

664. Classify the color of copper as a physical property or a chemical property. Explain your answer.

ANSWER: _____

665. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the highest frequency? You must explain your answer choice for full credit.



ANSWER: _____

666. A 975-mL sample of hydrogen bromide (HBr) is initially at 307 K. Its boiling point is -67.°C (206.15 K). The pressure is then reduced to 57% of its initial value. Determine whether the sample can remain entirely gaseous while its volume is held constant throughout the process. If it can, calculate the final temperature. Report the temperature rounded to the nearest °C. Justify your answer.

ANSWER: _____

667. Allison is preparing a deep-sea submersible. A rigid spherical air tank has an internal diameter of 82.0-cm. After filling, the pressure inside the tank is 193 atm and the temperature is 315 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

668. Will I_2 in CCl_4 dissolve appreciably? Offer brief reasoning for your answer. Answers without valid explanation earn no credit.

ANSWER: _____

669. A sample of silver has the properties listed below. Determine its density and justify your answer.

packing efficiency = 74%

molar mass = $107.868 \text{ g mol}^{-1}$

atomic radius = 144 pm

atoms per unit cell = 4

unit-cell edge length = 408.53 pm

ANSWER: _____

670. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 32 fs. What is the time in seconds? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

671. A civil engineer measures the length of an airport runway. The length is 2258 m. What is the length in kilometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

672. Calculate the osmotic pressure of a 0.0500 M aqueous solution of urea (CH_4N_2O) at 25°C .

ANSWER: _____

673. Anna conducted some measurements and determined that 0.12300 g of radium participated in a chemical reaction. Calculate the number of moles of radium that participated in the chemical reaction.

ANSWER: _____

674. Which substance has the higher normal boiling point: 1-propanol ($CH_3CH_2CH_2OH$) or propanal (CH_3CH_2CHO)? Justify your answer. Unsupported trend-recitation or memorized rules receive no credit.

ANSWER: _____

675. Which element is 82% (by mass) of ammonium bromide (NH_4Br)? Show your work.

ANSWER: _____

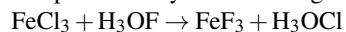
676. State the SI base unit used to measure length and give its symbol.

ANSWER: _____

677. A molecule's central atom is surrounded by 4 bonding groups and 2 lone pairs of electrons. Determine the electron-group arrangement, the molecular shape, and the ideal bond angle(s).

ANSWER: _____

678. A way to produce solid iron (III) fluoride is by combining aqueous iron (III) chloride with aqueous hydronium fluoride. The chemical reaction is represented by the following unbalanced chemical equation:



Knowing this, David pours 12.4 mL of a 0.0373-M solution of aqueous iron (III) chloride and then adds 15.5 mL of a 0.0183-M solution of aqueous hydronium fluoride. He then stirs the mixture and notices iron (III) fluoride precipitating out of the solution. What is the mass of iron (III) fluoride that he should expect the reaction to yield?

ANSWER: _____

679. Classify the following as a pure substance or a mixture: a crystal of pure table sugar (sucrose). You must explain your answer and include definitions.

ANSWER: _____

680. An unknown gas has a mass of 19.3 g and occupies 5.0 L at 124 kPa and 5.8°C. Determine the molar mass of the gas. Justify your answer.

ANSWER: _____

681. A nuclear chemist measures the mass of an artificial element. The mass is 48 pg. What is the mass in grams? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

682. For the central atom in NH_3 , determine its hybridization, electron-group geometry, molecular geometry, and the ideal bond angle(s). Justify your answer.

ANSWER: _____

683. Lauren has a 0.0773-M solution of caesium phosphate (Cs_3PO_4). If she needs to add 0.5568 g of caesium phosphate to a reaction flask, what volume of the solution should she pour? Be sure your answer has the correct number of significant figures.

ANSWER: _____

684. As HClO_2 is converted into $\text{Fe}(\text{ClO}_2)_3$ in the following reaction, determine whether chlorine is oxidized, reduced, or neither. Clearly explain why you chose your answer. Consider only the chlorine found in HClO_2 and $\text{Fe}(\text{ClO}_2)_3$.
- $$\text{Fe}_2\text{O}_3 + \text{HClO}_2 \rightarrow \text{H}_2\text{O} + \text{Fe}(\text{ClO}_2)_3$$

ANSWER: _____

685. Br_2 is sealed inside a rigid reaction vessel having a measured volume of 7.29 L. After the gas reaches 5.49°C , the pressure inside is measured to be 0.383 atm. Determine the amount of gas present. Justify your answer. Report the amount in SI units.

ANSWER: _____

686. Consider the reaction $\text{CO}(\text{g}) + 2\text{H}_2(\text{g}) \longrightarrow \text{CH}_3\text{OH}(\text{g})$. If 11.5 cm^3 of H_2 gas is completely consumed at the same temperature and pressure, what volume of CH_3OH gas is produced? Justify your answer.

ANSWER: _____

687. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is $5.73 \times 10^{-14}\text{ m}$. What is the wavelength in femtometers? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

688. A particular planet has a mass of $5.4 \times 10^{27}\text{ g}$ and an oil tanker has a mass of $7.62 \times 10^7\text{ kg}$. What is the mass of 1 mole of this particular planet? Ensure your answer has the correct number of significant figures.

ANSWER: _____

689. The reaction $2\text{N}_2\text{O}_5(\text{g}) \longrightarrow 4\text{NO}_2(\text{g}) + \text{O}_2(\text{g})$ occurs at constant temperature and pressure. If 6.9cm^3 of N_2O_5 gas reacts completely, what volume of NO_2 gas is consumed or produced? Report the numerical answer with units and justify your answer.

ANSWER: _____

690. How many grams of NH_3 can vaporize when 23.2-kJ of heat is absorbed during vaporization at -33°C ? Assume $\Delta H = 1370\text{J/g}$.

ANSWER: _____

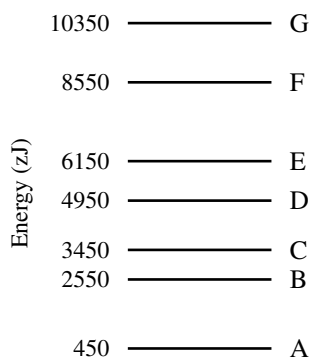
691. Measurements show that an unknown compound has the following composition:

Element	Mass %
Si	30.5 %
S	69.5 %

Write the empirical chemical formula of the unknown compound. You must show calculations to support your why you chose your answer.

ANSWER: _____

692. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the shortest wavelength? You must explain your answer choice for full credit.



ANSWER: _____

693. The chemical formula for ditellurium monocarbide is Te_2C . Calculate the molar mass of ditellurium monocarbide. Round your answer to 4 significant figures.

ANSWER: _____

694. A solid shows identical properties in every direction (isotropic) and has no sharp melting point. Is it more likely crystalline or amorphous? Justify.

ANSWER: _____

695. To help farmers and gardeners, commercial fertilizers have a big three-number "NPK" label on the bag that gives the amounts of three key plant nutrients: nitrogen, phosphorus, and potassium. For example, a bag of "10-10-10" fertilizer has 10% by mass nitrogen, 10% by mass potassium as potash (K_2O) and 10% by mass phosphorus as phosphate (P_2O_5).

ingredient	percent by mass
calcium nitrate ($\text{Ca}(\text{NO}_3)_2$)	23.0%
urea ($(\text{NH}_2)_2\text{CO}$)	20.0%
potassium sulfate (K_2SO_4)	16.0%
ammonium phosphate ($(\text{NH}_4)_3\text{PO}_4$)	3.0%
inert ingredients	38.0%

What should the second (potassium) number on the bag's label be? In other words, calculate the percent by mass of potassium in this fertilizer. Round your answer to the nearest tenth of a percent.

ANSWER: _____

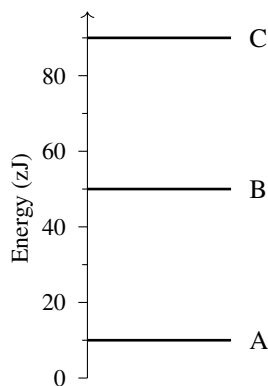
696. Several types of electromagnetic radiation (EMR) are: radio waves, ultraviolet light, violet light, and red light. Order these types of EMR from lowest frequency to highest frequency.

ANSWER: _____

697. Imagine an alternate universe in which the value of the Planck constant is $1.93 \times 10^{-32} \text{ J}\cdot\text{s}$. In that universe, is quantum mechanics necessary to accurately describe a $2.8 \times 10^{-11} \text{ m}$ -wide object that has a mass of $6.8 \times 10^{-11} \text{ kg}$ and is moving at $3.2 \times 10^{-5} \text{ m/s}$? Write "yes" if quantum mechanics is necessary and "no" if it is not, then explain why you made your choice. Answers lacking supporting explanation will earn no credit.

ANSWER: _____

698. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its second excited state? Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

699. A geophysicist measures the pressure in a borehole. The pressure is 6.531×10^7 Pa. What is the pressure in megapascals? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

700. As part of his duties as an analytical chemist, Daniel measures the amounts of elements E_1 and E_2 found in four samples collected from an unknown substance X. He then records these masses in a table in his lab notebook, reproduced below. It is known that X contains no elements other than E_1 and E_2 . Is X a pure substance or a mixture? If X is a pure substance, calculate the mass of element E_1 that Daniel would find in a new 10.0-g sample of X. Round your answer to two significant figures. Explain why you chose your answer.

sample	mass of E_1	mass of E_2
1	5.7 g	2.1 g
2	11.5 g	4.3 g
3	16.0 g	6.0 g
4	18.3 g	6.9 g

ANSWER: _____

701. Rank the following substances from highest to lowest vapor pressure at room temperature: N_2 , He, CO_2 , H_2O . For credit you must justify your answer.

ANSWER: _____

702. Rank the following noble gases from lowest to highest boiling point: He, Kr, Ar, Xe. All noble gases interact only via London dispersion forces.

ANSWER: _____

703. An unknown solid has the following properties: high melting point, brittle, conducts when molten, does not conduct as solid. Classify this solid as ionic, metallic, covalent-network, or molecular. For credit you must justify your answer choice.

ANSWER: _____

704. Ian must prepare 776 mL of a 1.65-mM working solution of aqueous ammonium oxide ((NH₄)₂O). He will do this by pouring some 0.07166-M aqueous ammonium oxide stock solution into a graduated cylinder and diluting it with distilled water. How much of the aqueous-ammonium oxide stock solution must Ian pour into the graduated cylinder? Report your answer in mL. Be sure your answer has the correct number of significant figures.

ANSWER: _____

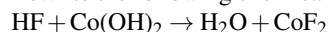
705. The N atom in NH₄⁺ has 4 regions of electron density. How many unhybridized *p* orbitals remain on that atom? Justify your answer.

ANSWER: _____

706. What is the packing efficiency (percent of space occupied by atoms) of a simple cubic (SC) unit cell?

ANSWER: _____

707. Rewrite the following chemical equation as a balanced equation using the smallest possible whole-number stoichiometric coefficients.



ANSWER: _____

708. Rank the following noble gases from highest to lowest boiling point: He, Xe, Ne, Kr. All noble gases interact only via London dispersion forces.

ANSWER: _____

709. Measurements show that the enthalpy of a mixture of gaseous reactants has decreased by 248 kJ during a certain chemical reaction, which was carried out at a constant pressure. Evidenced by the volume change, -298 kJ of work must have been done on the mixture during the reaction. Calculate the change in energy of the gas mixture during the reaction and report whether the reaction is endothermic, exothermic, or neither. You must explain your answer choice for full credit.

ANSWER: _____

710. The reaction $2\text{NaHCO}_3(\text{s}) \longrightarrow \text{Na}_2\text{CO}_3(\text{s}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{g})$ is carried out in a sealed, initially evacuated vessel with a volume of 12.5 L. After 6.74 g of NaHCO_3 completely reacts while all other reactants are present in excess at 9.72°C , determine the pressure of CO_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

711. In an ionic crystal a cation (Ca^{2+} , radius 100 pm) sits among anions (O^{2-} , radius 140 pm). Predict which type of hole the cation occupies and its coordination number. Justify with the radius ratio.

ANSWER: _____

712. An atom is sp^3d^2 hybridized. Which atomic orbitals remain unhybridized? Justify your answer.

ANSWER: _____

713. Sean needs to know the mass of a sample of H_3OI to 4 significant figures. He puts the sample on a digital scale. This is what the scale shows: 83.188 g. If this measurement is precise enough for Sean, round it to 4 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: _____

714. Does the central atom in OF_2 obey the octet rule? If not, classify the exception as electron-deficient or expanded octet, and state the number of electrons around the central atom.

ANSWER: _____

715. The extraction of aluminum metal from the aluminum hydroxide in bauxite ore by the Hall-Héroult process is one of the most remarkable success stories of late-19th-century chemistry, turning aluminum from a rare and precious metal into the cheap commodity it is today. In the first step, aluminum hydroxide reacts to form alumina (which is aluminum oxide) and water. In the second step, alumina and carbon react to form aluminum and carbon dioxide. Write the net chemical equation for the production of aluminum from aluminum hydroxide and carbon. Ensure that it is balanced.

ANSWER: _____

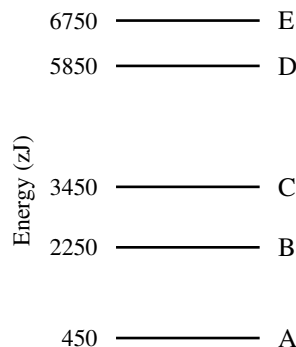
716. Allison needs to determine the mass of a sample of motor oil. The density of the sample is $7.4 \frac{\text{g}}{\text{mL}}$ and its volume is 8.87 mL. What is the mass of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

717. CO_2 at 40°C (above its critical temperature of 31°C) is compressed isothermally from 1 atm to 100 atm. Name the phase transition(s) that occur, in order, and justify your answer using the phase diagram.

ANSWER: _____

718. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the highest frequency? You must explain your answer choice for full credit.



ANSWER: _____

719. A particular planet has a mass of 7.857×10^{27} g and an oil tanker has a mass of 1.8×10^6 kg. What is the mass of 1 mole of this particular planet? Ensure your answer has the correct number of significant figures.

ANSWER: _____

720. A molecule's central atom is surrounded by 5 bonding groups and 0 lone pairs of electrons. Determine the electron-group arrangement, the molecular shape, and the ideal bond angle(s).

ANSWER: _____

721. A bonding interaction has a nodal plane that contains the internuclear axis. Determine whether it is a σ bond or a π bond. Justify your answer.

ANSWER: _____

722. A flexible gas bag contains 2.12 L of argon at -19.3°C while indoors. It is then taken outside, where the air temperature is 22.2°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

ANSWER: _____

723. Boric acid reacts with manganese (II) hydroxide to form water and manganese (II) borate. If 4.524 g of water is produced from the reaction of 7.320 g of boric acid and 24.7 g of manganese (II) hydroxide, calculate the percent yield of water. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

724. A 25.8-g sample of ice at 0°C absorbs 14.2 kJ of heat. Determine the final state and temperature. ($\Delta H_{\text{fus}} = 334 \text{ J/g}$, $c_{\text{water}} = 4.18 \text{ J g}^{-1}\text{°C}^{-1}$.)

ANSWER: _____

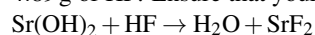
725. Classify sodium chloride (NaCl) as a molecular (covalent) solid, a covalent network solid, an ionic solid, or a metallic solid, and predict whether it has a relatively high or low melting point. Explain.

ANSWER: _____

726. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

ANSWER: _____

727. Given this unbalanced chemical equation, report the maximum mass of SrF_2 that could be produced by the mixing of 8.59 g of Sr(OH)_2 with 4.89 g of HF. Ensure that your answer has the correct number of significant figures.



ANSWER: _____

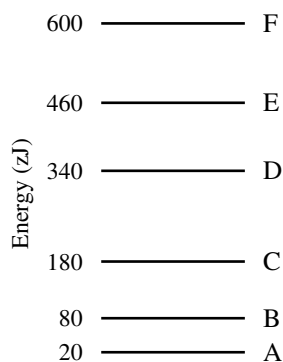
728. Which liquid is more viscous: ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) or water (H_2O)? Justify your answer.

ANSWER: _____

729. A gas mixture has a total pressure of 1.07 atm. The mole fraction of N_2 is 0.350. Determine the partial pressure of the gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

730. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the absorption line at the longest wavelength? You must explain your answer choice for full credit.



ANSWER: _____

731. Two isomers share the molecular formula C_5H_{12} : neopentane (compact/spherical) and n-pentane (straight-chain (elongated)). Which has the higher boiling point? Justify your answer in terms of intermolecular forces.

ANSWER: _____

732. It takes 158 kJ/mol to break a bond between two particular elements. Calculate the maximum wavelength of light that will break one of these bonds.

ANSWER: _____

733. Rank the following substances from highest to lowest vapor pressure at room temperature: H_2O , N_2 , He, CO_2 . For credit you must justify your answer.

ANSWER: _____

734. Calculate the wavelength of the line in the emission line spectrum of hydrogen caused by the transition of the electron from an orbital with $n = 12$ to an orbital with $n = 3$. Round your answer to 3 significant figures.

ANSWER: _____

735. Consider the following compounds: hydrosulfuric acid, carbon monoxide, oxygen difluoride, and nitrogen monoxide. Write the molecular formula for each of the listed compounds that has a dipole moment. If none have a dipole moment, write "None have a dipole moment". Draw each compound and explain your answer.

ANSWER: _____

736. A chemist notes that a reaction is carried out at 50 °F. Convert this temperature to degrees Celsius. Round your answer to the nearest integer.

ANSWER: _____

737. A lab technician measures the weight of a reagent. The weight is 3007 g. What is the weight in kilograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

738. Calculate the heat required to warm 37.8-g of ice from -12°C to 0°C . $c_{\text{ice}} = 2.09 \text{ J g}^{-1}\text{C}^{-1}$.

ANSWER: _____

739. A millworker measures the width of a plank. The width is 2.0 dm. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

740. Determine the electron-group arrangement, molecular shape, and ideal bond angle(s) for hydrosulfuric acid.

ANSWER: _____

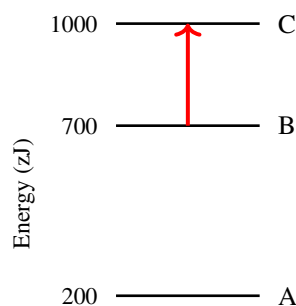
741. The vapor pressure of ethanol ($\text{C}_2\text{H}_5\text{OH}$) is 0.2064 atm at 60.00°C and 1.000 atm at 78.40°C . Calculate the molar enthalpy of vaporization.

ANSWER: _____

742. What is the packing efficiency (percent of space occupied by atoms) of a simple cubic (SC) unit cell?

ANSWER: _____

743. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from B to C, as shown by the arrow. Calculate the energy of the photon that would be absorbed or emitted. Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

744. At 25°C and 1.00 atm, the solubility of O_2 (O_2) in water is $1.3 \times 10^{-3} \text{ mol/L}$. What partial pressure of O_2 is required to achieve a solubility of $5.2 \times 10^{-3} \text{ mol/L}$? Use Henry's law: $P = C/K_{\text{H}}$.

ANSWER: _____

745. A diatomic molecule has a measured dipole moment of 2.45 D and a bond length of 102 pm. Calculate the percent ionic character of the bond. Report your answer with two digits after the decimal.

ANSWER: _____

746. Which substance has the higher normal boiling point: water (H_2O) or decane ($\text{C}_{10}\text{H}_{22}$)? Justify your answer. Unsupported trend-recitation or memorized rules receive no credit.

ANSWER: _____

747. What is the coordination number of each atom in a body-centered cubic (BCC) structure?

ANSWER: _____

748. Fill in the information missing from the following table. For each subshell, you must provide the principal quantum number n , the angular momentum quantum number l , and the maximum number of electrons that can fit in the subshell N_e . Credit is awarded only if the table is completed correctly. You may fill in the table directly; there is no need to redraw it.

Subshell	n	l	N_e
4s			
3s			
1s			

ANSWER: _____

749. The vapor pressure of benzene (C_6H_6) is 0.2421 atm at 40.00°C and 1.000 atm at 80.10°C . Calculate the molar enthalpy of vaporization.

ANSWER: _____

750. For the P atom in PH_3 , determine its hybridization, molecular geometry about that atom, and ideal bond angle. Justify your answer.

ANSWER: _____

751. Draw the Lewis electron-dot symbol for an atom of Fr.

ANSWER: _____

752. Samantha is comparing two unknown liquid samples. Sample A has a mass of 53 g and a volume of 42.7 mL. Sample B has a mass of 78.9 g and a volume of 9.8 mL. Which sample has the greater density? You must include correct work that supports your conclusion.

ANSWER: _____

753. As part of her routine quality-control checks, Ashley, an analytical chemist at Southern Union Superb Chemicals Corporation, must determine the concentration of a 2000-L fresh batch of hydroiodic acid solution. She takes a 50.00-mL sample of hydroiodic acid solution and titrates it with a sodium hydroxide solution, which she made by dissolving 4.112 g of solid sodium hydroxide in 1.10 L of distilled water. If Ashley finds that it takes 188 mL of the sodium hydroxide solution for the titration to reach the equivalence point, what molarity of the hydroiodic acid solution does Ashley calculate?

ANSWER: _____

754. Rank the following liquids from highest to lowest viscosity at 20°C: castor oil, HOCH₂CH₂OH, CH₃COCH₃, H₂O.

ANSWER: _____

755. Determine which bond is expected to be weaker: the N≡N bond in N₂ or the N–N bond in N₂H₄. Justify your answer from fundamental principles. Responses based only on memorized trends will not receive credit.

ANSWER: _____

756. Write the complete MO electron configuration for Li₂⁺. Determine its bond order and magnetic character. Justify each answer.

ANSWER: _____

757. A mixture contains particles with diameters of 700 nm. Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

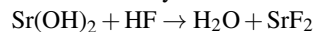
758. A sample of 27.7 g of aluminum reacts completely with excess hydrochloric acid to produce aluminum chloride and hydrogen gas in a sealed container. The container initially contains 0.328 mol of neon gas. The inert gas does not react, and no gas escapes. Determine the mole fraction of neon after the reaction is complete.

ANSWER: _____

759. A beam of light is passed through clear sugar water. Will the Tyndall effect be observed? Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

760. Given this unbalanced chemical equation, report the amount of Sr(OH)₂ needed to produce 0.6298 moles of H₂O. Report your answer in moles. Ensure your answer has the correct number of significant figures.

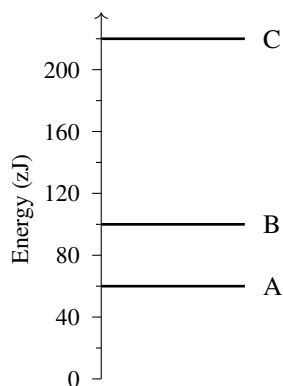


ANSWER: _____

761. Classify density as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: _____

762. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its first excited state? Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

763. A shipwright measures the length of a bolt. The length is 52.8 cm. What is the length in meters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

764. A high-energy physicist measures the time between collisions of atoms in a fluid. The time is 51 fs. What is the time in seconds? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

765. Experimental measurements show that the bond angle(s) in XeF_2 are 180° . Explain why this observation led chemists to describe the bonding on the Xe atom using hybrid orbitals.

ANSWER: _____

766. Explain why a supercritical fluid can dissolve solutes like a liquid yet diffuse through a solid matrix like a gas.

ANSWER: _____

767. On a heating curve of temperature versus heat added, one segment is a flat plateau at the boiling point. What is happening to the substance there, and which equation gives the heat? Justify.

ANSWER: _____

768. Rank the following substances in order of highest to lowest boiling point: He, Kr, Ar, Ne. Justify your ranking. Responses based only on memorized trends will not receive credit.

ANSWER: _____

769. Oxygen gas is collected over water. The measured pressure is 771 torr, and the vapor pressure of water is 29 torr. Determine the partial pressure of the oxygen gas. Report your answer in SI units. Justify your answer.

ANSWER: _____

770. Michael prepares a solution of sodium hypochlorite (NaClO) by measuring out 17.53 mmol of sodium hypochlorite into a 100.-mL volumetric flask and filling the flask to the mark with water. Calculate the concentration of the solution in mol/L. Be sure your answer has the correct number of significant figures.

ANSWER: _____

771. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$0.79706\text{ g} + 8.0462\text{ g}$$

ANSWER: _____

772. Iron (III) sulfide reacts with hydronium chloride to form hydronium sulfide and iron (III) chloride. Calculate the moles of hydronium chloride needed to produce 0.518 moles of hydronium sulfide. Ensure your answer has the correct number of significant figures.

ANSWER: _____

773. Rank the following noble gases from lowest to highest boiling point: Ne, He, Ar, Kr. All noble gases interact only via London dispersion forces.

ANSWER: _____

774. The molar enthalpy of sublimation of naphthalene (C_{10}H_8) is 62.36 kJ/mol and its molar enthalpy of vaporization is 43.3 kJ/mol. Calculate the molar enthalpy of fusion.

ANSWER: _____

775. A central atom in ClF_3 has 3 bonds and 2 lone pairs. Predict its hybridization, electron-group geometry, and molecular geometry. Explain how the number of electron groups and lone pairs determines each answer.

ANSWER: _____

776. An engineer measures the capacity of an electricity-generating plant. The capacity is 819 MW. What is the capacity in watts? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

777. Tyler needs to make 427.5 mL of aqueous magnesium iodide (MgI_2) working solution. He plans to do this by adding distilled water to a 85.0-mL stock solution of 0.075-M aqueous magnesium iodide. What will be the concentration of his working solution? Report your answer in mM. Be sure your answer has the correct number of significant figures.

ANSWER: _____

778. Arrange the following substances from highest to lowest boiling point: I_2 , Br_2 , Cl_2 , F_2 . Justify your ranking. Responses based only on memorized trends will not receive credit.

ANSWER: _____

779. The extraction of aluminum metal from the aluminum hydroxide in bauxite ore by the Hall-Héroult process is one of the most remarkable success stories of late-19th-century chemistry, turning aluminum from a rare and precious metal into the cheap commodity it is today. In the first step, aluminum hydroxide reacts to form alumina (which is aluminum oxide) and water. In the second step, alumina and carbon react to form aluminum and carbon dioxide. Write the net chemical equation for the production of aluminum from aluminum hydroxide and carbon. Ensure that it is balanced.

ANSWER: _____

780. The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

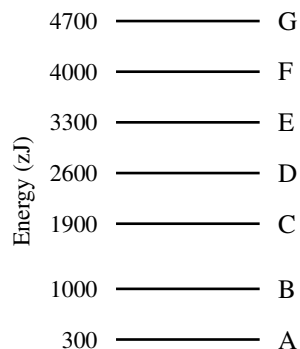
781. Write the chemical formula for selenic acid.

ANSWER: _____

782. For nitrogen (N_2): normal melting point -210°C , normal boiling point -196°C , triple point (-210°C , 0.124 atm), critical point (-147°C , 34 atm). What phase is present at -231°C and 0.0207 atm? Justify your answer.

ANSWER: _____

783. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. How many lines are in the emission line spectrum?



ANSWER: _____

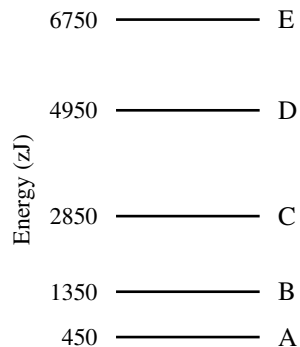
784. A botanist measures the thickness of a leaf. The thickness is 2.5 mm. What is the thickness in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

785. Of the following labels, identify all that can be applied to the chemical compound AsP: ionic, molecular, acid, and base. Clearly explain why you chose your label(s).

ANSWER: _____

786. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the highest frequency? You must explain your answer choice for full credit.



ANSWER: _____

787. An ionic solid has a face-centered cubic array of anions with cations in all of the octahedral holes. Name this structure type (with a common example) and justify.

ANSWER: _____

788. Which species is more strongly attracted to a magnetic field: H_2 or H_2^- ? Justify your conclusion by writing the molecular-orbital electron configuration of each species.

ANSWER: _____

789. Calculate the osmotic pressure of a 0.500M aqueous solution of NaCl (NaCl) at 25°C. Assume complete dissociation.

ANSWER: _____

790. Several types of electromagnetic radiation (EMR) are: yellow light, ultraviolet light, gamma rays, and radio waves. Order these types of EMR from shortest wavelength to longest wavelength.

ANSWER: _____

791. Rank the following ionic compounds from least negative to most negative (strongest) lattice energy: LiF, NaF, KF.

ANSWER: _____

792. Katherine is performing a combustion analysis on a compound with a molecular molar mass of 124.23 g/mol that is composed of carbon, hydrogen, and perhaps oxygen. A sample of 6.77 g of the compound is completely burned in excess oxygen and the mass of the products is carefully measured:

Product	Mass (g)
Carbon Dioxide	21.59 g
Water	7.85 g

Use this information to find the molecular formula of the compound.

ANSWER: _____

793. A botanist measures the thickness of a leaf. The thickness is 4.0 mm. What is the thickness in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

794. How much heat is absorbed when 33.1-g of liquid water vaporizes at 100°C? $\Delta H_{\text{vap}} = 2260 \text{ J/g}$.

ANSWER: _____

795. Blood plasma has a total dissolved-particle concentration of about 0.300 osmol/L. Classify each of the following as hypertonic, hypotonic, or isotonic with blood: 0.100 M MgCl_2 , 0.100 M Na_2SO_4 , 0.100 M CaCl_2 . Justify each classification.

ANSWER: _____

796. Write the name of the inorganic acid H_2CrO_4 .

ANSWER: _____

797. A cooking pot has a mass of $8.20 \times 10^0 \text{ kg}$ and a particular planet has a mass of $6.643 \times 10^{26} \text{ g}$. How many moles of cooking pots have a mass equal to the mass of this particular planet? Ensure your answer has the correct number of significant figures.

ANSWER: _____

798. Sulfuric acid is one of the most widely used chemicals in the world, particularly in the production of fertilizers. It is synthesized in a two-step contact process. In the first step, sulfur (S_8) reacts with oxygen to form sulfur dioxide. In the second step, sulfur dioxide, oxygen gas, and water combine to form sulfuric acid. Write the net chemical equation for the production of sulfuric acid from sulfur, oxygen, and water. Ensure that it is balanced.

ANSWER: _____

799. An atom of copper has a mass of $6.7 \times 10^{-23} \text{ g}$ and a brick has a mass of $9.795 \times 10^0 \text{ kg}$. What is the mass of 1 mole of bricks? Ensure your answer has the correct number of significant figures.

ANSWER: _____

800. Will O_2 in water dissolve appreciably? Offer brief reasoning for your answer. Answers without valid explanation earn no credit.

ANSWER: _____

801. Nicholas is instructed to dissolve 0.8697 g of sodium oxide in 233.2 mL of a 239-mM aqueous solution of chromium (III) chloride. Calculate the molarity of the sodium cation in the solution after the reaction has occurred. Assume the volume of the solution does not change when sodium is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

802. How many grams of H₂O can melt when 24.0-kJ of heat is absorbed during melting at 0°C? Assume $\Delta H = 334 \text{ J/g}$.

ANSWER: _____

803. Use the Born-Haber cycle to calculate ΔH_f° of KCl. $\Delta H_{\text{sub}}(\text{K}) = 89 \text{ kJ/mol}$, $IE_1(\text{K}) = 419 \text{ kJ/mol}$, $\frac{1}{2}D(\text{Cl}_2) = 121 \text{ kJ/mol}$, $EA(\text{Cl}) = -349 \text{ kJ/mol}$, $U(\text{KCl}) = -717 \text{ kJ/mol}$.

ANSWER: _____

804. Classify CO₂ as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: _____

805. Dissolving potassium chloride (KCl) is endothermic ($\Delta H_{\text{sol}} = +17.2 \text{ kJ/mol}$), yet it happens spontaneously. What makes this possible? Explain.

ANSWER: _____

806. A crystal produces an intense diffraction maximum for monochromatic X-rays under the conditions below. A crystal with plane spacing 995 pm produces the second-order diffraction maximum at a glancing angle of $\theta = 8.9^\circ$. Determine the wavelength of the incident X-rays. Calculate the requested quantity and justify your answer.

ANSWER: _____

807. Dissolving lithium chloride (LiCl) is exothermic ($\Delta H_{\text{sol}} = -37.0 \text{ kJ/mol}$). Identify the two factors that decide whether it dissolves spontaneously. Explain.

ANSWER: _____

808. Copper (II) permanganate reacts with ammonium iodide to form copper (II) iodide and ammonium permanganate. Write the balanced chemical equation for this reaction using the smallest possible whole-number stoichiometric coefficients.

ANSWER: _____

809. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$0.73744\text{ mL} + 48.\text{ mL}$$

ANSWER: _____

810. A geologist measures the temperature of a hot spring. The temperature of the spring is 40°C . Convert this temperature to kelvin. Report your answer with the correct number of significant figures.

ANSWER: _____

811. Which liquid is more viscous: glycerol ($\text{C}_3\text{H}_8\text{O}_3$) or ethanol ($\text{C}_2\text{H}_5\text{OH}$)? Justify your answer.

ANSWER: _____

812. Write the full electron configuration of a neutral beryllium atom. Then, using the octet rule, predict the charge of the monatomic ion that beryllium forms and identify the noble gas with which that ion is isoelectronic. Explain your reasoning.

ANSWER: _____

813. The partial pressure of O_2 in a gas mixture is 0.6072 atm . The mole fraction of O_2 is 0.180 . Determine the total pressure of the mixture. Report your answer in SI units. Justify your answer.

ANSWER: _____

814. For benzene (C_6H_6), $\Delta H_{\text{fus}} = 9.87\text{ kJ/mol}$ and $\Delta H_{\text{vap}} = 30.8\text{ kJ/mol}$. Explain why boiling absorbs so much more energy than melting.

ANSWER: _____

815. Provide the chemical formula for the molecular compound dichlorine heptoxide.

ANSWER: _____

816. A solution of heptane and octane is prepared. Based on the intermolecular forces present, predict whether the solution will show ideal behavior, a positive deviation, or a negative deviation from Raoult's law. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

817. Write the full electron configuration of a neutral phosphorus atom. Then, using the octet rule, predict the charge of the monatomic ion that phosphorus forms and identify the noble gas with which that ion is isoelectronic. Explain your reasoning.

ANSWER: _____

818. Jason has a 1.12-M solution of hydronium sulfate ($(\text{H}_3\text{O})_2\text{SO}_4$). If he pours 21.01 mL of hydronium sulfate into a reaction flask, what is the mass of hydronium sulfate that he has added to the flask? Be sure your answer has the correct number of significant figures.

ANSWER: _____

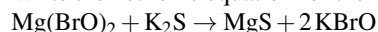
819. A nurse measures the weight of his patient's medicine. The weight is 0.0056 g. What is the weight in milligrams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

820. A monochromatic X-ray beam strikes a crystalline solid. Successive crystal planes produce a strong reflected beam under the conditions given below. Adjacent crystal planes are separated by 200 pm. Monochromatic X-rays of wavelength 154 pm produce the first-order diffraction maximum. Determine the glancing angle at which the maximum occurs. Calculate the requested quantity and justify your answer.

ANSWER: _____

821. Write the net ionic equation for the following chemical reaction, which takes place in aqueous solution:



ANSWER: _____

822. Samantha conducted some measurements and determined that 716. g of chlorine gas participated in a chemical reaction. Calculate the number of moles of chlorine gas that participated in the chemical reaction.

ANSWER: _____

823. Why do O₂, F₂, and Ne₂ NOT show *s-p* mixing, while Li₂ through N₂ do?

ANSWER: _____

824. Hydrosulfuric acid reacts with bismuth hydroxide to form water and bismuth sulfide. Calculate the mass of hydrosulfuric acid needed to produce 5.449 g of bismuth sulfide. Ensure your answer has the correct number of significant figures.

ANSWER: _____

825. There are two steps in the extraction of copper metal from chalcocite, a copper ore. In the first step, copper (I) sulfide and oxygen gas react to form copper (I) oxide and sulfur dioxide. In the second step, copper (I) oxide and carbon react to form copper and carbon monoxide. Write the net chemical equation for the production of copper from copper (I) sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

826. A chemist carefully measures the amount of heat needed to raise the temperature of a 1.44-kg sample of a pure substance from 1.57°C to 19.2°C. The experiment shows that 1.56 kJ of heat is needed. What can the chemist report as the value of specific heat capacity of the substance?

ANSWER: _____

827. A fixed amount of gas is initially at 7.2 atm and 9.80°C. It is then brought to a new state in which its temperature is decreased to -53°C and its volume is 60% larger than its initial volume. Determine the final pressure. Report your answer in SI units and justify your answer.

ANSWER: _____

828. Measurements show that the enthalpy of a mixture of gaseous reactants has decreased by 163.3 kJ during a certain chemical reaction, which was carried out at a constant pressure. Evidenced by the volume change, -178.8 kJ of work must have been done on the mixture during the reaction. Calculate the change in energy of the gas mixture during the reaction and report whether the reaction is endothermic, exothermic, or neither. You must explain your answer choice for full credit.

ANSWER: _____

829. Write the chemical formula of chromium (III) bisulfite dihydrate.

ANSWER: _____

830. A geophysicist measures the pressure in a borehole. The pressure is 2.920×10^7 Pa. What is the pressure in megapascals? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

831. Each week, Deskens Chemicals, Inc., a manufacturer of fine chemicals, uses 646 kg of compound A and 667 kg of compound B in a reaction that produces only one product, compound C. If the average percent yield of C in the company's reactors is 71.56%, calculate the average mass of compound C that the company actually ships each week.

ANSWER: _____

832. What is the packing efficiency (percent of space occupied by atoms) of a simple cubic (SC) unit cell?

ANSWER: _____

833. A gas-filled syringe has an internal diameter of 5.2-cm. The plunger is pushed inward with a constant force of 43.6-N. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

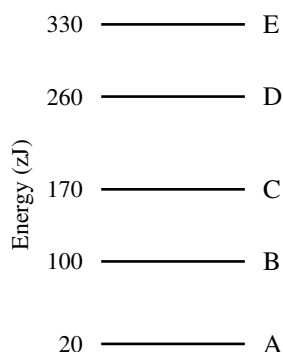
834. Rank the following species in order of increasing bond length: O_2 , O_2^+ , O_2^- , O_2^{2-} . Justify your ranking using MO theory.

ANSWER: _____

835. Mary conducted some measurements and determined that 0.420 moles of mercury participated in a chemical reaction. Calculate the mass of mercury that participated in the chemical reaction.

ANSWER: _____

836. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. How many lines are in the emission line spectrum?



ANSWER: _____

837. Tungsten crystallizes in a BCC unit cell with an atomic radius of 137 pm. Calculate the edge length of the unit cell in pm.

ANSWER: _____

838. Identify whether the following describes a random error or a systematic error, and explain your answer: Wind gusts cause repeated outdoor temperature measurements to vary randomly around the true value.

ANSWER: _____

839. On the potential-energy curve for I_2 , what happens to the potential energy when the two atoms are very far apart? Why does this happen? Justify your answer.

ANSWER: _____

840. The reaction $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g)$ occurs with all gases measured at the same temperature and pressure. If 11 mL of methane reacts completely, what volume of oxygen gas is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

841. A beam of light is passed through blood. Will the Tyndall effect be observed? Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

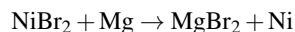
842. Identify the dominant type of intermolecular force present in pure N_2 . You must justify your answer.

ANSWER: _____

843. Write the chemical formula for perchloric acid.

ANSWER: _____

844. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.



ANSWER: _____

845. Suppose a popular FM-radio station broadcasts radio waves with a frequency of 98.2 MHz. Calculate the wavelength of these radio waves.

ANSWER: _____

846. Determine whether iron and bromine can form an ionic compound. If they can, provide the empirical formula of an ionic compound they form. Explain why you chose your answer.

ANSWER: _____

847. The vapor pressure of a volatile solvent is measured at 55°C. The sample is then heated to 66°C in a sealed container while liquid remains present. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

848. How many atoms are contained within one face-centered cubic (FCC) unit cell?

ANSWER: _____

849. A laboratory apparatus contains gas beneath a circular movable piston of diameter 6.62-cm. The gas column is 11.0-cm high before any compression. A constant downward force of 55.2-N is applied to the piston. Determine the pressure exerted on the gas. Report your answer in SI units with the correct number of significant figures and justify your answer.

ANSWER: _____

850. Classify the ability to rust of iron as a physical property or a chemical property. Explain your answer.

ANSWER: _____

851. A solution of H_2 (H_2) in water at 25°C is prepared at a partial pressure of 1.0 atm. If the partial pressure is increased to 3.0 atm, by what factor does the solubility change? Use Henry's law.

ANSWER: _____

852. An unknown solid has a melting point of 114°C , is a non-conductor as a solid, and is a non-conductor. Classify it as ionic, metallic, covalent-network, or molecular.

ANSWER: _____

853. Which compound has a lower lattice energy, K_2O or Na_2O ? Explain your choice.

ANSWER: _____

854. A gas sample is trapped inside a syringe with its tip sealed. Its initial volume is 6.5 L. After the piston is moved, the gas occupies 1.23 L and its pressure is measured to be 330 kPa. The temperature remains constant throughout. Determine the initial pressure. Justify your answer.

ANSWER: _____

855. Rank the following ionic compounds in order of increasing lattice energy (smallest to largest). Briefly justify your ranking using Coulomb's law.

NaCl , CaO , KCl

ANSWER: _____

856. A solution of H_2 (H_2) in water at 25°C is prepared at a partial pressure of 2.0 atm. If the partial pressure is increased to 8.0 atm, by what factor does the solubility change? Use Henry's law.

ANSWER: _____

857. Predict the products of the reaction below. In other words, write the products to complete the right-hand side of the chemical equation. Ensure the final equation is balanced.

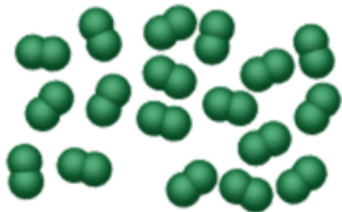
$\text{HCl} + \text{Pb}(\text{OH})_2 \rightarrow$

ANSWER: _____

858. In an ionic crystal a cation (Be^{2+} , radius 45 pm) sits among anions (O^{2-} , radius 140 pm). Predict which type of hole the cation occupies and its coordination number. Justify with the radius ratio.

ANSWER: _____

859. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? You must justify your answer choice.



ANSWER: _____

860. Magnesium hydroxide reacts with arsenous acid to form water and magnesium arsenite. Calculate the mass of magnesium hydroxide needed to produce 7.554 g of magnesium arsenite. Ensure your answer has the correct number of significant figures.

ANSWER: _____

861. A solution is prepared by dissolving 25.0 g of urea ($\text{CH}_4\text{N}_2\text{O}$, $M = 60.06 \text{ g/mol}$) in 100. g of water. Calculate the molality of the solution.

ANSWER: _____

862. An oil tanker has a mass of $5.3 \times 10^8 \text{ kg}$ and an atom of copper has a mass of $9.1 \times 10^{-22} \text{ g}$. What is the mass of 1 mole of copper atoms? Ensure your answer has the correct number of significant figures.

ANSWER: _____

863. Several types of electromagnetic radiation (EMR) are: blue light, infrared light, violet light, and yellow light. Order these types of EMR from shortest wavelength to longest wavelength.

ANSWER: _____

864. Provide the element symbol that represents sulfur.

ANSWER: _____

865. Andrew has a 4.95-M solution of hydronium phosphate ($(\text{H}_3\text{O})_3\text{PO}_4$). If he pours 5.55 mL of hydronium phosphate into a reaction flask, what is the mass of hydronium phosphate that he has added to the flask? Be sure your answer has the correct number of significant figures.

ANSWER: _____

866. How many grams of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) must be dissolved in 1.00×10^3 g of water to raise the boiling point by 0.768°C ? $K_b(\text{H}_2\text{O}) = 0.512^\circ\text{C kg/mol}$.

ANSWER: _____

867. Calculate the wavelength of the line in the emission line spectrum of hydrogen caused by the transition of the electron from an orbital with $n = 6$ to an orbital with $n = 4$. Round your answer to 3 significant figures.

ANSWER: _____

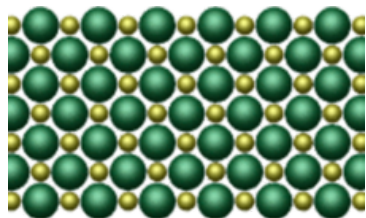
868. An optician measures the diameter of her patient's pupil. The diameter is 0.0052 m. What is the diameter in millimeters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

869. Write the chemical formula for acetic acid.

ANSWER: _____

870. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? You must justify your answer choice.



ANSWER: _____

871. Classify the following as a pure substance or a mixture: a crystal of pure table sugar (sucrose). You must explain your answer and include definitions.

ANSWER: _____

872. Zoe measures the amount of diarsenic tritelluride produced during a chemical reaction. She finds that 692. g is produced. Calculate the number of moles of diarsenic tritelluride produced. Ensure your answer has the correct number of significant figures.

ANSWER: _____

873. An atom forms both an sp^2 hybrid orbital and an sp^3 hybrid orbital. Without calculating any numerical energies, determine which hybrid orbital has the higher expectation value of energy and justify your answer.

ANSWER: _____

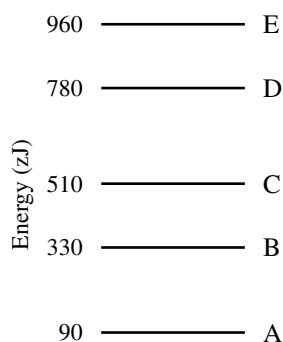
874. A liquid has a vapor pressure of 97 kPa at 28°C. The external pressure is 1.00 atm. Determine whether the liquid will boil. Report the vapor pressure and justify your answer.

ANSWER: _____

875. Nitric acid is often manufactured in a two-step process from the atmospheric gases nitrogen and oxygen plus hydrogen, prepared by reforming of natural gas. In the first step, nitrogen and hydrogen react to form ammonia. In the second step, ammonia and oxygen react to form nitric acid and water. Write the net chemical equation for the production of nitric acid from nitrogen, hydrogen, and oxygen. Ensure that it is balanced.

ANSWER: _____

876. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. How many lines are in the absorption line spectrum?

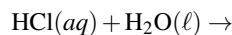


ANSWER: _____

877. Nick adds 169.4 mL of a 0.014-M solution of rubidium nitrite (RbNO_2) to a reaction flask. Calculate the amount of rubidium nitrite that he has added to the flask. Report your answer in moles and be sure your answer has the correct number of significant figures.

ANSWER: _____

878. Predict the products of the reaction below by completing the right-hand side of the chemical equation. Provide the full balanced equation with state symbols.



ANSWER: _____

879. The reaction $\text{H}_2(\text{g}) + \text{Cl}_2(\text{g}) \longrightarrow 2\text{HCl}(\text{g})$ occurs with all gases measured at the same temperature and pressure. If 9.7 mL of hydrogen gas reacts completely, what volume of hydrogen chloride is consumed or produced? Report the answer in SI units and justify your answer.

ANSWER: _____

880. A liquid is boiling at 47°C in a chamber where the external pressure is 28.8 kPa. It occupies a volume of 3 L. Determine whether the liquid's vapor pressure at this temperature is less than, equal to, or greater than the external pressure. Estimate the vapor pressure in SI units and justify your answer.

ANSWER: _____

881. Why does heating a sealed liquid toward its critical point make the meniscus between liquid and vapor fade and vanish?

ANSWER: _____

882. Hydrofluoric acid reacts with magnesium hydroxide to form water and magnesium fluoride. Suppose 5.57 g of hydrofluoric acid is mixed with 13.3 g of magnesium hydroxide. Calculate the maximum mass of magnesium fluoride that could be produced by the chemical reaction. Ensure that your answer has the correct number of significant figures.

ANSWER: _____

883. Can two atoms be joined by a π bond without also sharing a σ bond? Explain by describing how σ and π bonds are formed. Explain your reasoning completely. Answers without a correct explanation will not receive full credit.

ANSWER: _____

884. Kevin is preparing a deep-sea submersible. A rigid spherical air tank has an internal diameter of 65.1-cm. After filling, the pressure inside the tank is 123 atm and the temperature is 304 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

885. Explain why adding an electrolyte such as alum to a colloid makes the particles coagulate and settle. A complete answer must reference molecular structure or charge; assertion alone earns no credit.

ANSWER: _____

886. How many atoms are contained within one face-centered cubic (FCC) unit cell?

ANSWER: _____

887. A mixture of dinitrogen monoxide and carbon dioxide gas is expanded from a volume of 62 L to a volume of 81 L while the pressure is held constant at 88 atm. Calculate the work done on the gas mixture. Report your answer in SI unit of work and to 3 significant figures.

ANSWER: _____

888. A rigid container is evacuated and then refilled to the same temperature and pressure with a different ideal gas whose molar mass is larger. Determine whether the gas density increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

889. Two different orientations are possible for the same pair of bonding orbitals. One orientation produces greater overlap than the other. Predict which orientation produces the stronger covalent bond. Justify your answer.

ANSWER: _____

890. Classify the following as a physical or chemical change: Sharpening a pencil. Explain your answer.

ANSWER: _____

891. Two isomers share the molecular formula C_4H_{10} : n-butane (straight-chain) and isobutane (branched (compact)). Which has the higher boiling point? Justify your answer in terms of intermolecular forces.

ANSWER: _____

892. Katherine measures the density of a metal sample in her lab. She finds that the sample has a mass of 42.83 g and a volume of 7.60 mL. What is the density of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

893. Write the chemical formula of copper (II) oxalate tetrahydrate.

ANSWER: _____

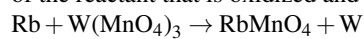
894. The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

895. An ionic compound crystallizes with a face-centered cubic array of anions and cations in half of the tetrahedral holes. What is the cation:anion ratio (the formula unit) in the compound? Justify by counting ions per cell.

ANSWER: _____

896. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reactant that is oxidized and the formula of the reactant that is reduced. Provide an explanation for why you made these choices.



ANSWER: _____

897. Classify the bond between lithium (Li) and fluorine (F) as nonpolar covalent, polar covalent, or ionic. If the bond is polar or ionic, identify which atom carries the partial negative charge (δ^-). Justify your answer using the electronegativity difference.

ANSWER: _____

898. A sample of iron has the properties listed below. Determine its density and justify your answer.

unit-cell edge length = 286.65 pm

crystal structure = BCC

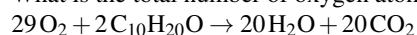
electron configuration = $[\text{Ar}]3d^64s^2$

molar mass = $55.845 \text{ g mol}^{-1}$

coordination number = 8

ANSWER: _____

899. What is the total number of oxygen atoms shown on the reactant side of this chemical equation?



ANSWER: _____

900. State the SI base unit used to measure amount of substance and give its symbol.

ANSWER: _____

901. An unknown gas has a mass of 18 g and occupies 3.15 L at 130 kPa and 5.2°C. Determine the molar mass of the gas. Justify your answer.

ANSWER: _____

902. A solution of cyclohexane and benzene is prepared. Based on the intermolecular forces present, predict whether the solution will show ideal behavior, a positive deviation, or a negative deviation from Raoult's law. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: _____

903. An unknown compound has a molar mass of 58 g/mol. A sample is vaporized at 96.°C. At a pressure of 0.940 atm, the gas occupies 720 mL. Determine the mass of the sample. Assume the gas behaves ideally under these conditions. Report your answer in SI units where applicable. Justify your answer.

ANSWER: _____

904. Solid iodine (I_2) has its triple point at 114°C and 0.12 atm. It is held at a constant external pressure of 0.0225 atm and slowly warmed. Does the solid melt or sublime? Justify your answer.

ANSWER: _____

905. Ethan is instructed to dissolve 2.755 g of barium iodide in 140.7 mL of a 57.78-mM aqueous solution of hydronium fluoride. Calculate the molarity of the hydronium cation in the solution after the reaction has occurred. Assume the volume of the solution does not change when hydronium is dissolved in it. Ensure your answer has the correct number of significant figures.

ANSWER: _____

906. Two Period-2 atoms each contribute one 2s orbital and three 2p orbitals to molecular-orbital formation. How many molecular orbitals are produced? Justify your answer.

ANSWER: _____

907. Calculate the mole fractions of acetone ($\text{C}_3\text{H}_6\text{O}$) and water in a solution prepared by mixing 9.21 g of acetone with 50.0 g of water.

ANSWER: _____

908. Classify mayonnaise as a type of colloid: identify its dispersed phase and dispersion medium, then name the colloid type.

ANSWER: _____

909. Identify the hybridization of each carbon atom in $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$ (acetone). Explain your answer.

ANSWER: _____

910. A nurse measures the weight of his patient's medicine. The weight is 0.0086 g. What is the weight in milligrams? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

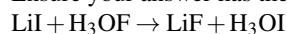
911. Iron is extracted from its oxide ores in a two-step process that has been practiced for centuries. In the first step, iron (III) oxide reacts with carbon monoxide to produce iron (II) oxide and carbon dioxide. In the second step, iron (II) oxide further reacts with carbon monoxide to produce elemental iron and more carbon dioxide. Write the net chemical equation for the production of iron from iron (III) oxide and carbon monoxide. Ensure that it is balanced.

ANSWER: _____

912. A solution of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is 25.0% by mass. Calculate the molality of the solution. ($\mathcal{M}(\text{glucose}) = 180.16 \text{ g/mol}$)

ANSWER: _____

913. Given this unbalanced chemical equation, report the amount of LiI needed to produce 0.2913 moles of H_3OI . Report your answer in moles. Ensure your answer has the correct number of significant figures.



ANSWER: _____

914. The chemical formula for copper (II) chlorite is $\text{Cu}(\text{ClO}_2)_2$. Calculate the molar mass of copper (II) chlorite. Round your answer to 5 significant figures.

ANSWER: _____

915. Calculate the number of oxygen atoms in a 3.4-g sample of potassium phosphate (K_3PO_4).

ANSWER: _____

916. The extraction of zinc from its sulfide ores involves a two-step roasting and reduction process. In the first step, zinc sulfide reacts with oxygen to form zinc oxide and sulfur dioxide. In the second step, zinc oxide is reduced by carbon to form zinc metal and carbon monoxide. Write the net chemical equation for the production of zinc from zinc sulfide, oxygen, and carbon. Ensure that it is balanced.

ANSWER: _____

917. The reaction $2\text{HgO} \longrightarrow 2\text{Hg} + \text{O}_2$ occurs with all gaseous substances at the same temperature and pressure. A reaction proceeds until 13 mL of O_2 has been consumed. What volume of Hg corresponds to this amount? Report the answer using the same volume unit. Justify your answer.

ANSWER: _____

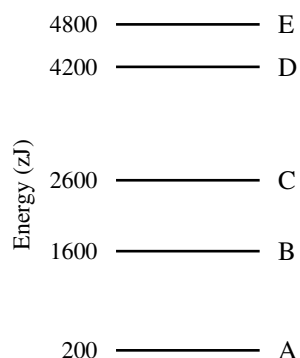
918. Two molecular orbitals are formed from the same pair of atomic orbitals. One has less electron density between the nuclei than the other. Which molecular orbital has the higher energy? Justify your answer.

ANSWER: _____

919. A solution contains more than the equilibrium maximum at a given temperature. Classify the solution as saturated, unsaturated, or supersaturated. You must explain your answer choice for credit.

ANSWER: _____

920. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the longest wavelength? You must explain your answer choice for full credit.



ANSWER: _____

921. Patricia measures the density of a liquid reagent in her lab. She finds that the sample has a mass of 130g and a volume of 16.6mL. What is the density of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

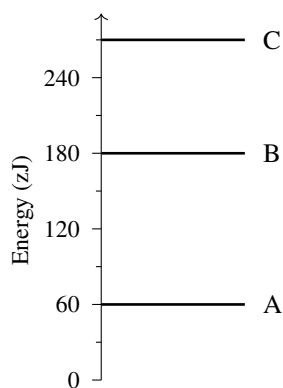
922. A 14-L tank at 120°C and 1.5 atm contains 0.45 mol of sulfur trioxide (SO_3) and 0.27 mol of sulfur hexafluoride (SF_6). Determine the mole fraction of sulfur trioxide in the mixture. Justify your answer.

ANSWER: _____

923. Will benzene in water dissolve appreciably? Offer brief reasoning for your answer. Answers without valid explanation earn no credit.

ANSWER: _____

924. This energy diagram shows the allowed energy levels of an electron in a certain atom. What is the energy of the electron in its second excited state? Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

925. Does the central atom in NF_3 obey the octet rule? If not, classify the exception as electron-deficient or expanded octet, and state the number of electrons around the central atom.

ANSWER: _____

926. Two solutions are prepared: solution A is a 0.200m solution of NaCl (NaCl) in water, and solution B is a 0.200m solution of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) in water. Which solution has the greater freezing point depression? Calculate both values. (Assume complete dissociation for ionic solutes.)

ANSWER: _____

927. A European researcher reports an experimental temperature for a U.S. publication. The temperature is 40°C . Convert this temperature to degrees Fahrenheit and round to the nearest integer.

ANSWER: _____

928. A solution is prepared by dissolving 30.0 g of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$, $M = 342.3 \text{ g/mol}$) in 100. g of water. Calculate the mole fraction of sucrose.

ANSWER: _____

929. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.315 m. What is the thickness in centimeters? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

930. Rewrite the following chemical equation as a balanced equation using the smallest possible whole-number stoichiometric coefficients.
 $\text{H}_2\text{CrO}_4 + \text{Mg}(\text{OH})_2 \rightarrow \text{H}_2\text{O} + \text{MgCrO}_4$

ANSWER: _____

931. The mass enthalpy of vaporization of water (H_2O) is 2260 J/g and $M = 18.015 \text{ g/mol}$. Find the molar enthalpy of vaporization of water.

ANSWER: _____

932. A 0.500 M solution of NaCl (NaCl) has a density of 1.02 g/mL. Calculate the molality of the solution. ($M = 58.44 \text{ g/mol}$)

ANSWER: _____

933. Classify the following as an element or a compound: a piece of aluminum foil made of pure aluminum. You must explain your answer using correct reasoning.

ANSWER: _____

934. Which species is predicted to be more stable by MO theory: C_2 or B_2 ? Justify using bond orders.

ANSWER: _____

935. Suppose that there is heat transfer of 60 J to a system, while the system does 59 J of work. Later, there is heat transfer of 47 J out of the system while 36 J of work is done on the system. What is the net change in internal energy of the system?

ANSWER: _____

936. Rank these alcohols from most to least soluble in water: CH_3OH , $\text{C}_2\text{H}_5\text{OH}$, $\text{C}_5\text{H}_{11}\text{OH}$, $\text{C}_6\text{H}_{13}\text{OH}$. Explain the trend.

ANSWER: _____

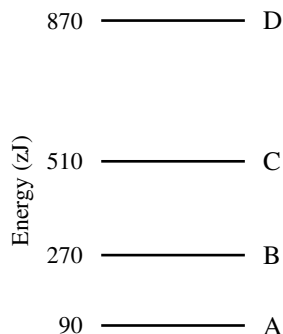
937. Calculate the mass percent of bromine in calcium bromide (CaBr_2). Round your answer to the nearest percentage.

ANSWER: _____

938. A cell biologist measures the width of a cell. The width is $70\ \mu\text{m}$. What is the width in meters? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

939. This energy diagram shows the allowed energy levels of an electron in a certain atom or molecule. Which transition causes the emission line at the lowest frequency? You must explain your answer choice for full credit.



ANSWER: _____

940. State the type(s) of bonding present in Na_2CO_3 . Explain your answer.

ANSWER: _____

941. The species CO_3^{2-} exhibits resonance. Draw all of its equivalent resonance structures. What is the bond order of each C–O bond in the resonance hybrid?

ANSWER: _____

942. Which species is predicted to be more stable by MO theory: N_2 or O_2 ? Justify using bond orders.

ANSWER: _____

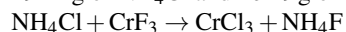
943. Provide the name and empirical formula of an ionic compound that is formed by the silver cation and the sulfide anion.

ANSWER: _____

944. Two substances have intermolecular potential-energy curves of identical shape but different well depths. Which forms the solid with the higher melting point? Justify.

ANSWER: _____

945. Given this unbalanced chemical equation, calculate and report the percent yield of NH_4F if 7.35 g of NH_4F is produced from the reaction of 19.11 g of NH_4Cl and 20.29 g of CrF_3 . Ensure that your answer has the correct number of significant figures.



ANSWER: _____

946. A solution is prepared by dissolving 1.00 g of hemoglobin in enough water to make 50.0 mL of solution. The osmotic pressure at 25°C is 7.60×10^{-3} atm. Calculate the molar mass of hemoglobin. Use $\Pi = MRT$, so $M = \Pi/(RT)$, then $\mathcal{M} = (m/V)/M$.

ANSWER: _____

947. A student draws benzene with alternating short and long carbon-carbon bonds and claims this represents the actual molecule. Is the claim correct? Justify your answer.

ANSWER: _____

948. Provide the number of protons and the number of neutrons there are in an atom of ^{27}Al .

ANSWER: _____

949. Rank the following bonds in order of increasing bond length: $\text{O}=\text{O}$, $\text{O}-\text{O}$. Support your ranking with a brief explanation for credit. Credit is not awarded for simply citing a memorized trend.

ANSWER: _____

950. A central atom in SiH_4 has 4 bonds and 0 lone pairs. Predict its hybridization, electron-group geometry, and molecular geometry. Explain how the number of electron groups and lone pairs determines each answer.

ANSWER: _____

951. What major chemical species are present in an aqueous solution of 0.0341-M silver fluoride?

ANSWER: _____

952. Classify sodium chloride (NaCl) as a molecular (covalent) solid, a covalent network solid, an ionic solid, or a metallic solid, and predict whether it has a relatively high or low melting point. Explain.

ANSWER: _____

953. State the type(s) of bonding present in CaCl_2 . Explain your answer.

ANSWER: _____

954. An oil tanker has a mass of $5.29 \times 10^8 \text{ kg}$ and an atom of oxygen has a mass of $6.165 \times 10^{-21} \text{ g}$. How many moles of oil tankers have a mass equal to the mass of an atom? Ensure your answer has the correct number of significant figures.

ANSWER: _____

955. Sean weighs out 2.675 g of nitrous acid (HNO_2) into a 40.-mL volumetric flask and dilutes the solution to the mark with distilled water. He plans to titrate the acid with 0.3226-M magnesium hydroxide solution. Calculate the volume of magnesium hydroxide solution that Sean must add to reach the final equivalence point. Ensure your answer has the correct number of significant figures.

ANSWER: _____

956. The vapor pressure of ethanol ($\text{C}_2\text{H}_5\text{OH}$) is 0.2064 atm at 60.00°C and 1.000 atm at 78.40°C . Calculate the molar enthalpy of vaporization.

ANSWER: _____

957. Manganese commonly occurs in nature as a mineral. The extraction of manganese from the carbonite mineral rhodochrosite involves a two-step process. In the first step, manganese (II) carbonate and oxygen react to form manganese (IV) oxide and carbon dioxide. In the second step, manganese (IV) oxide and aluminum react to form manganese and aluminum oxide. Write the net chemical equation for the production of manganese from manganese (II) carbonate, oxygen, and aluminum. Ensure that it is balanced.

ANSWER: _____

958. Classify AgBr as either an ionic or a molecular compound. Explain why you chose your answer.

ANSWER: _____

959. Construct the valence molecular-orbital energy-level diagram for B_2 and fill in its electrons. Determine the bond order and whether B_2 is paramagnetic or diamagnetic. Justify your conclusions.

ANSWER: _____

960. An atom can form at most 0π bonds. What is its hybridization? Justify your answer.

ANSWER: _____

961. Copper crystallizes in a FCC unit cell with an atomic radius of 128 pm. Calculate the edge length of the unit cell in pm.

ANSWER: _____

962. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is 7.50×10^{-14} m. What is the wavelength in femtometers? Report your answer with units. Report your answer in scientific notation.

ANSWER: _____

963. Dylan is preparing a deep-sea submersible. A rigid spherical air tank has an internal diameter of 66.9-cm. After filling, the pressure inside the tank is 184 atm and the temperature is 281 K. Determine the amount of air stored in the tank. Report your answer in SI units and justify your answer.

ANSWER: _____

964. A student argues that an isolated carbon atom should already contain four sp^3 hybrid orbitals before it forms any bonds. Is the claim correct? Justify your answer.

ANSWER: _____

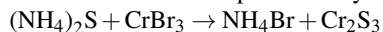
965. Classify frozen argon, a solid noble gas (Ar) as ionic, metallic, covalent network, or molecular, and predict its melting point (high or low) and electrical conductivity. Justify your answer.

ANSWER: _____

966. The chemical formula for iron (III) sulfide is Fe_2S_3 . Calculate the molar mass of iron (III) sulfide. Round your answer to 4 significant figures.

ANSWER: _____

967. A way to produce solid chromium (III) sulfide is by combining aqueous ammonium sulfide with aqueous chromium (III) bromide. The chemical reaction is represented by the following unbalanced chemical equation:



Knowing this, Justin pours 20.6 mL of a 0.71-M solution of aqueous ammonium sulfide and then adds 10.8 mL of a 0.166-M solution of aqueous chromium (III) bromide. He then stirs the mixture and notices chromium (III) sulfide precipitating out of the solution. What is the mass of chromium (III) sulfide that he should expect the reaction to yield?

ANSWER: _____

968. Rank the following bonds in order of increasing bond length: O=O, O–O. Support your ranking with a brief explanation for credit. Credit is not awarded for simply citing a memorized trend.

ANSWER: _____

969. For ammonia (NH_3), $\Delta H_{\text{fus}} = 5.66 \text{ kJ/mol}$ and $\Delta H_{\text{vap}} = 23.4 \text{ kJ/mol}$. Explain why boiling absorbs so much more energy than melting.

ANSWER: _____

970. Equal masses (1.00 g each) of $\text{C}_6\text{H}_{12}\text{O}_6$, AlCl_3 , LiCl are each dissolved in 1.00 kg of water. Which solution has the greatest boiling point elevation?

ANSWER: _____

971. Determine which bond is expected to have the smallest bond dissociation energy from the following bonds: the C–O bond in ethanol ($\text{C}_2\text{H}_5\text{OH}$), the N–O bond in nitrate (NO_3^-), the S=O bond in sulfurous acid (H_2SO_3), the C–C bond in ethyne (C_2H_2), the P–O bond in phosphate (PO_4^{3-}), and the N–O bond in nitrogen dioxide (NO_2). Justify your answer from fundamental principles. Responses based on memorized trends or rules will not receive credit.

ANSWER: _____

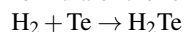
972. Identify the hybridization of each carbon atom in $\text{CH}_3-\text{C}(=\text{O})-\text{CH}_3$ (acetone). Explain your answer.

ANSWER: _____

973. The vapor pressure of a volatile solvent is measured at 55°C . The sample is then heated to 68°C in a sealed container while liquid remains present. Determine whether the vapor pressure increases, decreases, or remains unchanged. Justify your answer.

ANSWER: _____

974. Determine whether the following reaction is an oxidation-reduction (redox) reaction. If the reaction is a redox reaction, write down the formula of the reducing agent and the formula of the oxidizing agent. Provide an explanation for why you made these choices.



ANSWER: _____

975. Calculate the molar solubility of CH_4 (CH_4) in water at 25°C when the partial pressure of CH_4 above the solution is 4.0 atm. Use $K_H = 1.4 \times 10^{-3} \text{ mol}/(\text{L}\cdot\text{atm})$ (Henry's law: $C = K_H P$).

ANSWER: _____

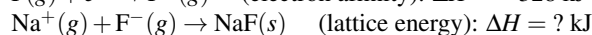
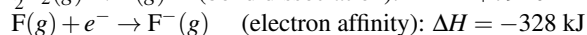
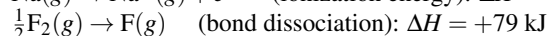
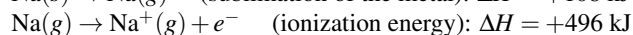
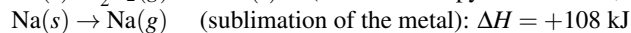
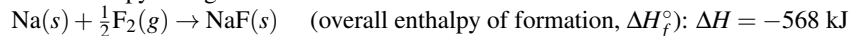
976. List the following electron subshells in order of increasing energy. Assume these subshells are in an atom with many electrons.
2s, 5s, 4p, 1s, 2p

ANSWER: _____

977. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is $8.76 \times 10^{-5} \text{ g}$. What is the mass in micrograms? Report your answer with units. Report your answer in standard notation, as a decimal.

ANSWER: _____

978. Use a Born-Haber cycle (Hess's law) to determine the missing enthalpy change for the formation of $\text{NaF}(s)$ from its elements. The steps and their enthalpy changes are:



ANSWER: _____

979. Calculate the heat required to completely vaporize 14.4-g of methanol (CH_3OH) at its normal boiling point of 65°C . $\Delta H_{\text{vap}} = 35.2 \text{ kJ/mol}$, $M = 32.042 \text{ g/mol}$.

ANSWER: _____

980. A 0.500M solution of NaCl (NaCl) has a density of 1.02 g/mL. Calculate the molality of the solution. ($M = 58.44 \text{ g/mol}$)

ANSWER: _____

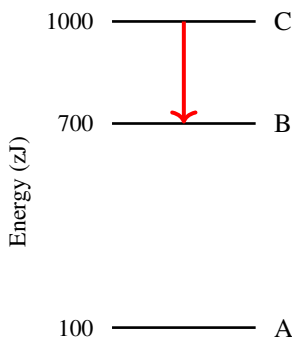
981. A solid shows identical properties in every direction (isotropic) and has no sharp melting point. Is it more likely crystalline or amorphous? Justify.

ANSWER: _____

982. A refrigeration technician notes that a storage unit is set to 131 °F. Convert this temperature to degrees Celsius. Round your answer to the nearest integer.

ANSWER: _____

983. This energy diagram shows the allowed energy levels of an electron in a certain atom. Suppose the electron makes a transition from C to B, as shown by the arrow. Calculate the wavelength of the photon that would be absorbed or emitted. Report your answer to three significant figures. Note that the SI prefix zepto (z) means 10^{-21} .



ANSWER: _____

984. Does the following process produce a new substance? A copper wire is bent into a coil. You must explain your answer with valid reasoning.

ANSWER: _____

985. A flexible latex balloon contains 2.49L of helium at -14°C while indoors. It is then taken outdoors, where the air temperature is 29°C . Assuming the balloon remains flexible and reaches thermal equilibrium with its surroundings, determine its final volume. Justify your answer.

ANSWER: _____

986. Which species is predicted to be more stable by MO theory: N_2 or O_2 ? Justify using bond orders.

ANSWER: _____

987. Identify the main group to which X belongs, given the ionic compound SrX_2 . Explain your answer.

ANSWER: _____

988. Ethanol ($\text{C}_2\text{H}_5\text{OH}$, density 0.790 g/mL) is shaken with an equal volume of water (1.00 g/mL). Will two layers form? If so, which liquid ends up on top? Explain.

ANSWER: _____

989. A system consists of pure water at its triple point. Determine its degrees of freedom using the Gibbs phase rule, then state what that result means. Justify your answer.

ANSWER: _____

990. Iron is extracted from its oxide ores in a two-step process that has been practiced for centuries. In the first step, iron (III) oxide reacts with carbon monoxide to produce iron (II) oxide and carbon dioxide. In the second step, iron (II) oxide further reacts with carbon monoxide to produce elemental iron and more carbon dioxide. Write the net chemical equation for the production of iron from iron (III) oxide and carbon monoxide. Ensure that it is balanced.

ANSWER: _____

991. A red blood cell (internal particle concentration ≈ 0.300 osmol/L) is placed in a 0.500 M solution of urea ($\text{CH}_4\text{N}_2\text{O}$). Which way does water move across the membrane, and what happens to the cell? Explain your reasoning; answers without justification earn no credit.

ANSWER: _____

992. Ethanol ($\text{C}_2\text{H}_5\text{OH}$) has a normal boiling point of 78°C and $\Delta H_{\text{vap}} = 38.6$ kJ/mol. At what temperature does it boil at high altitude, where the external pressure is 0.65 atm? ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$.)

ANSWER: _____

993. The reaction $\text{Fe(s)} + 2\text{HCl(aq)} \longrightarrow \text{FeCl}_2\text{(aq)} + \text{H}_2\text{(g)}$ is carried out in a sealed, initially evacuated vessel with a volume of 8.65 L. After 6.10 g of HCl completely reacts while all other reactants are present in excess at 7.4°C, determine the pressure of H_2 . Ignore the volume occupied by any solids or liquids. Report your answer in SI units and justify your answer.

ANSWER: _____

994. Ian measures the density of a metal sample in his lab. He finds that the sample has a mass of 187.2 g and a volume of 50.37 mL. What is the density of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

995. What major chemical species are present in an aqueous solution of 0.23-M chromium (III) sulfite?

ANSWER: _____

996. Suppose one carbon atom in benzene could not contribute an unhybridized p orbital. Would benzene still contain a single delocalized π system extending around the entire ring? Justify your answer.

ANSWER: _____

997. A European researcher reports an experimental temperature for a U.S. publication. The temperature is -25 °C. Convert this temperature to degrees Fahrenheit and round to the nearest integer.

ANSWER: _____

998. The following information is available for nickel. Determine the density of nickel and justify your answer.
- unit-cell edge length = 352.40 pm
 - crystal structure = FCC
 - atomic radius = 124 pm
 - atoms per unit cell = 4
 - molar mass = 58.693 g mol⁻¹
 - coordination number = 12

ANSWER: _____

999. Phosphoric acid, which most commonly finds uses as a rust inhibitor, a food additive, and an etching agent for dental and orthopedic use, can be synthesized using a two-step thermal process. In the first step, tetratomic phosphorus and diatomic oxygen react to form diphosphorus pentoxide. In the second step, diphosphorus pentoxide and water react to form phosphoric acid. Write the net chemical equation for the production of phosphoric acid from tetratomic phosphorus, diatomic oxygen, and water. Ensure that it is balanced.

ANSWER: _____

1000. Using MO theory, rank the following diatomic molecules in order of increasing bond order: N_2 , B_2 , Li_2 . Justify your ranking by explaining how the MO electron occupancies lead to the observed trend.

ANSWER: _____